TYPE(REFERENCE_MANUAL) :: CODATA = V2.5.2

! Fortran library for fundamental physical constants

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June 1, 2026

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Preface

This document provides a reference manual for the *codata* library. The documentation is PDF-centric and authored in LaTeX in order to provide a stable, high-quality printout. The PDF edition preserves the intended layout, typography, equations, figures and cross-references, ensuring a consistent reading experience across platforms and over time. A HTML version of the documentation is rendered with *make4ht*.

In addition, POSIX-style man pages are generated from the sources using the *prep(1)* preprocessor and are provided as both terminal-native documentation in the binaries and as a PDF document.

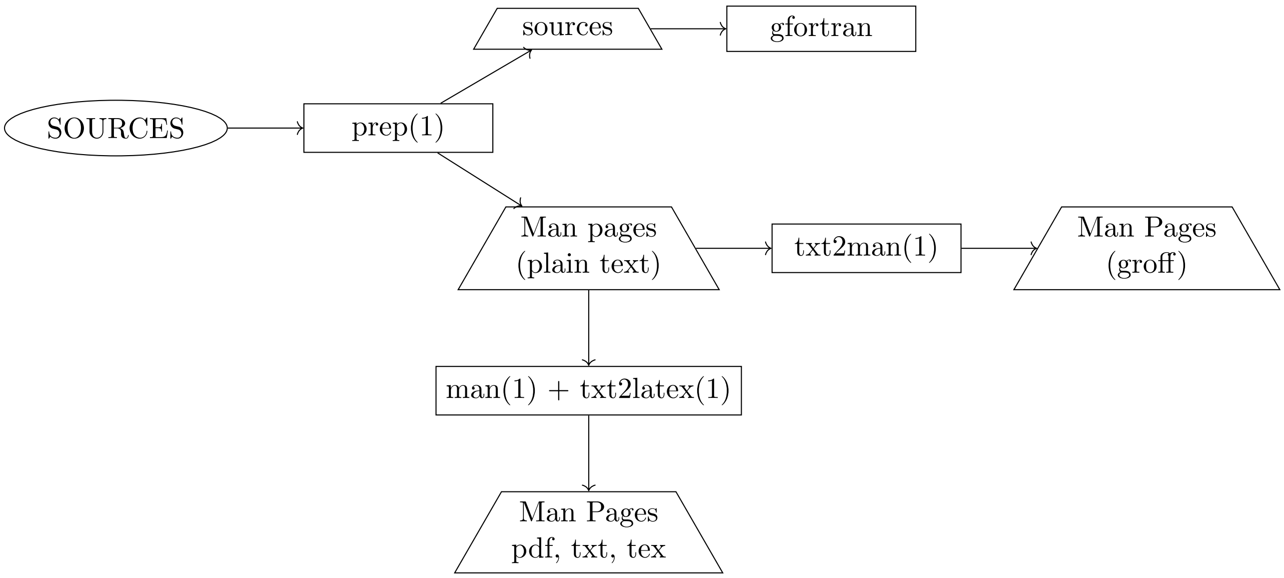
Additional HTML versions are also available for users who prefer quick navigation, keyword search, or rapid access to specific sections. However, those versions provide only the documentation of the API generated from the docstrings using *Sphinx* and *FORD*. They are provided as a convenient complement to the present PDF documentation, and should not be considered as a replacement for the present reference manual. The *Sphinx* HTML version presents the APIs for Fortran, C, and Python — whereas, the *FORD* HTML version is Fortran-specific and is primarily intended for Fortran users.

Related Documentation:

- Reference Manual [PDF]: `https://milanskocic.github.io/codata/latex/codata-refman.pdf`
- Reference Manual [HTML]: `https://milanskocic.github.io/codata`
- Man Pages [GROFF, TXT, PDF]: `https://milanskocic.github.io/codata/man/`
- APIs with Sphinx [HTML]: `https://milanskocic.github.io/codata/sphinx`
- API with FORD [HTML]: `https://milanskocic.github.io/codata/ford`

Related resources:

- Binary releases: `https://github.com/MilanSkocic/codata/releases`
- *prep(1)*: `https://github.com/urbanjost/prep`
- *make4ht*: `https://github.com/michal-h21/make4ht`
- Sphinx: `https://github.com/fortran-lang/sphinx-fortran-domain`
- FORD: `https://github.com/Fortran-FOSS-Programmers/ford`


SOURCES
prep(1)
sources
gfortran
Man pages
(plain text)
txt2man(1)
Man Pages
(groff)
man(1) + txt2latex(1)
Man Pages
pdf, txt, tex

Part I

Introduction

Part II

Installation

Part III

Theoretical Background

Part IV

API

Part V

Examples and Tutorials

Chapter 1

Examples

```fortran
program example_in_f
use codata
implicit none
print '(A)', '########## EXAMPLE IN FORTRAN ##########'
print '(A)', '# VERSION'
print *, "version = ", version()
print '(A)', '# CONSTANTS'
print *, "c = ",  SPEED_OF_LIGHT_IN_VACUUM%value
print '(A)', '# UNCERTAINTY'
print *, "u(c) = ", SPEED_OF_LIGHT_IN_VACUUM%uncertainty
print '(A)', '# OLDER VALUES'
print '(A, F23.16)', "Mu_2022(latest) = ", MOLAR_MASS_CONSTANT%value
print '(A, F23.16)', "Mu_2018 = ", MOLAR_MASS_CONSTANT_2018%value
print '(A, F23.16)', "Mu_2014 = ",  MOLAR_MASS_CONSTANT_2014%value
print '(A, F23.16)', "Mu_2010 = ",  MOLAR_MASS_CONSTANT_2010%value
end program
```

```c
#include <stdio.h>
#include "codata.h"
int main(void){
printf("########## EXAMPLE IN C ##########\n");
printf("%s\n","# VERSION");
printf("version = %s\n", codata_version());
printf("%s\n","# CONSTANTS");
printf("c = %f\n", SPEED_OF_LIGHT_IN_VACUUM.value);
printf("%s\n","# UNCERTAINTY");
printf("u(c) = %f\n", SPEED_OF_LIGHT_IN_VACUUM.uncertainty);
printf("%s\n","# OLDER VALUES");
printf("Mu_2022(latest) = %23.16f\n", MOLAR_MASS_CONSTANT.value);
printf("Mu_2018 = %23.16f\n", MOLAR_MASS_CONSTANT_2018.value);
printf("Mu_2014 = %23.16f\n", MOLAR_MASS_CONSTANT_2014.value);
printf("Mu_2010 = %23.16f\n", MOLAR_MASS_CONSTANT_2010.value);
return 0;
}
```

```python
import sys
sys.path.insert(0, "../py/src/")
import pycodata
print("########## EXAMPLE IN PYTHON ##########")
print("# VERSION")
```

```
print(f"version = {pycodata.__version__}")
print("# Constants")
print("c =", pycodata.SPEED_OF_LIGHT_IN_VACUUM["value"])
print("# UNCERTAINTY")
print("u(c) = ", pycodata.SPEED_OF_LIGHT_IN_VACUUM["uncertainty"])
print("# OLDER VALUES")
print("Mu_2022 = ", pycodata.MOLAR_MASS_CONSTANT["value"])
print("Mu_2018 = ", pycodata.MOLAR_MASS_CONSTANT_2018["value"])
print("Mu_2014 = ", pycodata.MOLAR_MASS_CONSTANT_2014["value"])
print("Mu_2010 = ", pycodata.MOLAR_MASS_CONSTANT_2010["value"])
```

Part VI

Man Pages

Chapter 2

codata(3)

NAME

codata - libray for fundamental physical constants

LIBRARY

codata (-libcodata, -lcodata)

SYNOPSIS

`use codata include "codata.h" import pycodata`

DESCRIPTION

codata is a Fortran library providing the fundamental physical constants according to CODATA (https://www.nist.gov/programs-projects/codata-values-fundamental-physical-constants). A C API allows usage from C, or can be used as a basis for other wrappers. A python wrapper allows easy usage from Python.

The latest codata constants (2022) https://pml.nist.gov/cuu/Constants were integrated in stdlib https://github.com/fortran-lang/stdlib/releases/tag/ since version 0.7.0. The constants are implemented as derived type which carries the name, the value, the uncertainty and the unit. This library is complementary to the constants defined in the stdlib by providing older values for the constants. The latest values (2022) do not have the year as a suffix in their name whereas older values (2010, 2014, 2018) feature the year as a suffix in their name.

All codata (physical) constants are defined as a derived type codata_constant_type. All the codata constants are provided as double precision reals. The names are quite long and can be aliased with shorter names. The derived type codata_constant_type defines 4 members and 2 procedures.

```
type :: codata_constant_type
    character(len=64) :: name
    real(dp) :: value
    real(dp) :: uncertainty
    character(len=32) :: unit
contains
    procedure :: print
    procedure :: to_real_sp
```

```
        procedure :: to_real_dp
        generic :: to_real => to_real_sp, to_real_dp
    end type codata_constant_type
```

A module level interface to_real is available for getting the constant value or uncertainty of a constant.

o to_real(self, mold, uncertainty) result(r) Get the constant value or uncertainty. Warning: Some constants cannot be converted to single precision sp reals due to the value of the exponents.

o class(codata_constant_type), intent(in) :: self Codata constant

o real(sp), intent(in) :: mold Should be of type real single or double precision

o logical, intent(in), optional :: uncertainty Set to true if the uncertainty is required. Default to .false..

The C API exposes a structure codata_constant_type that defines the same members as in Fortran. to_real is not exposed in the C API.

```
    type, bind(C) :: capi_constant_type
        character(kind=c_char) :: name(65)
        real(c_double) :: value
        real(c_double) :: uncertainty
        character(kind=c_char) :: unit(33)
    end type capi_constant_type

    typedef struct codata_constant_type{
        char name[65];
        double value;
        double uncertainty;
        char unit[33];
    }cct;
```

The Python wrapper encapsulates the members in a dictionnary with the keys being name, value, uncertainty and unit. to_real is not exposed in the Python wrapper.

NOTES

To use codata within your fpm(1) (https://github.com/fortran-lang/fpm) project, add the following lines to your file:

```
[dependencies]
codata = { git="https://github.com/MilanSkocic/codata.git" }
```

EXAMPLE

Example in Fortran

```
program example_in_f
use codata
implicit none
print '(A)', '########## EXAMPLE IN FORTRAN ##########'
print '(A)', '# VERSION'
print *, "version = ", version()
print '(A)', '# CONSTANTS'
print *, "c = ",  SPEED_OF_LIGHT_IN_VACUUM%value
print '(A)', '# UNCERTAINTY'
print *, "u(c) = ", SPEED_OF_LIGHT_IN_VACUUM%uncertainty
print '(A)', '# OLDER VALUES'
print '(A, F23.16)', "Mu_2022(latest) = ", MOLAR_MASS_CONSTANT%value
print '(A, F23.16)', "Mu_2018 = ", MOLAR_MASS_CONSTANT_2018%value
print '(A, F23.16)', "Mu_2014 = ",  MOLAR_MASS_CONSTANT_2014%value
print '(A, F23.16)', "Mu_2010 = ",  MOLAR_MASS_CONSTANT_2010%value
end program
```

Example in C

```
#include <stdio.h>
#include "codata.h"
int main(void){
printf("########## EXAMPLE IN C ##########\n");
printf("%s\n","# VERSION");
printf("version = %s\n", codata_version());
printf("%s\n","# CONSTANTS");
printf("c = %f\n", SPEED_OF_LIGHT_IN_VACUUM.value);
printf("%s\n","# UNCERTAINTY");
printf("u(c) = %f\n", SPEED_OF_LIGHT_IN_VACUUM.uncertainty);
printf("%s\n","# OLDER VALUES");
printf("Mu_2022(latest) = %23.16f\n", MOLAR_MASS_CONSTANT.value);
printf("Mu_2018 = %23.16f\n", MOLAR_MASS_CONSTANT_2018.value);
printf("Mu_2014 = %23.16f\n", MOLAR_MASS_CONSTANT_2014.value);
printf("Mu_2010 = %23.16f\n", MOLAR_MASS_CONSTANT_2010.value);
return 0;
}
```

Example in Python

```
import sys
sys.path.insert(0, "../py/src/")
import pycodata
print("########## EXAMPLE IN PYTHON ##########")
print("# VERSION")
print(f"version = {pycodata.__version__}")
print("# Constants")
print("c =", pycodata.SPEED_OF_LIGHT_IN_VACUUM["value"])
```

```
print("# UNCERTAINTY")
print("u(c) = ", pycodata.SPEED_OF_LIGHT_IN_VACUUM["uncertainty"])
print("# OLDER VALUES")
print("Mu_2022 = ", pycodata.MOLAR_MASS_CONSTANT["value"])
print("Mu_2018 = ", pycodata.MOLAR_MASS_CONSTANT_2018["value"])
print("Mu_2014 = ", pycodata.MOLAR_MASS_CONSTANT_2014["value"])
print("Mu_2010 = ", pycodata.MOLAR_MASS_CONSTANT_2010["value"])
```

SEE ALSO

gsl(3), codata(1), fpm(1)

Chapter 3

codata_2010(3)

NAME

codata - libray for fundamental physical constants

LIBRARY

codata (-libcodata, -lcodata)

SYNOPSIS

```
use codata
   include "codata.h"
   import pycodata
```

DESCRIPTION

codata is a Fortran library providing the fundamental physical constants according to CODATA
List of available constants:

- LATTICE_SPACING_OF_SILICON_2010
- ALPHA_PARTICLE_ELECTRON_MASS_RATIO_2010
- ALPHA_PARTICLE_MASS_2010
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_2010
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- ALPHA_PARTICLE_MASS_IN_U_2010
- ALPHA_PARTICLE_MOLAR_MASS_2010
- ALPHA_PARTICLE_PROTON_MASS_RATIO_2010
- ANGSTROM_STAR_2010
- ATOMIC_MASS_CONSTANT_2010
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_2010
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_IN_MEV_2010

- ATOMIC_MASS_UNIT_ELECTRON_VOLT_RELATIONSHIP_2010
- ATOMIC_MASS_UNIT_HARTREE_RELATIONSHIP_2010
- ATOMIC_MASS_UNIT_HERTZ_RELATIONSHIP_2010
- ATOMIC_MASS_UNIT_INVERSE_METER_RELATIONSHIP_2010
- ATOMIC_MASS_UNIT_JOULE_RELATIONSHIP_2010
- ATOMIC_MASS_UNIT_KELVIN_RELATIONSHIP_2010
- ATOMIC_MASS_UNIT_KILOGRAM_RELATIONSHIP_2010
- ATOMIC_UNIT_OF_1ST_HYPERPOLARIZABILITY_2010
- ATOMIC_UNIT_OF_2ND_HYPERPOLARIZABILITY_2010
- ATOMIC_UNIT_OF_ACTION_2010
- ATOMIC_UNIT_OF_CHARGE_2010
- ATOMIC_UNIT_OF_CHARGE_DENSITY_2010
- ATOMIC_UNIT_OF_CURRENT_2010
- ATOMIC_UNIT_OF_ELECTRIC_DIPOLE_MOM_2010
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_2010
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_GRADIENT_2010
- ATOMIC_UNIT_OF_ELECTRIC_POLARIZABILITY_2010
- ATOMIC_UNIT_OF_ELECTRIC_POTENTIAL_2010
- ATOMIC_UNIT_OF_ELECTRIC_QUADRUPOLE_MOM_2010
- ATOMIC_UNIT_OF_ENERGY_2010
- ATOMIC_UNIT_OF_FORCE_2010
- ATOMIC_UNIT_OF_LENGTH_2010
- ATOMIC_UNIT_OF_MAG_DIPOLE_MOM_2010
- ATOMIC_UNIT_OF_MAG_FLUX_DENSITY_2010
- ATOMIC_UNIT_OF_MAGNETIZABILITY_2010
- ATOMIC_UNIT_OF_MASS_2010
- ATOMIC_UNIT_OF_MOMUM_2010
- ATOMIC_UNIT_OF_PERMITTIVITY_2010
- ATOMIC_UNIT_OF_TIME_2010
- ATOMIC_UNIT_OF_VELOCITY_2010
- AVOGADRO_CONSTANT_2010
- BOHR_MAGNETON_2010

- BOHR_MAGNETON_IN_EV_T_2010
- BOHR_MAGNETON_IN_HZ_T_2010
- BOHR_MAGNETON_IN_INVERSE_METERS_PER_TESLA_2010
- BOHR_MAGNETON_IN_K_T_2010
- BOHR_RADIUS_2010
- BOLTZMANN_CONSTANT_2010
- BOLTZMANN_CONSTANT_IN_EV_K_2010
- BOLTZMANN_CONSTANT_IN_HZ_K_2010
- BOLTZMANN_CONSTANT_IN_INVERSE_METERS_PER_KELVIN_2010
- CHARACTERISTIC_IMPEDANCE_OF_VACUUM_2010
- CLASSICAL_ELECTRON_RADIUS_2010
- COMPTON_WAVELENGTH_2010
- COMPTON_WAVELENGTH_OVER_2_PI_2010
- CONDUCTANCE_QUANTUM_2010
- CONVENTIONAL_VALUE_OF_JOSEPHSON_CONSTANT_2010
- CONVENTIONAL_VALUE_OF_VON_KLITZING_CONSTANT_2010
- CU_X_UNIT_2010
- DEUTERON_ELECTRON_MAG_MOM_RATIO_2010
- DEUTERON_ELECTRON_MASS_RATIO_2010
- DEUTERON_G_FACTOR_2010
- DEUTERON_MAG_MOM_2010
- DEUTERON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- DEUTERON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- DEUTERON_MASS_2010
- DEUTERON_MASS_ENERGY_EQUIVALENT_2010
- DEUTERON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- DEUTERON_MASS_IN_U_2010
- DEUTERON_MOLAR_MASS_2010
- DEUTERON_NEUTRON_MAG_MOM_RATIO_2010
- DEUTERON_PROTON_MAG_MOM_RATIO_2010
- DEUTERON_PROTON_MASS_RATIO_2010
- DEUTERON_RMS_CHARGE_RADIUS_2010

- ELECTRIC_CONSTANT_2010
- ELECTRON_CHARGE_TO_MASS_QUOTIENT_2010
- ELECTRON_DEUTERON_MAG_MOM_RATIO_2010
- ELECTRON_DEUTERON_MASS_RATIO_2010
- ELECTRON_G_FACTOR_2010
- ELECTRON_GYROMAG_RATIO_2010
- ELECTRON_GYROMAG_RATIO_OVER_2_PI_2010
- ELECTRON_HELION_MASS_RATIO_2010
- ELECTRON_MAG_MOM_2010
- ELECTRON_MAG_MOM_ANOMALY_2010
- ELECTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- ELECTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- ELECTRON_MASS_2010
- ELECTRON_MASS_ENERGY_EQUIVALENT_2010
- ELECTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- ELECTRON_MASS_IN_U_2010
- ELECTRON_MOLAR_MASS_2010
- ELECTRON_MUON_MAG_MOM_RATIO_2010
- ELECTRON_MUON_MASS_RATIO_2010
- ELECTRON_NEUTRON_MAG_MOM_RATIO_2010
- ELECTRON_NEUTRON_MASS_RATIO_2010
- ELECTRON_PROTON_MAG_MOM_RATIO_2010
- ELECTRON_PROTON_MASS_RATIO_2010
- ELECTRON_TAU_MASS_RATIO_2010
- ELECTRON_TO_ALPHA_PARTICLE_MASS_RATIO_2010
- ELECTRON_TO_SHIELDED_HELION_MAG_MOM_RATIO_2010
- ELECTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2010
- ELECTRON_TRITON_MASS_RATIO_2010
- ELECTRON_VOLT_2010
- ELECTRON_VOLT_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- ELECTRON_VOLT_HARTREE_RELATIONSHIP_2010
- ELECTRON_VOLT_HERTZ_RELATIONSHIP_2010

- ELECTRON_VOLT_INVERSE_METER_RELATIONSHIP_2010
- ELECTRON_VOLT_JOULE_RELATIONSHIP_2010
- ELECTRON_VOLT_KELVIN_RELATIONSHIP_2010
- ELECTRON_VOLT_KILOGRAM_RELATIONSHIP_2010
- ELEMENTARY_CHARGE_2010
- ELEMENTARY_CHARGE_OVER_H_2010
- FARADAY_CONSTANT_2010
- FARADAY_CONSTANT_FOR_CONVENTIONAL_ELECTRIC_CURRENT_2010
- FERMI_COUPLING_CONSTANT_2010
- FINE_STRUCTURE_CONSTANT_2010
- FIRST_RADIATION_CONSTANT_2010
- FIRST_RADIATION_CONSTANT_FOR_SPECTRAL_RADIANCE_2010
- HARTREE_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
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- HARTREE_HERTZ_RELATIONSHIP_2010
- HARTREE_INVERSE_METER_RELATIONSHIP_2010
- HARTREE_JOULE_RELATIONSHIP_2010
- HARTREE_KELVIN_RELATIONSHIP_2010
- HARTREE_KILOGRAM_RELATIONSHIP_2010
- HELION_ELECTRON_MASS_RATIO_2010
- HELION_G_FACTOR_2010
- HELION_MAG_MOM_2010
- HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- HELION_MASS_2010
- HELION_MASS_ENERGY_EQUIVALENT_2010
- HELION_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- HELION_MASS_IN_U_2010
- HELION_MOLAR_MASS_2010
- HELION_PROTON_MASS_RATIO_2010

- HERTZ_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- HERTZ_ELECTRON_VOLT_RELATIONSHIP_2010
- HERTZ_HARTREE_RELATIONSHIP_2010
- HERTZ_INVERSE_METER_RELATIONSHIP_2010
- HERTZ_JOULE_RELATIONSHIP_2010
- HERTZ_KELVIN_RELATIONSHIP_2010
- HERTZ_KILOGRAM_RELATIONSHIP_2010
- INVERSE_FINE_STRUCTURE_CONSTANT_2010
- INVERSE_METER_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- INVERSE_METER_ELECTRON_VOLT_RELATIONSHIP_2010
- INVERSE_METER_HARTREE_RELATIONSHIP_2010
- INVERSE_METER_HERTZ_RELATIONSHIP_2010
- INVERSE_METER_JOULE_RELATIONSHIP_2010
- INVERSE_METER_KELVIN_RELATIONSHIP_2010
- INVERSE_METER_KILOGRAM_RELATIONSHIP_2010
- INVERSE_OF_CONDUCTANCE_QUANTUM_2010
- JOSEPHSON_CONSTANT_2010
- JOULE_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- JOULE_ELECTRON_VOLT_RELATIONSHIP_2010
- JOULE_HARTREE_RELATIONSHIP_2010
- JOULE_HERTZ_RELATIONSHIP_2010
- JOULE_INVERSE_METER_RELATIONSHIP_2010
- JOULE_KELVIN_RELATIONSHIP_2010
- JOULE_KILOGRAM_RELATIONSHIP_2010
- KELVIN_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- KELVIN_ELECTRON_VOLT_RELATIONSHIP_2010
- KELVIN_HARTREE_RELATIONSHIP_2010
- KELVIN_HERTZ_RELATIONSHIP_2010
- KELVIN_INVERSE_METER_RELATIONSHIP_2010
- KELVIN_JOULE_RELATIONSHIP_2010
- KELVIN_KILOGRAM_RELATIONSHIP_2010
- KILOGRAM_ATOMIC_MASS_UNIT_RELATIONSHIP_2010

- KILOGRAM_ELECTRON_VOLT_RELATIONSHIP_2010
- KILOGRAM_HARTREE_RELATIONSHIP_2010
- KILOGRAM_HERTZ_RELATIONSHIP_2010
- KILOGRAM_INVERSE_METER_RELATIONSHIP_2010
- KILOGRAM_JOULE_RELATIONSHIP_2010
- KILOGRAM_KELVIN_RELATIONSHIP_2010
- LATTICE_PARAMETER_OF_SILICON_2010
- LOSCHMIDT_CONSTANT_273_15_K_100_KPA_2010
- LOSCHMIDT_CONSTANT_273_15_K_101_325_KPA_2010
- MAG_CONSTANT_2010
- MAG_FLUX_QUANTUM_2010
- MOLAR_GAS_CONSTANT_2010
- MOLAR_MASS_CONSTANT_2010
- MOLAR_MASS_OF_CARBON_12_2010
- MOLAR_PLANCK_CONSTANT_2010
- MOLAR_PLANCK_CONSTANT_TIMES_C_2010
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_100_KPA_2010
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_101_325_KPA_2010
- MOLAR_VOLUME_OF_SILICON_2010
- MO_X_UNIT_2010
- MUON_COMPTON_WAVELENGTH_2010
- MUON_COMPTON_WAVELENGTH_OVER_2_PI_2010
- MUON_ELECTRON_MASS_RATIO_2010
- MUON_G_FACTOR_2010
- MUON_MAG_MOM_2010
- MUON_MAG_MOM_ANOMALY_2010
- MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- MUON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- MUON_MASS_2010
- MUON_MASS_ENERGY_EQUIVALENT_2010
- MUON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- MUON_MASS_IN_U_2010

- MUON_MOLAR_MASS_2010
- MUON_NEUTRON_MASS_RATIO_2010
- MUON_PROTON_MAG_MOM_RATIO_2010
- MUON_PROTON_MASS_RATIO_2010
- MUON_TAU_MASS_RATIO_2010
- NATURAL_UNIT_OF_ACTION_2010
- NATURAL_UNIT_OF_ACTION_IN_EV_S_2010
- NATURAL_UNIT_OF_ENERGY_2010
- NATURAL_UNIT_OF_ENERGY_IN_MEV_2010
- NATURAL_UNIT_OF_LENGTH_2010
- NATURAL_UNIT_OF_MASS_2010
- NATURAL_UNIT_OF_MOMUM_2010
- NATURAL_UNIT_OF_MOMUM_IN_MEV_C_2010
- NATURAL_UNIT_OF_TIME_2010
- NATURAL_UNIT_OF_VELOCITY_2010
- NEUTRON_COMPTON_WAVELENGTH_2010
- NEUTRON_COMPTON_WAVELENGTH_OVER_2_PI_2010
- NEUTRON_ELECTRON_MAG_MOM_RATIO_2010
- NEUTRON_ELECTRON_MASS_RATIO_2010
- NEUTRON_G_FACTOR_2010
- NEUTRON_GYROMAG_RATIO_2010
- NEUTRON_GYROMAG_RATIO_OVER_2_PI_2010
- NEUTRON_MAG_MOM_2010
- NEUTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- NEUTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- NEUTRON_MASS_2010
- NEUTRON_MASS_ENERGY_EQUIVALENT_2010
- NEUTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- NEUTRON_MASS_IN_U_2010
- NEUTRON_MOLAR_MASS_2010
- NEUTRON_MUON_MASS_RATIO_2010
- NEUTRON_PROTON_MAG_MOM_RATIO_2010

- NEUTRON_PROTON_MASS_DIFFERENCE_2010
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_2010
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- NEUTRON_PROTON_MASS_DIFFERENCE_IN_U_2010
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- WIEN_WAVELENGTH_DISPLACEMENT_LAW_CONSTANT_2010

SEE ALSO

codata(3), codata_2010(3), codata_2014(3), codata_2018(3), codata_2022(3)

Chapter 4

codata_2014(3)

NAME

codata - libray for fundamental physical constants

LIBRARY

codata (-libcodata, -lcodata)

SYNOPSIS

```
use codata
   include "codata.h"
   import pycodata
```

DESCRIPTION

codata is a Fortran library providing the fundamental physical constants according to CODATA
List of available constants:

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- TRITON_MAG_MOM_2014
- TRITON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- TRITON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- TRITON_MASS_2014
- TRITON_MASS_ENERGY_EQUIVALENT_2014
- TRITON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- TRITON_MASS_IN_U_2014
- TRITON_MOLAR_MASS_2014
- TRITON_PROTON_MASS_RATIO_2014
- UNIFIED_ATOMIC_MASS_UNIT_2014
- VON_KLITZING_CONSTANT_2014

- WEAK_MIXING_ANGLE_2014
- WIEN_FREQUENCY_DISPLACEMENT_LAW_CONSTANT_2014
- WIEN_WAVELENGTH_DISPLACEMENT_LAW_CONSTANT_2014

SEE ALSO

codata(3), codata_2010(3), codata_2014(3), codata_2018(3), codata_2022(3)

Chapter 5

codata_2018(3)

NAME

codata - libray for fundamental physical constants

LIBRARY

codata (-libcodata, -lcodata)

SYNOPSIS

```
use codata
   include "codata.h"
   import pycodata
```

DESCRIPTION

codata is a Fortran library providing the fundamental physical constants according to CODATA
List of available constants:

- ALPHA_PARTICLE_ELECTRON_MASS_RATIO_2018
- ALPHA_PARTICLE_MASS_2018
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_2018
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- ALPHA_PARTICLE_MASS_IN_U_2018
- ALPHA_PARTICLE_MOLAR_MASS_2018
- ALPHA_PARTICLE_PROTON_MASS_RATIO_2018
- ALPHA_PARTICLE_RELATIVE_ATOMIC_MASS_2018
- ANGSTROM_STAR_2018
- ATOMIC_MASS_CONSTANT_2018
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_2018
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_IN_MEV_2018

- ATOMIC_MASS_UNIT_ELECTRON_VOLT_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_HARTREE_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_HERTZ_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_INVERSE_METER_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_JOULE_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_KELVIN_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_KILOGRAM_RELATIONSHIP_2018
- ATOMIC_UNIT_OF_1ST_HYPERPOLARIZABILITY_2018
- ATOMIC_UNIT_OF_2ND_HYPERPOLARIZABILITY_2018
- ATOMIC_UNIT_OF_ACTION_2018
- ATOMIC_UNIT_OF_CHARGE_2018
- ATOMIC_UNIT_OF_CHARGE_DENSITY_2018
- ATOMIC_UNIT_OF_CURRENT_2018
- ATOMIC_UNIT_OF_ELECTRIC_DIPOLE_MOM_2018
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_2018
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_GRADIENT_2018
- ATOMIC_UNIT_OF_ELECTRIC_POLARIZABILITY_2018
- ATOMIC_UNIT_OF_ELECTRIC_POTENTIAL_2018
- ATOMIC_UNIT_OF_ELECTRIC_QUADRUPOLE_MOM_2018
- ATOMIC_UNIT_OF_ENERGY_2018
- ATOMIC_UNIT_OF_FORCE_2018
- ATOMIC_UNIT_OF_LENGTH_2018
- ATOMIC_UNIT_OF_MAG_DIPOLE_MOM_2018
- ATOMIC_UNIT_OF_MAG_FLUX_DENSITY_2018
- ATOMIC_UNIT_OF_MAGNETIZABILITY_2018
- ATOMIC_UNIT_OF_MASS_2018
- ATOMIC_UNIT_OF_MOMENTUM_2018
- ATOMIC_UNIT_OF_PERMITTIVITY_2018
- ATOMIC_UNIT_OF_TIME_2018
- ATOMIC_UNIT_OF_VELOCITY_2018
- AVOGADRO_CONSTANT_2018
- BOHR_MAGNETON_2018

- BOHR_MAGNETON_IN_EV_T_2018
- BOHR_MAGNETON_IN_HZ_T_2018
- BOHR_MAGNETON_IN_INVERSE_METER_PER_TESLA_2018
- BOHR_MAGNETON_IN_K_T_2018
- BOHR_RADIUS_2018
- BOLTZMANN_CONSTANT_2018
- BOLTZMANN_CONSTANT_IN_EV_K_2018
- BOLTZMANN_CONSTANT_IN_HZ_K_2018
- BOLTZMANN_CONSTANT_IN_INVERSE_METER_PER_KELVIN_2018
- CHARACTERISTIC_IMPEDANCE_OF_VACUUM_2018
- CLASSICAL_ELECTRON_RADIUS_2018
- COMPTON_WAVELENGTH_2018
- CONDUCTANCE_QUANTUM_2018
- CONVENTIONAL_VALUE_OF_AMPERE_90_2018
- CONVENTIONAL_VALUE_OF_COULOMB_90_2018
- CONVENTIONAL_VALUE_OF_FARAD_90_2018
- CONVENTIONAL_VALUE_OF_HENRY_90_2018
- CONVENTIONAL_VALUE_OF_JOSEPHSON_CONSTANT_2018
- CONVENTIONAL_VALUE_OF_OHM_90_2018
- CONVENTIONAL_VALUE_OF_VOLT_90_2018
- CONVENTIONAL_VALUE_OF_VON_KLITZING_CONSTANT_2018
- CONVENTIONAL_VALUE_OF_WATT_90_2018
- COPPER_X_UNIT_2018
- DEUTERON_ELECTRON_MAG_MOM_RATIO_2018
- DEUTERON_ELECTRON_MASS_RATIO_2018
- DEUTERON_G_FACTOR_2018
- DEUTERON_MAG_MOM_2018
- DEUTERON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- DEUTERON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- DEUTERON_MASS_2018
- DEUTERON_MASS_ENERGY_EQUIVALENT_2018
- DEUTERON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018

- DEUTERON_MASS_IN_U_2018
- DEUTERON_MOLAR_MASS_2018
- DEUTERON_NEUTRON_MAG_MOM_RATIO_2018
- DEUTERON_PROTON_MAG_MOM_RATIO_2018
- DEUTERON_PROTON_MASS_RATIO_2018
- DEUTERON_RELATIVE_ATOMIC_MASS_2018
- DEUTERON_RMS_CHARGE_RADIUS_2018
- ELECTRON_CHARGE_TO_MASS_QUOTIENT_2018
- ELECTRON_DEUTERON_MAG_MOM_RATIO_2018
- ELECTRON_DEUTERON_MASS_RATIO_2018
- ELECTRON_G_FACTOR_2018
- ELECTRON_GYROMAG_RATIO_2018
- ELECTRON_GYROMAG_RATIO_IN_MHZ_T_2018
- ELECTRON_HELION_MASS_RATIO_2018
- ELECTRON_MAG_MOM_2018
- ELECTRON_MAG_MOM_ANOMALY_2018
- ELECTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- ELECTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- ELECTRON_MASS_2018
- ELECTRON_MASS_ENERGY_EQUIVALENT_2018
- ELECTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- ELECTRON_MASS_IN_U_2018
- ELECTRON_MOLAR_MASS_2018
- ELECTRON_MUON_MAG_MOM_RATIO_2018
- ELECTRON_MUON_MASS_RATIO_2018
- ELECTRON_NEUTRON_MAG_MOM_RATIO_2018
- ELECTRON_NEUTRON_MASS_RATIO_2018
- ELECTRON_PROTON_MAG_MOM_RATIO_2018
- ELECTRON_PROTON_MASS_RATIO_2018
- ELECTRON_RELATIVE_ATOMIC_MASS_2018
- ELECTRON_TAU_MASS_RATIO_2018
- ELECTRON_TO_ALPHA_PARTICLE_MASS_RATIO_2018

- ELECTRON_TO_SHIELDED_HELION_MAG_MOM_RATIO_2018
- ELECTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2018
- ELECTRON_TRITON_MASS_RATIO_2018
- ELECTRON_VOLT_2018
- ELECTRON_VOLT_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- ELECTRON_VOLT_HARTREE_RELATIONSHIP_2018
- ELECTRON_VOLT_HERTZ_RELATIONSHIP_2018
- ELECTRON_VOLT_INVERSE_METER_RELATIONSHIP_2018
- ELECTRON_VOLT_JOULE_RELATIONSHIP_2018
- ELECTRON_VOLT_KELVIN_RELATIONSHIP_2018
- ELECTRON_VOLT_KILOGRAM_RELATIONSHIP_2018
- ELEMENTARY_CHARGE_2018
- ELEMENTARY_CHARGE_OVER_H_BAR_2018
- FARADAY_CONSTANT_2018
- FERMI_COUPLING_CONSTANT_2018
- FINE_STRUCTURE_CONSTANT_2018
- FIRST_RADIATION_CONSTANT_2018
- FIRST_RADIATION_CONSTANT_FOR_SPECTRAL_RADIANCE_2018
- HARTREE_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- HARTREE_ELECTRON_VOLT_RELATIONSHIP_2018
- HARTREE_ENERGY_2018
- HARTREE_ENERGY_IN_EV_2018
- HARTREE_HERTZ_RELATIONSHIP_2018
- HARTREE_INVERSE_METER_RELATIONSHIP_2018
- HARTREE_JOULE_RELATIONSHIP_2018
- HARTREE_KELVIN_RELATIONSHIP_2018
- HARTREE_KILOGRAM_RELATIONSHIP_2018
- HELION_ELECTRON_MASS_RATIO_2018
- HELION_G_FACTOR_2018
- HELION_MAG_MOM_2018
- HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018

- HELION_MASS_2018
- HELION_MASS_ENERGY_EQUIVALENT_2018
- HELION_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- HELION_MASS_IN_U_2018
- HELION_MOLAR_MASS_2018
- HELION_PROTON_MASS_RATIO_2018
- HELION_RELATIVE_ATOMIC_MASS_2018
- HELION_SHIELDING_SHIFT_2018
- HERTZ_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- HERTZ_ELECTRON_VOLT_RELATIONSHIP_2018
- HERTZ_HARTREE_RELATIONSHIP_2018
- HERTZ_INVERSE_METER_RELATIONSHIP_2018
- HERTZ_JOULE_RELATIONSHIP_2018
- HERTZ_KELVIN_RELATIONSHIP_2018
- HERTZ_KILOGRAM_RELATIONSHIP_2018
- HYPERFINE_TRANSITION_FREQUENCY_OF_CS_133_2018
- INVERSE_FINE_STRUCTURE_CONSTANT_2018
- INVERSE_METER_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- INVERSE_METER_ELECTRON_VOLT_RELATIONSHIP_2018
- INVERSE_METER_HARTREE_RELATIONSHIP_2018
- INVERSE_METER_HERTZ_RELATIONSHIP_2018
- INVERSE_METER_JOULE_RELATIONSHIP_2018
- INVERSE_METER_KELVIN_RELATIONSHIP_2018
- INVERSE_METER_KILOGRAM_RELATIONSHIP_2018
- INVERSE_OF_CONDUCTANCE_QUANTUM_2018
- JOSEPHSON_CONSTANT_2018
- JOULE_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- JOULE_ELECTRON_VOLT_RELATIONSHIP_2018
- JOULE_HARTREE_RELATIONSHIP_2018
- JOULE_HERTZ_RELATIONSHIP_2018
- JOULE_INVERSE_METER_RELATIONSHIP_2018
- JOULE_KELVIN_RELATIONSHIP_2018

- JOULE_KILOGRAM_RELATIONSHIP_2018
- KELVIN_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- KELVIN_ELECTRON_VOLT_RELATIONSHIP_2018
- KELVIN_HARTREE_RELATIONSHIP_2018
- KELVIN_HERTZ_RELATIONSHIP_2018
- KELVIN_INVERSE_METER_RELATIONSHIP_2018
- KELVIN_JOULE_RELATIONSHIP_2018
- KELVIN_KILOGRAM_RELATIONSHIP_2018
- KILOGRAM_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- KILOGRAM_ELECTRON_VOLT_RELATIONSHIP_2018
- KILOGRAM_HARTREE_RELATIONSHIP_2018
- KILOGRAM_HERTZ_RELATIONSHIP_2018
- KILOGRAM_INVERSE_METER_RELATIONSHIP_2018
- KILOGRAM_JOULE_RELATIONSHIP_2018
- KILOGRAM_KELVIN_RELATIONSHIP_2018
- LATTICE_PARAMETER_OF_SILICON_2018
- LATTICE_SPACING_OF_IDEAL_SI_220_2018
- LOSCHMIDT_CONSTANT_273_15_K_100_KPA_2018
- LOSCHMIDT_CONSTANT_273_15_K_101_325_KPA_2018
- LUMINOUS_EFFICACY_2018
- MAG_FLUX_QUANTUM_2018
- MOLAR_GAS_CONSTANT_2018
- MOLAR_MASS_CONSTANT_2018
- MOLAR_MASS_OF_CARBON_12_2018
- MOLAR_PLANCK_CONSTANT_2018
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_100_KPA_2018
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_101_325_KPA_2018
- MOLAR_VOLUME_OF_SILICON_2018
- MOLYBDENUM_X_UNIT_2018
- MUON_COMPTON_WAVELENGTH_2018
- MUON_ELECTRON_MASS_RATIO_2018
- MUON_G_FACTOR_2018

- MUON_MAG_MOM_2018
- MUON_MAG_MOM_ANOMALY_2018
- MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- MUON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- MUON_MASS_2018
- MUON_MASS_ENERGY_EQUIVALENT_2018
- MUON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- MUON_MASS_IN_U_2018
- MUON_MOLAR_MASS_2018
- MUON_NEUTRON_MASS_RATIO_2018
- MUON_PROTON_MAG_MOM_RATIO_2018
- MUON_PROTON_MASS_RATIO_2018
- MUON_TAU_MASS_RATIO_2018
- NATURAL_UNIT_OF_ACTION_2018
- NATURAL_UNIT_OF_ACTION_IN_EV_S_2018
- NATURAL_UNIT_OF_ENERGY_2018
- NATURAL_UNIT_OF_ENERGY_IN_MEV_2018
- NATURAL_UNIT_OF_LENGTH_2018
- NATURAL_UNIT_OF_MASS_2018
- NATURAL_UNIT_OF_MOMENTUM_2018
- NATURAL_UNIT_OF_MOMENTUM_IN_MEV_C_2018
- NATURAL_UNIT_OF_TIME_2018
- NATURAL_UNIT_OF_VELOCITY_2018
- NEUTRON_COMPTON_WAVELENGTH_2018
- NEUTRON_ELECTRON_MAG_MOM_RATIO_2018
- NEUTRON_ELECTRON_MASS_RATIO_2018
- NEUTRON_G_FACTOR_2018
- NEUTRON_GYROMAG_RATIO_2018
- NEUTRON_GYROMAG_RATIO_IN_MHZ_T_2018
- NEUTRON_MAG_MOM_2018
- NEUTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- NEUTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018

- NEUTRON_MASS_2018
- NEUTRON_MASS_ENERGY_EQUIVALENT_2018
- NEUTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- NEUTRON_MASS_IN_U_2018
- NEUTRON_MOLAR_MASS_2018
- NEUTRON_MUON_MASS_RATIO_2018
- NEUTRON_PROTON_MAG_MOM_RATIO_2018
- NEUTRON_PROTON_MASS_DIFFERENCE_2018
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_2018
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_IN_MEV_2018
- NEUTRON_PROTON_MASS_DIFFERENCE_IN_U_2018
- NEUTRON_PROTON_MASS_RATIO_2018
- NEUTRON_RELATIVE_ATOMIC_MASS_2018
- NEUTRON_TAU_MASS_RATIO_2018
- NEUTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2018
- NEWTONIAN_CONSTANT_OF_GRAVITATION_2018
- NEWTONIAN_CONSTANT_OF_GRAVITATION_OVER_H_BAR_C_2018
- NUCLEAR_MAGNETON_2018
- NUCLEAR_MAGNETON_IN_EV_T_2018
- NUCLEAR_MAGNETON_IN_INVERSE_METER_PER_TESLA_2018
- NUCLEAR_MAGNETON_IN_K_T_2018
- NUCLEAR_MAGNETON_IN_MHZ_T_2018
- PLANCK_CONSTANT_2018
- PLANCK_CONSTANT_IN_EV_HZ_2018
- PLANCK_LENGTH_2018
- PLANCK_MASS_2018
- PLANCK_MASS_ENERGY_EQUIVALENT_IN_GEV_2018
- PLANCK_TEMPERATURE_2018
- PLANCK_TIME_2018
- PROTON_CHARGE_TO_MASS_QUOTIENT_2018
- PROTON_COMPTON_WAVELENGTH_2018
- PROTON_ELECTRON_MASS_RATIO_2018

- PROTON_G_FACTOR_2018
- PROTON_GYROMAG_RATIO_2018
- PROTON_GYROMAG_RATIO_IN_MHZ_T_2018
- PROTON_MAG_MOM_2018
- PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- PROTON_MAG_SHIELDING_CORRECTION_2018
- PROTON_MASS_2018
- PROTON_MASS_ENERGY_EQUIVALENT_2018
- PROTON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- PROTON_MASS_IN_U_2018
- PROTON_MOLAR_MASS_2018
- PROTON_MUON_MASS_RATIO_2018
- PROTON_NEUTRON_MAG_MOM_RATIO_2018
- PROTON_NEUTRON_MASS_RATIO_2018
- PROTON_RELATIVE_ATOMIC_MASS_2018
- PROTON_RMS_CHARGE_RADIUS_2018
- PROTON_TAU_MASS_RATIO_2018
- QUANTUM_OF_CIRCULATION_2018
- QUANTUM_OF_CIRCULATION_TIMES_2_2018
- REDUCED_COMPTON_WAVELENGTH_2018
- REDUCED_MUON_COMPTON_WAVELENGTH_2018
- REDUCED_NEUTRON_COMPTON_WAVELENGTH_2018
- REDUCED_PLANCK_CONSTANT_2018
- REDUCED_PLANCK_CONSTANT_IN_EV_S_2018
- REDUCED_PLANCK_CONSTANT_TIMES_C_IN_MEV_FM_2018
- REDUCED_PROTON_COMPTON_WAVELENGTH_2018
- REDUCED_TAU_COMPTON_WAVELENGTH_2018
- RYDBERG_CONSTANT_2018
- RYDBERG_CONSTANT_TIMES_C_IN_HZ_2018
- RYDBERG_CONSTANT_TIMES_HC_IN_EV_2018
- RYDBERG_CONSTANT_TIMES_HC_IN_J_2018

- SACKUR_TETRODE_CONSTANT_1_K_100_KPA_2018
- SACKUR_TETRODE_CONSTANT_1_K_101_325_KPA_2018
- SECOND_RADIATION_CONSTANT_2018
- SHIELDED_HELION_GYROMAG_RATIO_2018
- SHIELDED_HELION_GYROMAG_RATIO_IN_MHZ_T_2018
- SHIELDED_HELION_MAG_MOM_2018
- SHIELDED_HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- SHIELDED_HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- SHIELDED_HELION_TO_PROTON_MAG_MOM_RATIO_2018
- SHIELDED_HELION_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2018
- SHIELDED_PROTON_GYROMAG_RATIO_2018
- SHIELDED_PROTON_GYROMAG_RATIO_IN_MHZ_T_2018
- SHIELDED_PROTON_MAG_MOM_2018
- SHIELDED_PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- SHIELDED_PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- SHIELDING_DIFFERENCE_OF_D_AND_P_IN_HD_2018
- SHIELDING_DIFFERENCE_OF_T_AND_P_IN_HT_2018
- SPEED_OF_LIGHT_IN_VACUUM_2018
- STANDARD_ACCELERATION_OF_GRAVITY_2018
- STANDARD_ATMOSPHERE_2018
- STANDARD_STATE_PRESSURE_2018
- STEFAN_BOLTZMANN_CONSTANT_2018
- TAU_COMPTON_WAVELENGTH_2018
- TAU_ELECTRON_MASS_RATIO_2018
- TAU_ENERGY_EQUIVALENT_2018
- TAU_MASS_2018
- TAU_MASS_ENERGY_EQUIVALENT_2018
- TAU_MASS_IN_U_2018
- TAU_MOLAR_MASS_2018
- TAU_MUON_MASS_RATIO_2018
- TAU_NEUTRON_MASS_RATIO_2018
- TAU_PROTON_MASS_RATIO_2018

- THOMSON_CROSS_SECTION_2018
- TRITON_ELECTRON_MASS_RATIO_2018
- TRITON_G_FACTOR_2018
- TRITON_MAG_MOM_2018
- TRITON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- TRITON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- TRITON_MASS_2018
- TRITON_MASS_ENERGY_EQUIVALENT_2018
- TRITON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- TRITON_MASS_IN_U_2018
- TRITON_MOLAR_MASS_2018
- TRITON_PROTON_MASS_RATIO_2018
- TRITON_RELATIVE_ATOMIC_MASS_2018
- TRITON_TO_PROTON_MAG_MOM_RATIO_2018
- UNIFIED_ATOMIC_MASS_UNIT_2018
- VACUUM_ELECTRIC_PERMITTIVITY_2018
- VACUUM_MAG_PERMEABILITY_2018
- VON_KLITZING_CONSTANT_2018
- WEAK_MIXING_ANGLE_2018
- WIEN_FREQUENCY_DISPLACEMENT_LAW_CONSTANT_2018
- WIEN_WAVELENGTH_DISPLACEMENT_LAW_CONSTANT_2018
- W_TO_Z_MASS_RATIO_2018

SEE ALSO

codata(3), codata_2010(3), codata_2014(3), codata_2018(3), codata_2022(3)

Chapter 6

codata_2022(3)

NAME

codata - libray for fundamental physical constants

LIBRARY

codata (-libcodata, -lcodata)

SYNOPSIS

```
use codata
   include "codata.h"
   import pycodata
```

DESCRIPTION

codata is a Fortran library providing the fundamental physical constants according to CODATA
List of available constants:

- ALPHA_PARTICLE_ELECTRON_MASS_RATIO
- ALPHA_PARTICLE_MASS
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_IN_MEV
- ALPHA_PARTICLE_MASS_IN_U
- ALPHA_PARTICLE_MOLAR_MASS
- ALPHA_PARTICLE_PROTON_MASS_RATIO
- ALPHA_PARTICLE_RELATIVE_ATOMIC_MASS
- ALPHA_PARTICLE_RMS_CHARGE_RADIUS
- ANGSTROM_STAR
- ATOMIC_MASS_CONSTANT
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT

- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_IN_MEV
- ATOMIC_MASS_UNIT_ELECTRON_VOLT_RELATIONSHIP
- ATOMIC_MASS_UNIT_HARTREE_RELATIONSHIP
- ATOMIC_MASS_UNIT_HERTZ_RELATIONSHIP
- ATOMIC_MASS_UNIT_INVERSE_METER_RELATIONSHIP
- ATOMIC_MASS_UNIT_JOULE_RELATIONSHIP
- ATOMIC_MASS_UNIT_KELVIN_RELATIONSHIP
- ATOMIC_MASS_UNIT_KILOGRAM_RELATIONSHIP
- ATOMIC_UNIT_OF_1ST_HYPERPOLARIZABILITY
- ATOMIC_UNIT_OF_2ND_HYPERPOLARIZABILITY
- ATOMIC_UNIT_OF_ACTION
- ATOMIC_UNIT_OF_CHARGE
- ATOMIC_UNIT_OF_CHARGE_DENSITY
- ATOMIC_UNIT_OF_CURRENT
- ATOMIC_UNIT_OF_ELECTRIC_DIPOLE_MOM
- ATOMIC_UNIT_OF_ELECTRIC_FIELD
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_GRADIENT
- ATOMIC_UNIT_OF_ELECTRIC_POLARIZABILITY
- ATOMIC_UNIT_OF_ELECTRIC_POTENTIAL
- ATOMIC_UNIT_OF_ELECTRIC_QUADRUPOLE_MOM
- ATOMIC_UNIT_OF_ENERGY
- ATOMIC_UNIT_OF_FORCE
- ATOMIC_UNIT_OF_LENGTH
- ATOMIC_UNIT_OF_MAG_DIPOLE_MOM
- ATOMIC_UNIT_OF_MAG_FLUX_DENSITY
- ATOMIC_UNIT_OF_MAGNETIZABILITY
- ATOMIC_UNIT_OF_MASS
- ATOMIC_UNIT_OF_MOMENTUM
- ATOMIC_UNIT_OF_PERMITTIVITY
- ATOMIC_UNIT_OF_TIME
- ATOMIC_UNIT_OF_VELOCITY
- AVOGADRO_CONSTANT

- BOHR_MAGNETON
- BOHR_MAGNETON_IN_EV_T
- BOHR_MAGNETON_IN_HZ_T
- BOHR_MAGNETON_IN_INVERSE_METER_PER_TESLA
- BOHR_MAGNETON_IN_K_T
- BOHR_RADIUS
- BOLTZMANN_CONSTANT
- BOLTZMANN_CONSTANT_IN_EV_K
- BOLTZMANN_CONSTANT_IN_HZ_K
- BOLTZMANN_CONSTANT_IN_INVERSE_METER_PER_KELVIN
- CHARACTERISTIC_IMPEDANCE_OF_VACUUM
- CLASSICAL_ELECTRON_RADIUS
- COMPTON_WAVELENGTH
- CONDUCTANCE_QUANTUM
- CONVENTIONAL_VALUE_OF_AMPERE_90
- CONVENTIONAL_VALUE_OF_COULOMB_90
- CONVENTIONAL_VALUE_OF_FARAD_90
- CONVENTIONAL_VALUE_OF_HENRY_90
- CONVENTIONAL_VALUE_OF_JOSEPHSON_CONSTANT
- CONVENTIONAL_VALUE_OF_OHM_90
- CONVENTIONAL_VALUE_OF_VOLT_90
- CONVENTIONAL_VALUE_OF_VON_KLITZING_CONSTANT
- CONVENTIONAL_VALUE_OF_WATT_90
- COPPER_X_UNIT
- DEUTERON_ELECTRON_MAG_MOM_RATIO
- DEUTERON_ELECTRON_MASS_RATIO
- DEUTERON_G_FACTOR
- DEUTERON_MAG_MOM
- DEUTERON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- DEUTERON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- DEUTERON_MASS
- DEUTERON_MASS_ENERGY_EQUIVALENT

- DEUTERON_MASS_ENERGY_EQUIVALENT_IN_MEV
- DEUTERON_MASS_IN_U
- DEUTERON_MOLAR_MASS
- DEUTERON_NEUTRON_MAG_MOM_RATIO
- DEUTERON_PROTON_MAG_MOM_RATIO
- DEUTERON_PROTON_MASS_RATIO
- DEUTERON_RELATIVE_ATOMIC_MASS
- DEUTERON_RMS_CHARGE_RADIUS
- ELECTRON_CHARGE_TO_MASS_QUOTIENT
- ELECTRON_DEUTERON_MAG_MOM_RATIO
- ELECTRON_DEUTERON_MASS_RATIO
- ELECTRON_G_FACTOR
- ELECTRON_GYROMAG_RATIO
- ELECTRON_GYROMAG_RATIO_IN_MHZ_T
- ELECTRON_HELION_MASS_RATIO
- ELECTRON_MAG_MOM
- ELECTRON_MAG_MOM_ANOMALY
- ELECTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- ELECTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- ELECTRON_MASS
- ELECTRON_MASS_ENERGY_EQUIVALENT
- ELECTRON_MASS_ENERGY_EQUIVALENT_IN_MEV
- ELECTRON_MASS_IN_U
- ELECTRON_MOLAR_MASS
- ELECTRON_MUON_MAG_MOM_RATIO
- ELECTRON_MUON_MASS_RATIO
- ELECTRON_NEUTRON_MAG_MOM_RATIO
- ELECTRON_NEUTRON_MASS_RATIO
- ELECTRON_PROTON_MAG_MOM_RATIO
- ELECTRON_PROTON_MASS_RATIO
- ELECTRON_RELATIVE_ATOMIC_MASS
- ELECTRON_TAU_MASS_RATIO

- ELECTRON_TO_ALPHA_PARTICLE_MASS_RATIO
- ELECTRON_TO_SHIELDED_HELION_MAG_MOM_RATIO
- ELECTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO
- ELECTRON_TRITON_MASS_RATIO
- ELECTRON_VOLT
- ELECTRON_VOLT_ATOMIC_MASS_UNIT_RELATIONSHIP
- ELECTRON_VOLT_HARTREE_RELATIONSHIP
- ELECTRON_VOLT_HERTZ_RELATIONSHIP
- ELECTRON_VOLT_INVERSE_METER_RELATIONSHIP
- ELECTRON_VOLT_JOULE_RELATIONSHIP
- ELECTRON_VOLT_KELVIN_RELATIONSHIP
- ELECTRON_VOLT_KILOGRAM_RELATIONSHIP
- ELEMENTARY_CHARGE
- ELEMENTARY_CHARGE_OVER_H_BAR
- FARADAY_CONSTANT
- FERMI_COUPLING_CONSTANT
- FINE_STRUCTURE_CONSTANT
- FIRST_RADIATION_CONSTANT
- FIRST_RADIATION_CONSTANT_FOR_SPECTRAL_RADIANCE
- HARTREE_ATOMIC_MASS_UNIT_RELATIONSHIP
- HARTREE_ELECTRON_VOLT_RELATIONSHIP
- HARTREE_ENERGY
- HARTREE_ENERGY_IN_EV
- HARTREE_HERTZ_RELATIONSHIP
- HARTREE_INVERSE_METER_RELATIONSHIP
- HARTREE_JOULE_RELATIONSHIP
- HARTREE_KELVIN_RELATIONSHIP
- HARTREE_KILOGRAM_RELATIONSHIP
- HELION_ELECTRON_MASS_RATIO
- HELION_G_FACTOR
- HELION_MAG_MOM
- HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO

- HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- HELION_MASS
- HELION_MASS_ENERGY_EQUIVALENT
- HELION_MASS_ENERGY_EQUIVALENT_IN_MEV
- HELION_MASS_IN_U
- HELION_MOLAR_MASS
- HELION_PROTON_MASS_RATIO
- HELION_RELATIVE_ATOMIC_MASS
- HELION_SHIELDING_SHIFT
- HERTZ_ATOMIC_MASS_UNIT_RELATIONSHIP
- HERTZ_ELECTRON_VOLT_RELATIONSHIP
- HERTZ_HARTREE_RELATIONSHIP
- HERTZ_INVERSE_METER_RELATIONSHIP
- HERTZ_JOULE_RELATIONSHIP
- HERTZ_KELVIN_RELATIONSHIP
- HERTZ_KILOGRAM_RELATIONSHIP
- HYPERFINE_TRANSITION_FREQUENCY_OF_CS_133
- INVERSE_FINE_STRUCTURE_CONSTANT
- INVERSE_METER_ATOMIC_MASS_UNIT_RELATIONSHIP
- INVERSE_METER_ELECTRON_VOLT_RELATIONSHIP
- INVERSE_METER_HARTREE_RELATIONSHIP
- INVERSE_METER_HERTZ_RELATIONSHIP
- INVERSE_METER_JOULE_RELATIONSHIP
- INVERSE_METER_KELVIN_RELATIONSHIP
- INVERSE_METER_KILOGRAM_RELATIONSHIP
- INVERSE_OF_CONDUCTANCE_QUANTUM
- JOSEPHSON_CONSTANT
- JOULE_ATOMIC_MASS_UNIT_RELATIONSHIP
- JOULE_ELECTRON_VOLT_RELATIONSHIP
- JOULE_HARTREE_RELATIONSHIP
- JOULE_HERTZ_RELATIONSHIP
- JOULE_INVERSE_METER_RELATIONSHIP

- JOULE_KELVIN_RELATIONSHIP
- JOULE_KILOGRAM_RELATIONSHIP
- KELVIN_ATOMIC_MASS_UNIT_RELATIONSHIP
- KELVIN_ELECTRON_VOLT_RELATIONSHIP
- KELVIN_HARTREE_RELATIONSHIP
- KELVIN_HERTZ_RELATIONSHIP
- KELVIN_INVERSE_METER_RELATIONSHIP
- KELVIN_JOULE_RELATIONSHIP
- KELVIN_KILOGRAM_RELATIONSHIP
- KILOGRAM_ATOMIC_MASS_UNIT_RELATIONSHIP
- KILOGRAM_ELECTRON_VOLT_RELATIONSHIP
- KILOGRAM_HARTREE_RELATIONSHIP
- KILOGRAM_HERTZ_RELATIONSHIP
- KILOGRAM_INVERSE_METER_RELATIONSHIP
- KILOGRAM_JOULE_RELATIONSHIP
- KILOGRAM_KELVIN_RELATIONSHIP
- LATTICE_PARAMETER_OF_SILICON
- LATTICE_SPACING_OF_IDEAL_SI_220
- LOSCHMIDT_CONSTANT_273_15_K_100_KPA
- LOSCHMIDT_CONSTANT_273_15_K_101_325_KPA
- LUMINOUS_EFFICACY
- MAG_FLUX_QUANTUM
- MOLAR_GAS_CONSTANT
- MOLAR_MASS_CONSTANT
- MOLAR_MASS_OF_CARBON_12
- MOLAR_PLANCK_CONSTANT
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_100_KPA
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_101_325_KPA
- MOLAR_VOLUME_OF_SILICON
- MOLYBDENUM_X_UNIT
- MUON_COMPTON_WAVELENGTH
- MUON_ELECTRON_MASS_RATIO

- MUON_G_FACTOR
- MUON_MAG_MOM
- MUON_MAG_MOM_ANOMALY
- MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- MUON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- MUON_MASS
- MUON_MASS_ENERGY_EQUIVALENT
- MUON_MASS_ENERGY_EQUIVALENT_IN_MEV
- MUON_MASS_IN_U
- MUON_MOLAR_MASS
- MUON_NEUTRON_MASS_RATIO
- MUON_PROTON_MAG_MOM_RATIO
- MUON_PROTON_MASS_RATIO
- MUON_TAU_MASS_RATIO
- NATURAL_UNIT_OF_ACTION
- NATURAL_UNIT_OF_ACTION_IN_EV_S
- NATURAL_UNIT_OF_ENERGY
- NATURAL_UNIT_OF_ENERGY_IN_MEV
- NATURAL_UNIT_OF_LENGTH
- NATURAL_UNIT_OF_MASS
- NATURAL_UNIT_OF_MOMENTUM
- NATURAL_UNIT_OF_MOMENTUM_IN_MEV_C
- NATURAL_UNIT_OF_TIME
- NATURAL_UNIT_OF_VELOCITY
- NEUTRON_COMPTON_WAVELENGTH
- NEUTRON_ELECTRON_MAG_MOM_RATIO
- NEUTRON_ELECTRON_MASS_RATIO
- NEUTRON_G_FACTOR
- NEUTRON_GYROMAG_RATIO
- NEUTRON_GYROMAG_RATIO_IN_MHZ_T
- NEUTRON_MAG_MOM
- NEUTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO

- NEUTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- NEUTRON_MASS
- NEUTRON_MASS_ENERGY_EQUIVALENT
- NEUTRON_MASS_ENERGY_EQUIVALENT_IN_MEV
- NEUTRON_MASS_IN_U
- NEUTRON_MOLAR_MASS
- NEUTRON_MUON_MASS_RATIO
- NEUTRON_PROTON_MAG_MOM_RATIO
- NEUTRON_PROTON_MASS_DIFFERENCE
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_IN_MEV
- NEUTRON_PROTON_MASS_DIFFERENCE_IN_U
- NEUTRON_PROTON_MASS_RATIO
- NEUTRON_RELATIVE_ATOMIC_MASS
- NEUTRON_TAU_MASS_RATIO
- NEUTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO
- NEWTONIAN_CONSTANT_OF_GRAVITATION
- NEWTONIAN_CONSTANT_OF_GRAVITATION_OVER_H_BAR_C
- NUCLEAR_MAGNETON
- NUCLEAR_MAGNETON_IN_EV_T
- NUCLEAR_MAGNETON_IN_INVERSE_METER_PER_TESLA
- NUCLEAR_MAGNETON_IN_K_T
- NUCLEAR_MAGNETON_IN_MHZ_T
- PLANCK_CONSTANT
- PLANCK_CONSTANT_IN_EV_HZ
- PLANCK_LENGTH
- PLANCK_MASS
- PLANCK_MASS_ENERGY_EQUIVALENT_IN_GEV
- PLANCK_TEMPERATURE
- PLANCK_TIME
- PROTON_CHARGE_TO_MASS_QUOTIENT
- PROTON_COMPTON_WAVELENGTH

- PROTON_ELECTRON_MASS_RATIO
- PROTON_G_FACTOR
- PROTON_GYROMAG_RATIO
- PROTON_GYROMAG_RATIO_IN_MHZ_T
- PROTON_MAG_MOM
- PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- PROTON_MAG_SHIELDING_CORRECTION
- PROTON_MASS
- PROTON_MASS_ENERGY_EQUIVALENT
- PROTON_MASS_ENERGY_EQUIVALENT_IN_MEV
- PROTON_MASS_IN_U
- PROTON_MOLAR_MASS
- PROTON_MUON_MASS_RATIO
- PROTON_NEUTRON_MAG_MOM_RATIO
- PROTON_NEUTRON_MASS_RATIO
- PROTON_RELATIVE_ATOMIC_MASS
- PROTON_RMS_CHARGE_RADIUS
- PROTON_TAU_MASS_RATIO
- QUANTUM_OF_CIRCULATION
- QUANTUM_OF_CIRCULATION_TIMES_2
- REDUCED_COMPTON_WAVELENGTH
- REDUCED_MUON_COMPTON_WAVELENGTH
- REDUCED_NEUTRON_COMPTON_WAVELENGTH
- REDUCED_PLANCK_CONSTANT
- REDUCED_PLANCK_CONSTANT_IN_EV_S
- REDUCED_PLANCK_CONSTANT_TIMES_C_IN_MEV_FM
- REDUCED_PROTON_COMPTON_WAVELENGTH
- REDUCED_TAU_COMPTON_WAVELENGTH
- RYDBERG_CONSTANT
- RYDBERG_CONSTANT_TIMES_C_IN_HZ
- RYDBERG_CONSTANT_TIMES_HC_IN_EV

- RYDBERG_CONSTANT_TIMES_HC_IN_J
- SACKUR_TETRODE_CONSTANT_1_K_100_KPA
- SACKUR_TETRODE_CONSTANT_1_K_101_325_KPA
- SECOND_RADIATION_CONSTANT
- SHIELDED_HELION_GYROMAG_RATIO
- SHIELDED_HELION_GYROMAG_RATIO_IN_MHZ_T
- SHIELDED_HELION_MAG_MOM
- SHIELDED_HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- SHIELDED_HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- SHIELDED_HELION_TO_PROTON_MAG_MOM_RATIO
- SHIELDED_HELION_TO_SHIELDED_PROTON_MAG_MOM_RATIO
- SHIELDED_PROTON_GYROMAG_RATIO
- SHIELDED_PROTON_GYROMAG_RATIO_IN_MHZ_T
- SHIELDED_PROTON_MAG_MOM
- SHIELDED_PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- SHIELDED_PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- SHIELDING_DIFFERENCE_OF_D_AND_P_IN_HD
- SHIELDING_DIFFERENCE_OF_T_AND_P_IN_HT
- SPEED_OF_LIGHT_IN_VACUUM
- STANDARD_ACCELERATION_OF_GRAVITY
- STANDARD_ATMOSPHERE
- STANDARD_STATE_PRESSURE
- STEFAN_BOLTZMANN_CONSTANT
- TAU_COMPTON_WAVELENGTH
- TAU_ELECTRON_MASS_RATIO
- TAU_ENERGY_EQUIVALENT
- TAU_MASS
- TAU_MASS_ENERGY_EQUIVALENT
- TAU_MASS_IN_U
- TAU_MOLAR_MASS
- TAU_MUON_MASS_RATIO
- TAU_NEUTRON_MASS_RATIO

- TAU_PROTON_MASS_RATIO
- THOMSON_CROSS_SECTION
- TRITON_ELECTRON_MASS_RATIO
- TRITON_G_FACTOR
- TRITON_MAG_MOM
- TRITON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- TRITON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- TRITON_MASS
- TRITON_MASS_ENERGY_EQUIVALENT
- TRITON_MASS_ENERGY_EQUIVALENT_IN_MEV
- TRITON_MASS_IN_U
- TRITON_MOLAR_MASS
- TRITON_PROTON_MASS_RATIO
- TRITON_RELATIVE_ATOMIC_MASS
- TRITON_TO_PROTON_MAG_MOM_RATIO
- UNIFIED_ATOMIC_MASS_UNIT
- VACUUM_ELECTRIC_PERMITTIVITY
- VACUUM_MAG_PERMEABILITY
- VON_KLITZING_CONSTANT
- WEAK_MIXING_ANGLE
- WIEN_FREQUENCY_DISPLACEMENT_LAW_CONSTANT
- WIEN_WAVELENGTH_DISPLACEMENT_LAW_CONSTANT
- W_TO_Z_MASS_RATIO

SEE ALSO

codata(3), codata_2010(3), codata_2014(3), codata_2018(3), codata_2022(3)

Chapter 7

codata(1)

NAME

codata - Command line for codata

SYNOPSIS

```
codata [OPTIONS] [REGEX_PATTERN ... ]
```

DESCRIPTION

codata is a command line interface which prints all the codata constants.

The current values are from 2022. Older values can be retrieved if needed and the output can be filtered with REGEX PATTERNS.

OPTIONS

–year, -y YEAR Year of the codata constants: 2022, 2018, 2014, 2010

–pattern, -p PATTERN Regex pattern for filtering the constants.

–value, -a Show only the value.

–error, -e Show only the uncertainty.

–usage Show usage text and exit.

–help Show help text and exit.

–verbose Display additional information.

–version Show version information and exit.

NOTES

You may replace the default options from a file if your first options begin with @file. Initial options will then be read from the "response file" "file.rsp" in the current directory.

If "file" does not exist or cannot be read, then an error occurs and the program stops. Each line of the file is prefixed with "options" and interpreted as a separate argument. The file itself may not contain @file arguments. That is, it is not processed recursively.

For more information on response files see https://urbanjost.github.io/M_CLI2/set_args.3m_cli2.html

EXAMPLE

Minimal example

```
codata
codata -y 2018 molar electron
codata -y 2014 -p 'molar.*gas','electron.*eV'
codata '[B,b]oltzmann.*eV'
```

SEE ALSO

codata(3)

Part VII

Appendices

Chapter 8

Raw Data

https://physics.nist.gov/cuu/Constants/index.html

8.1 2010 Fundamental Physical Constants — Complete Listing

2010 Fundamental Physical Constants --- Complete Listing

From: http://physics.nist.gov/constants

Quantity	Value	Uncertainty	Unit
{220} lattice spacing of silicon	192.015 5714 e-12	0.000 0032 e-12	m
alpha particle-electron mass ratio	7294.299 5361	0.000 0029	
alpha particle mass	6.644 656 75 e-27	0.000 000 29 e-27	kg
alpha particle mass energy equivalent	5.971 919 67 e-10	0.000 000 26 e-10	J
alpha particle mass energy equivalent in MeV	3727.379 240	0.000 082	MeV
alpha particle mass in u	4.001 506 179 125	0.000 000 000 062	u
alpha particle molar mass	4.001 506 179 125 e-3	0.000 000 000 062 e-3	kg mol^-1
alpha particle-proton mass ratio	3.972 599 689 33	0.000 000 000 36	
Angstrom star	1.000 014 95 e-10	0.000 000 90 e-10	m
atomic mass constant	1.660 538 921 e-27	0.000 000 073 e-27	kg
atomic mass constant energy equivalent	1.492 417 954 e-10	0.000 000 066 e-10	J
atomic mass constant energy equivalent in MeV	931.494 061	0.000 021	MeV
atomic mass unit-electron volt relationship	931.494 061 e6	0.000 021 e6	eV
atomic mass unit-hartree relationship	3.423 177 6845 e7	0.000 000 0024 e7	E_h
atomic mass unit-hertz relationship	2.252 342 7168 e23	0.000 000 0016 e23	Hz
atomic mass unit-inverse meter relationship	7.513 006 6042 e14	0.000 000 0053 e14	m^-1
atomic mass unit-joule relationship	1.492 417 954 e-10	0.000 000 066 e-10	J
atomic mass unit-kelvin relationship	1.080 954 08 e13	0.000 000 98 e13	K
atomic mass unit-kilogram relationship	1.660 538 921 e-27	0.000 000 073 e-27	kg
atomic unit of 1st hyperpolarizability	3.206 361 449 e-53	0.000 000 071 e-53	C^3 m^3 J^-2
atomic unit of 2nd hyperpolarizability	6.235 380 54 e-65	0.000 000 28 e-65	C^4 m^4 J^-3
atomic unit of action	1.054 571 726 e-34	0.000 000 047 e-34	J s
atomic unit of charge	1.602 176 565 e-19	0.000 000 035 e-19	C
atomic unit of charge density	1.081 202 338 e12	0.000 000 024 e12	C m^-3
atomic unit of current	6.623 617 95 e-3	0.000 000 15 e-3	A
atomic unit of electric dipole mom.	8.478 353 26 e-30	0.000 000 19 e-30	C m
atomic unit of electric field	5.142 206 52 e11	0.000 000 11 e11	V m^-1
atomic unit of electric field gradient	9.717 362 00 e21	0.000 000 21 e21	V m^-2
atomic unit of electric polarizability	1.648 777 2754 e-41	0.000 000 0016 e-41	C^2 m^2 J^-1
atomic unit of electric potential	27.211 385 05	0.000 000 60	V
atomic unit of electric quadrupole mom.	4.486 551 331 e-40	0.000 000 099 e-40	C m^2
atomic unit of energy	4.359 744 34 e-18	0.000 000 19 e-18	J
atomic unit of force	8.238 722 78 e-8	0.000 000 36 e-8	N
atomic unit of length	0.529 177 210 92 e-10	0.000 000 000 17 e-10	m
atomic unit of mag. dipole mom.	1.854 801 936 e-23	0.000 000 041 e-23	J T^-1
atomic unit of mag. flux density	2.350 517 464 e5	0.000 000 052 e5	T
atomic unit of magnetizability	7.891 036 607 e-29	0.000 000 013 e-29	J T^-2
atomic unit of mass	9.109 382 91 e-31	0.000 000 40 e-31	kg
atomic unit of mom.um	1.992 851 740 e-24	0.000 000 088 e-24	kg m s^-1
atomic unit of permittivity	1.112 650 056... e-10	(exact)	F m^-1
atomic unit of time	2.418 884 326 502 e-17	0.000 000 000 012 e-17	s
atomic unit of velocity	2.187 691 263 79 e6	0.000 000 000 71 e6	m s^-1
Avogadro constant	6.022 141 29 e23	0.000 000 27 e23	mol^-1
Bohr magneton	927.400 968 e-26	0.000 020 e-26	J T^-1
Bohr magneton in eV/T	5.788 381 8066 e-5	0.000 000 0038 e-5	eV T^-1
Bohr magneton in Hz/T	13.996 245 55 e9	0.000 000 31 e9	Hz T^-1
Bohr magneton in inverse meters per tesla	46.686 4498	0.000 0010	m^-1 T^-1
Bohr magneton in K/T	0.671 713 88	0.000 000 61	K T^-1
Bohr radius	0.529 177 210 92 e-10	0.000 000 000 17 e-10	m
Boltzmann constant	1.380 6488 e-23	0.000 0013 e-23	J K^-1
Boltzmann constant in eV/K	8.617 3324 e-5	0.000 0078 e-5	eV K^-1
Boltzmann constant in Hz/K	2.083 6618 e10	0.000 0019 e10	Hz K^-1
Boltzmann constant in inverse meters per kelvin	69.503 476	0.000 063	m^-1 K^-1
characteristic impedance of vacuum	376.730 313 461...	(exact)	ohm
classical electron radius	2.817 940 3267 e-15	0.000 000 0027 e-15	m
Compton wavelength	2.426 310 2389 e-12	0.000 000 0016 e-12	m
Compton wavelength over 2 pi	386.159 268 00 e-15	0.000 000 25 e-15	m
conductance quantum	7.748 091 7346 e-5	0.000 000 0025 e-5	S
conventional value of Josephson constant	483 597.9 e9	(exact)	Hz V^-1
conventional value of von Klitzing constant	25 812.807	(exact)	ohm
Cu x unit	1.002 076 97 e-13	0.000 000 28 e-13	m
deuteron-electron mag. mom. ratio	-4.664 345 537 e-4	0.000 000 039 e-4	
deuteron-electron mass ratio	3670.482 9652	0.000 0015	
deuteron g factor	0.857 438 2308	0.000 000 0072	
deuteron mag. mom.	0.433 073 489 e-26	0.000 000 010 e-26	J T^-1
deuteron mag. mom. to Bohr magneton ratio	0.466 975 4556 e-3	0.000 000 0039 e-3	
deuteron mag. mom. to nuclear magneton ratio	0.857 438 2308	0.000 000 0072	
deuteron mass	3.343 583 48 e-27	0.000 000 15 e-27	kg
deuteron mass energy equivalent	3.005 062 97 e-10	0.000 000 13 e-10	J
deuteron mass energy equivalent in MeV	1875.612 859	0.000 041	MeV
deuteron mass in u	2.013 553 212 712	0.000 000 000 077	u
deuteron molar mass	2.013 553 212 712 e-3	0.000 000 000 077 e-3	kg mol^-1
deuteron-neutron mag. mom. ratio	-0.448 206 52	0.000 000 11	
deuteron-proton mag. mom. ratio	0.307 012 2070	0.000 000 0024	
deuteron-proton mass ratio	1.999 007 500 97	0.000 000 000 18	
deuteron rms charge radius	2.1424 e-15	0.0021 e-15	m
electric constant	8.854 187 817... e-12	(exact)	F m^-1
electron charge to mass quotient	-1.758 820 088 e11	0.000 000 039 e11	C kg^-1
electron-deuteron mag. mom. ratio	-2143.923 498	0.000 018	
electron-deuteron mass ratio	2.724 437 1095 e-4	0.000 000 0011 e-4	
electron g factor	-2.002 319 304 361 53	0.000 000 000 000 53	
electron gyromag. ratio	1.760 859 708 e11	0.000 000 039 e11	s^-1 T^-1
electron gyromag. ratio over 2 pi	28 024.952 66	0.000 62	MHz T^-1
electron-helion mass ratio	1.819 543 0761 e-4	0.000 000 0017 e-4	
electron mag. mom.	-928.476 430 e-26	0.000 021 e-26	J T^-1
electron mag. mom. anomaly	1.159 652 180 76 e-3	0.000 000 000 27 e-3	
electron mag. mom. to Bohr magneton ratio	-1.001 159 652 180 76	0.000 000 000 000 27	
electron mag. mom. to nuclear magneton ratio	-1838.281 970 90	0.000 000 75	
electron mass	9.109 382 91 e-31	0.000 000 40 e-31	kg
electron mass energy equivalent	8.187 105 06 e-14	0.000 000 36 e-14	J

electron mass energy equivalent in MeV	0.510 998 928	0.000 000 011	MeV
electron mass in u	5.485 799 0946 e-4	0.000 000 0022 e-4	u
electron molar mass	5.485 799 0946 e-7	0.000 000 0022 e-7	kg mol^-1
electron-muon mag. mom. ratio	206.766 9896	0.000 0052	
electron-muon mass ratio	4.836 331 66 e-3	0.000 000 12 e-3	
electron-neutron mag. mom. ratio	960.920 50	0.000 23	
electron-neutron mass ratio	5.438 673 4461 e-4	0.000 000 0032 e-4	
electron-proton mag. mom. ratio	-658.210 6848	0.000 0054	
electron-proton mass ratio	5.446 170 2178 e-4	0.000 000 0022 e-4	
electron-tau mass ratio	2.875 92 e-4	0.000 26 e-4	
electron to alpha particle mass ratio	1.370 933 555 78 e-4	0.000 000 000 55 e-4	
electron to shielded helion mag. mom. ratio	864.058 257	0.000 010	
electron to shielded proton mag. mom. ratio	-658.227 5971	0.000 0072	
electron-triton mass ratio	1.819 200 0653 e-4	0.000 000 0017 e-4	
electron volt	1.602 176 565 e-19	0.000 000 035 e-19	J
electron volt-atomic mass unit relationship	1.073 544 150 e-9	0.000 000 024 e-9	u
electron volt-hartree relationship	3.674 932 379 e-2	0.000 000 081 e-2	E_h
electron volt-hertz relationship	2.417 989 348 e14	0.000 000 053 e14	Hz
electron volt-inverse meter relationship	8.065 544 29 e5	0.000 000 18 e5	m^-1
electron volt-joule relationship	1.602 176 565 e-19	0.000 000 035 e-19	J
electron volt-kelvin relationship	1.160 4519 e4	0.000 0011 e4	K
electron volt-kilogram relationship	1.782 661 845 e-36	0.000 000 039 e-36	kg
elementary charge	1.602 176 565 e-19	0.000 000 035 e-19	C
elementary charge over h	2.417 989 348 e14	0.000 000 053 e14	A J^-1
Faraday constant	96 485.3365	0.0021	C mol^-1
Faraday constant for conventional electric current	96 485.3321	0.0043	C_90 mol^-1
Fermi coupling constant	1.166 364 e-5	0.000 005 e-5	GeV^-2
fine-structure constant	7.297 352 5698 e-3	0.000 000 0024 e-3	
first radiation constant	3.741 771 53 e-16	0.000 000 17 e-16	W m^2
first radiation constant for spectral radiance	1.191 042 869 e-16	0.000 000 053 e-16	W m^2 sr^-1
hartree-atomic mass unit relationship	2.921 262 3246 e-8	0.000 000 0021 e-8	u
hartree-electron volt relationship	27.211 385 05	0.000 000 60	eV
Hartree energy	4.359 744 34 e-18	0.000 000 19 e-18	J
Hartree energy in eV	27.211 385 05	0.000 000 60	eV
hartree-hertz relationship	6.579 683 920 729 e15	0.000 000 000 033 e15	Hz
hartree-inverse meter relationship	2.194 746 313 708 e7	0.000 000 000 011 e7	m^-1
hartree-joule relationship	4.359 744 34 e-18	0.000 000 19 e-18	J
hartree-kelvin relationship	3.157 7504 e5	0.000 0029 e5	K
hartree-kilogram relationship	4.850 869 79 e-35	0.000 000 21 e-35	kg
helion-electron mass ratio	5495.885 2754	0.000 0050	
helion g factor	-4.255 250 613	0.000 000 050	
helion mag. mom.	-1.074 617 486 e-26	0.000 000 027 e-26	J T^-1
helion mag. mom. to Bohr magneton ratio	-1.158 740 958 e-3	0.000 000 014 e-3	
helion mag. mom. to nuclear magneton ratio	-2.127 625 306	0.000 000 025	
helion mass	5.006 412 34 e-27	0.000 000 22 e-27	kg
helion mass energy equivalent	4.499 539 02 e-10	0.000 000 20 e-10	J
helion mass energy equivalent in MeV	2808.391 482	0.000 062	MeV
helion mass in u	3.014 932 2468	0.000 000 0025	u
helion molar mass	3.014 932 2468 e-3	0.000 000 0025 e-3	kg mol^-1
helion-proton mass ratio	2.993 152 6707	0.000 000 0025	
hertz-atomic mass unit relationship	4.439 821 6689 e-24	0.000 000 0031 e-24	u
hertz-electron volt relationship	4.135 667 516 e-15	0.000 000 091 e-15	eV
hertz-hartree relationship	1.519 829 846 0045 e-16	0.000 000 000 0076 e-16	E_h
hertz-inverse meter relationship	3.335 640 951... e-9	(exact)	m^-1
hertz-joule relationship	6.626 069 57 e-34	0.000 000 29 e-34	J
hertz-kelvin relationship	4.799 2434 e-11	0.000 0044 e-11	K
hertz-kilogram relationship	7.372 496 68 e-51	0.000 000 33 e-51	kg
inverse fine-structure constant	137.035 999 074	0.000 000 044	
inverse meter-atomic mass unit relationship	1.331 025 051 20 e-15	0.000 000 000 94 e-15	u
inverse meter-electron volt relationship	1.239 841 930 e-6	0.000 000 027 e-6	eV
inverse meter-hartree relationship	4.556 335 252 755 e-8	0.000 000 000 023 e-8	E_h
inverse meter-hertz relationship	299 792 458	(exact)	Hz
inverse meter-joule relationship	1.986 445 684 e-25	0.000 000 088 e-25	J
inverse meter-kelvin relationship	1.438 7770 e-2	0.000 0013 e-2	K
inverse meter-kilogram relationship	2.210 218 902 e-42	0.000 000 098 e-42	kg
inverse of conductance quantum	12 906.403 7217	0.000 0042	ohm
Josephson constant	483 597.870 e9	0.011 e9	Hz V^-1
joule-atomic mass unit relationship	6.700 535 85 e9	0.000 000 30 e9	u
joule-electron volt relationship	6.241 509 34 e18	0.000 000 14 e18	eV
joule-hartree relationship	2.293 712 48 e17	0.000 000 10 e17	E_h
joule-hertz relationship	1.509 190 311 e33	0.000 000 067 e33	Hz
joule-inverse meter relationship	5.034 117 01 e24	0.000 000 22 e24	m^-1
joule-kelvin relationship	7.242 9716 e22	0.000 0066 e22	K
joule-kilogram relationship	1.112 650 056... e-17	(exact)	kg
kelvin-atomic mass unit relationship	9.251 0868 e-14	0.000 0084 e-14	u
kelvin-electron volt relationship	8.617 3324 e-5	0.000 0078 e-5	eV
kelvin-hartree relationship	3.166 8114 e-6	0.000 0029 e-6	E_h
kelvin-hertz relationship	2.083 6618 e10	0.000 0019 e10	Hz
kelvin-inverse meter relationship	69.503 476	0.000 063	m^-1
kelvin-joule relationship	1.380 6488 e-23	0.000 0013 e-23	J
kelvin-kilogram relationship	1.536 1790 e-40	0.000 0014 e-40	kg
kilogram-atomic mass unit relationship	6.022 141 29 e26	0.000 000 27 e26	u
kilogram-electron volt relationship	5.609 588 85 e35	0.000 000 12 e35	eV
kilogram-hartree relationship	2.061 485 968 e34	0.000 000 091 e34	E_h
kilogram-hertz relationship	1.356 392 608 e50	0.000 000 060 e50	Hz
kilogram-inverse meter relationship	4.524 438 73 e41	0.000 000 20 e41	m^-1
kilogram-joule relationship	8.987 551 787... e16	(exact)	J
kilogram-kelvin relationship	6.509 6582 e39	0.000 0059 e39	K
lattice parameter of silicon	543.102 0504 e-12	0.000 0089 e-12	m
Loschmidt constant (273.15 K, 100 kPa)	2.651 6462 e25	0.000 0024 e25	m^-3
Loschmidt constant (273.15 K, 101.325 kPa)	2.686 7805 e25	0.000 0024 e25	m^-3
mag. constant	12.566 370 614... e-7	(exact)	N A^-2
mag. flux quantum	2.067 833 758 e-15	0.000 000 046 e-15	Wb
molar gas constant	8.314 4621	0.000 0075	J mol^-1 K^-1
molar mass constant	1 e-3	(exact)	kg mol^-1
molar mass of carbon-12	12 e-3	(exact)	kg mol^-1
molar Planck constant	3.990 312 7176 e-10	0.000 000 0028 e-10	J s mol^-1
molar Planck constant times c	0.119 626 565 779	0.000 000 000 084	J m mol^-1
molar volume of ideal gas (273.15 K, 100 kPa)	22.710 953 e-3	0.000 021 e-3	m^3 mol^-1
molar volume of ideal gas (273.15 K, 101.325 kPa)	22.413 968 e-3	0.000 020 e-3	m^3 mol^-1
molar volume of silicon	12.058 833 01 e-6	0.000 000 80 e-6	m^3 mol^-1
Mo x unit	1.002 099 52 e-13	0.000 000 53 e-13	m
muon Compton wavelength	11.734 441 03 e-15	0.000 000 30 e-15	m

muon Compton wavelength over 2 pi	1.867 594 294 e-15	0.000 000 047 e-15	m
muon-electron mass ratio	206.768 2843	0.000 0052	
muon g factor	-2.002 331 8418	0.000 000 0013	
muon mag. mom.	-4.490 448 07 e-26	0.000 000 15 e-26	J T^-1
muon mag. mom. anomaly	1.165 920 91 e-3	0.000 000 63 e-3	
muon mag. mom. to Bohr magneton ratio	-4.841 970 44 e-3	0.000 000 12 e-3	
muon mag. mom. to nuclear magneton ratio	-8.890 596 97	0.000 000 22	
muon mass	1.883 531 475 e-28	0.000 000 096 e-28	kg
muon mass energy equivalent	1.692 833 667 e-11	0.000 000 086 e-11	J
muon mass energy equivalent in MeV	105.658 3715	0.000 0035	MeV
muon mass in u	0.113 428 9267	0.000 000 0029	u
muon molar mass	0.113 428 9267 e-3	0.000 000 0029 e-3	kg mol^-1
muon-neutron mass ratio	0.112 454 5177	0.000 000 0028	
muon-proton mag. mom. ratio	-3.183 345 107	0.000 000 084	
muon-proton mass ratio	0.112 609 5272	0.000 000 0028	
muon-tau mass ratio	5.946 49 e-2	0.000 54 e-2	
natural unit of action	1.054 571 726 e-34	0.000 000 047 e-34	J s
natural unit of action in eV s	6.582 119 28 e-16	0.000 000 15 e-16	eV s
natural unit of energy	8.187 105 06 e-14	0.000 000 36 e-14	J
natural unit of energy in MeV	0.510 998 928	0.000 000 011	MeV
natural unit of length	386.159 268 00 e-15	0.000 000 25 e-15	m
natural unit of mass	9.109 382 91 e-31	0.000 000 40 e-31	kg
natural unit of mom.um	2.730 924 29 e-22	0.000 000 12 e-22	kg m s^-1
natural unit of mom.um in MeV/c	0.510 998 928	0.000 000 011	MeV/c
natural unit of time	1.288 088 668 33 e-21	0.000 000 000 83 e-21	s
natural unit of velocity	299 792 458	(exact)	m s^-1
neutron Compton wavelength	1.319 590 9068 e-15	0.000 000 0011 e-15	m
neutron Compton wavelength over 2 pi	0.210 019 415 68 e-15	0.000 000 000 17 e-15	m
neutron-electron mag. mom. ratio	1.040 668 82 e-3	0.000 000 25 e-3	
neutron-electron mass ratio	1838.683 6605	0.000 0011	
neutron g factor	-3.826 085 45	0.000 000 90	
neutron gyromag. ratio	1.832 471 79 e8	0.000 000 43 e8	s^-1 T^-1
neutron gyromag. ratio over 2 pi	29.164 6943	0.000 0069	MHz T^-1
neutron mag. mom.	-0.966 236 47 e-26	0.000 000 23 e-26	J T^-1
neutron mag. mom. to Bohr magneton ratio	-1.041 875 63 e-3	0.000 000 25 e-3	
neutron mag. mom. to nuclear magneton ratio	-1.913 042 72	0.000 000 45	
neutron mass	1.674 927 351 e-27	0.000 000 074 e-27	kg
neutron mass energy equivalent	1.505 349 631 e-10	0.000 000 066 e-10	J
neutron mass energy equivalent in MeV	939.565 379	0.000 021	MeV
neutron mass in u	1.008 664 916 00	0.000 000 000 43	u
neutron molar mass	1.008 664 916 00 e-3	0.000 000 000 43 e-3	kg mol^-1
neutron-muon mass ratio	8.892 484 00	0.000 000 22	
neutron-proton mag. mom. ratio	-0.684 979 34	0.000 000 16	
neutron-proton mass difference	2.305 573 92 e-30	0.000 000 76 e-30	
neutron-proton mass difference energy equivalent	2.072 146 50 e-13	0.000 000 68 e-13	
neutron-proton mass difference energy equivalent in MeV	1.293 332 17	0.000 000 42	
neutron-proton mass difference in u	0.001 388 449 19	0.000 000 000 45	
neutron-proton mass ratio	1.001 378 419 17	0.000 000 000 45	
neutron-tau mass ratio	0.528 790	0.000 048	
neutron to shielded proton mag. mom. ratio	-0.684 996 94	0.000 000 16	
Newtonian constant of gravitation	6.673 84 e-11	0.000 80 e-11	m^3 kg^-1 s^-2
Newtonian constant of gravitation over h-bar c	6.708 37 e-39	0.000 80 e-39	(GeV/c^2)^-2
nuclear magneton	5.050 783 53 e-27	0.000 000 11 e-27	J T^-1
nuclear magneton in eV/T	3.152 451 2605 e-8	0.000 000 0022 e-8	eV T^-1
nuclear magneton in inverse meters per tesla	2.542 623 527 e-2	0.000 000 056 e-2	m^-1 T^-1
nuclear magneton in K/T	3.658 2682 e-4	0.000 0033 e-4	K T^-1
nuclear magneton in MHz/T	7.622 593 57	0.000 000 17	MHz T^-1
Planck constant	6.626 069 57 e-34	0.000 000 29 e-34	J s
Planck constant in eV s	4.135 667 516 e-15	0.000 000 091 e-15	eV s
Planck constant over 2 pi	1.054 571 726 e-34	0.000 000 047 e-34	J s
Planck constant over 2 pi in eV s	6.582 119 28 e-16	0.000 000 15 e-16	eV s
Planck constant over 2 pi times c in MeV fm	197.326 9718	0.000 0044	MeV fm
Planck length	1.616 199 e-35	0.000 097 e-35	m
Planck mass	2.176 51 e-8	0.000 13 e-8	kg
Planck mass energy equivalent in GeV	1.220 932 e19	0.000 073 e19	GeV
Planck temperature	1.416 833 e32	0.000 085 e32	K
Planck time	5.391 06 e-44	0.000 32 e-44	s
proton charge to mass quotient	9.578 833 58 e7	0.000 000 21 e7	C kg^-1
proton Compton wavelength	1.321 409 856 23 e-15	0.000 000 000 94 e-15	m
proton Compton wavelength over 2 pi	0.210 308 910 47 e-15	0.000 000 000 15 e-15	m
proton-electron mass ratio	1836.152 672 45	0.000 000 75	
proton g factor	5.585 694 713	0.000 000 046	
proton gyromag. ratio	2.675 222 005 e8	0.000 000 063 e8	s^-1 T^-1
proton gyromag. ratio over 2 pi	42.577 4806	0.000 0010	MHz T^-1
proton mag. mom.	1.410 606 743 e-26	0.000 000 033 e-26	J T^-1
proton mag. mom. to Bohr magneton ratio	1.521 032 210 e-3	0.000 000 012 e-3	
proton mag. mom. to nuclear magneton ratio	2.792 847 356	0.000 000 023	
proton mag. shielding correction	25.694 e-6	0.014 e-6	
proton mass	1.672 621 777 e-27	0.000 000 074 e-27	kg
proton mass energy equivalent	1.503 277 484 e-10	0.000 000 066 e-10	J
proton mass energy equivalent in MeV	938.272 046	0.000 021	MeV
proton mass in u	1.007 276 466 812	0.000 000 000 090	u
proton molar mass	1.007 276 466 812 e-3	0.000 000 000 090 e-3	kg mol^-1
proton-muon mass ratio	8.880 243 31	0.000 000 22	
proton-neutron mag. mom. ratio	-1.459 898 06	0.000 000 34	
proton-neutron mass ratio	0.998 623 478 26	0.000 000 000 45	
proton rms charge radius	0.8775 e-15	0.0051 e-15	m
proton-tau mass ratio	0.528 063	0.000 048	
quantum of circulation	3.636 947 5520 e-4	0.000 000 0024 e-4	m^2 s^-1
quantum of circulation times 2	7.273 895 1040 e-4	0.000 000 0047 e-4	m^2 s^-1
Rydberg constant	10 973 731.568 539	0.000 055	m^-1
Rydberg constant times c in Hz	3.289 841 960 364 e15	0.000 000 000 017 e15	Hz
Rydberg constant times hc in eV	13.605 692 53	0.000 000 30	eV
Rydberg constant times hc in J	2.179 872 171 e-18	0.000 000 096 e-18	J
Sackur-Tetrode constant (1 K, 100 kPa)	-1.151 7078	0.000 0023	
Sackur-Tetrode constant (1 K, 101.325 kPa)	-1.164 8708	0.000 0023	
second radiation constant	1.438 7770 e-2	0.000 0013 e-2	m K
shielded helion gyromag. ratio	2.037 894 659 e8	0.000 000 051 e8	s^-1 T^-1
shielded helion gyromag. ratio over 2 pi	32.434 100 84	0.000 000 81	MHz T^-1
shielded helion mag. mom.	-1.074 553 044 e-26	0.000 000 027 e-26	J T^-1
shielded helion mag. mom. to Bohr magneton ratio	-1.158 671 471 e-3	0.000 000 014 e-3	
shielded helion mag. mom. to nuclear magneton ratio	-2.127 497 718	0.000 000 025	
shielded helion to proton mag. mom. ratio	-0.761 766 558	0.000 000 011	

shielded helion to shielded proton mag. mom. ratio	-0.761 786 1313	0.000 000 0033	
shielded proton gyromag. ratio	2.675 153 268 e8	0.000 000 066 e8	s^-1 T^-1
shielded proton gyromag. ratio over 2 pi	42.576 3866	0.000 0010	MHz T^-1
shielded proton mag. mom.	1.410 570 499 e-26	0.000 000 035 e-26	J T^-1
shielded proton mag. mom. to Bohr magneton ratio	1.520 993 128 e-3	0.000 000 017 e-3	
shielded proton mag. mom. to nuclear magneton ratio	2.792 775 598	0.000 000 030	
speed of light in vacuum	299 792 458	(exact)	m s^-1
standard acceleration of gravity	9.806 65	(exact)	m s^-2
standard atmosphere	101 325	(exact)	Pa
standard-state pressure	100 000	(exact)	Pa
Stefan-Boltzmann constant	5.670 373 e-8	0.000 021 e-8	W m^-2 K^-4
tau Compton wavelength	0.697 787 e-15	0.000 063 e-15	m
tau Compton wavelength over 2 pi	0.111 056 e-15	0.000 010 e-15	m
tau-electron mass ratio	3477.15	0.31	
tau mass	3.167 47 e-27	0.000 29 e-27	kg
tau mass energy equivalent	2.846 78 e-10	0.000 26 e-10	J
tau mass energy equivalent in MeV	1776.82	0.16	MeV
tau mass in u	1.907 49	0.000 17	u
tau molar mass	1.907 49 e-3	0.000 17 e-3	kg mol^-1
tau-muon mass ratio	16.8167	0.0015	
tau-neutron mass ratio	1.891 11	0.000 17	
tau-proton mass ratio	1.893 72	0.000 17	
Thomson cross section	0.665 245 8734 e-28	0.000 000 0013 e-28	m^2
triton-electron mass ratio	5496.921 5267	0.000 0050	
triton g factor	5.957 924 896	0.000 000 076	
triton mag. mom.	1.504 609 447 e-26	0.000 000 038 e-26	J T^-1
triton mag. mom. to Bohr magneton ratio	1.622 393 657 e-3	0.000 000 021 e-3	
triton mag. mom. to nuclear magneton ratio	2.978 962 448	0.000 000 038	
triton mass	5.007 356 30 e-27	0.000 000 22 e-27	kg
triton mass energy equivalent	4.500 387 41 e-10	0.000 000 20 e-10	J
triton mass energy equivalent in MeV	2808.921 005	0.000 062	MeV
triton mass in u	3.015 500 7134	0.000 000 0025	u
triton molar mass	3.015 500 7134 e-3	0.000 000 0025 e-3	kg mol^-1
triton-proton mass ratio	2.993 717 0308	0.000 000 0025	
unified atomic mass unit	1.660 538 921 e-27	0.000 000 073 e-27	kg
von Klitzing constant	25 812.807 4434	0.000 0084	ohm
weak mixing angle	0.2223	0.0021	
Wien frequency displacement law constant	5.878 9254 e10	0.000 0053 e10	Hz K^-1
Wien wavelength displacement law constant	2.897 7721 e-3	0.000 0026 e-3	m K

8.2 2010 Fundamental Physical Constants — Complete Listing

2014 Fundamental Physical Constants --- Complete Listing

From: http://physics.nist.gov/constants

Quantity	Value	Uncertainty	Unit
{220} lattice spacing of silicon	192.015 5714 e-12	0.000 0032 e-12	m
alpha particle-electron mass ratio	7294.299 541 36	0.000 000 24	
alpha particle mass	6.644 657 230 e-27	0.000 000 082 e-27	kg
alpha particle mass energy equivalent	5.971 920 097 e-10	0.000 000 073 e-10	J
alpha particle mass energy equivalent in MeV	3727.379 378	0.000 023	MeV
alpha particle mass in u	4.001 506 179 127	0.000 000 000 063	u
alpha particle molar mass	4.001 506 179 127 e-3	0.000 000 000 063 e-3	kg mol^-1
alpha particle-proton mass ratio	3.972 599 689 07	0.000 000 000 36	
Angstrom star	1.000 014 95 e-10	0.000 000 90 e-10	m
atomic mass constant	1.660 539 040 e-27	0.000 000 020 e-27	kg
atomic mass constant energy equivalent	1.492 418 062 e-10	0.000 000 018 e-10	J
atomic mass constant energy equivalent in MeV	931.494 0954	0.000 0057	MeV
atomic mass unit-electron volt relationship	931.494 0954 e6	0.000 0057 e6	eV
atomic mass unit-hartree relationship	3.423 177 6902 e7	0.000 000 0016 e7	E_h
atomic mass unit-hertz relationship	2.252 342 7206 e23	0.000 000 0010 e23	Hz
atomic mass unit-inverse meter relationship	7.513 006 6166 e14	0.000 000 0034 e14	m^-1
atomic mass unit-joule relationship	1.492 418 062 e-10	0.000 000 018 e-10	J
atomic mass unit-kelvin relationship	1.080 954 38 e13	0.000 000 62 e13	K
atomic mass unit-kilogram relationship	1.660 539 040 e-27	0.000 000 020 e-27	kg
atomic unit of 1st hyperpolarizability	3.206 361 329 e-53	0.000 000 020 e-53	C^3 m^3 J^-2
atomic unit of 2nd hyperpolarizability	6.235 380 085 e-65	0.000 000 077 e-65	C^4 m^4 J^-3
atomic unit of action	1.054 571 800 e-34	0.000 000 013 e-34	J s
atomic unit of charge	1.602 176 6208 e-19	0.000 000 0098 e-19	C
atomic unit of charge density	1.081 202 3770 e12	0.000 000 0067 e12	C m^-3
atomic unit of current	6.623 618 183 e-3	0.000 000 041 e-3	A
atomic unit of electric dipole mom.	8.478 353 552 e-30	0.000 000 052 e-30	C m
atomic unit of electric field	5.142 206 707 e11	0.000 000 032 e11	V m^-1
atomic unit of electric field gradient	9.717 362 356 e21	0.000 000 060 e21	V m^-2
atomic unit of electric polarizability	1.648 777 2731 e-41	0.000 000 0011 e-41	C^2 m^2 J^-1
atomic unit of electric potential	27.211 386 02	0.000 000 17	V
atomic unit of electric quadrupole mom.	4.486 551 484 e-40	0.000 000 028 e-40	C m^2
atomic unit of energy	4.359 744 650 e-18	0.000 000 054 e-18	J
atomic unit of force	8.238 723 36 e-8	0.000 000 10 e-8	N
atomic unit of length	0.529 177 210 67 e-10	0.000 000 000 12 e-10	m
atomic unit of mag. dipole mom.	1.854 801 999 e-23	0.000 000 011 e-23	J T^-1
atomic unit of mag. flux density	2.350 517 550 e5	0.000 000 014 e5	T
atomic unit of magnetizability	7.891 036 5886 e-29	0.000 000 0090 e-29	J T^-2
atomic unit of mass	9.109 383 56 e-31	0.000 000 11 e-31	kg
atomic unit of mom.um	1.992 851 882 e-24	0.000 000 024 e-24	kg m s^-1
atomic unit of permittivity	1.112 650 056... e-10	(exact)	F m^-1
atomic unit of time	2.418 884 326 509 e-17	0.000 000 000 014 e-17	s
atomic unit of velocity	2.187 691 262 77 e6	0.000 000 000 50 e6	m s^-1
Avogadro constant	6.022 140 857 e23	0.000 000 074 e23	mol^-1
Bohr magneton	927.400 9994 e-26	0.000 0057 e-26	J T^-1
Bohr magneton in eV/T	5.788 381 8012 e-5	0.000 000 0026 e-5	eV T^-1
Bohr magneton in Hz/T	13.996 245 042 e9	0.000 000 086 e9	Hz T^-1
Bohr magneton in inverse meters per tesla	46.686 448 14	0.000 000 29	m^-1 T^-1
Bohr magneton in K/T	0.671 714 05	0.000 000 39	K T^-1
Bohr radius	0.529 177 210 67 e-10	0.000 000 000 12 e-10	m
Boltzmann constant	1.380 648 52 e-23	0.000 000 79 e-23	J K^-1
Boltzmann constant in eV/K	8.617 3303 e-5	0.000 0050 e-5	eV K^-1
Boltzmann constant in Hz/K	2.083 6612 e10	0.000 0012 e10	Hz K^-1
Boltzmann constant in inverse meters per kelvin	69.503 457	0.000 040	m^-1 K^-1
characteristic impedance of vacuum	376.730 313 461...	(exact)	ohm
classical electron radius	2.817 940 3227 e-15	0.000 000 0019 e-15	m
Compton wavelength	2.426 310 2367 e-12	0.000 000 0011 e-12	m
Compton wavelength over 2 pi	386.159 267 64 e-15	0.000 000 18 e-15	m
conductance quantum	7.748 091 7310 e-5	0.000 000 0018 e-5	S
conventional value of Josephson constant	483 597.9 e9	(exact)	Hz V^-1
conventional value of von Klitzing constant	25 812.807	(exact)	ohm
Cu x unit	1.002 076 97 e-13	0.000 000 28 e-13	m
deuteron-electron mag. mom. ratio	-4.664 345 535 e-4	0.000 000 026 e-4	
deuteron-electron mass ratio	3670.482 967 85	0.000 000 13	
deuteron g factor	0.857 438 2311	0.000 000 0048	
deuteron mag. mom.	0.433 073 5040 e-26	0.000 000 0036 e-26	J T^-1
deuteron mag. mom. to Bohr magneton ratio	0.466 975 4554 e-3	0.000 000 0026 e-3	
deuteron mag. mom. to nuclear magneton ratio	0.857 438 2311	0.000 000 0048	
deuteron mass	3.343 583 719 e-27	0.000 000 041 e-27	kg
deuteron mass energy equivalent	3.005 063 183 e-10	0.000 000 037 e-10	J
deuteron mass energy equivalent in MeV	1875.612 928	0.000 012	MeV
deuteron mass in u	2.013 553 212 745	0.000 000 000 040	u
deuteron molar mass	2.013 553 212 745 e-3	0.000 000 000 040 e-3	kg mol^-1
deuteron-neutron mag. mom. ratio	-0.448 206 52	0.000 000 11	
deuteron-proton mag. mom. ratio	0.307 012 2077	0.000 000 0015	
deuteron-proton mass ratio	1.999 007 500 87	0.000 000 000 19	
deuteron rms charge radius	2.1413 e-15	0.0025 e-15	m
electric constant	8.854 187 817... e-12	(exact)	F m^-1
electron charge to mass quotient	-1.758 820 024 e11	0.000 000 011 e11	C kg^-1
electron-deuteron mag. mom. ratio	-2143.923 499	0.000 012	
electron-deuteron mass ratio	2.724 437 107 484 e-4	0.000 000 000 096 e-4	
electron g factor	-2.002 319 304 361 82	0.000 000 000 000 52	
electron gyromag. ratio	1.760 859 644 e11	0.000 000 011 e11	s^-1 T^-1
electron gyromag. ratio over 2 pi	28 024.951 64	0.000 17	MHz T^-1
electron-helion mass ratio	1.819 543 074 854 e-4	0.000 000 000 088 e-4	
electron mag. mom.	-928.476 4620 e-26	0.000 0057 e-26	J T^-1
electron mag. mom. anomaly	1.159 652 180 91 e-3	0.000 000 000 26 e-3	
electron mag. mom. to Bohr magneton ratio	-1.001 159 652 180 91	0.000 000 000 000 26	
electron mag. mom. to nuclear magneton ratio	-1838.281 972 34	0.000 000 17	
electron mass	9.109 383 56 e-31	0.000 000 11 e-31	kg
electron mass energy equivalent	8.187 105 65 e-14	0.000 000 10 e-14	J

electron mass energy equivalent in MeV	0.510 998 9461	0.000 000 0031	MeV
electron mass in u	5.485 799 090 70 e-4	0.000 000 000 16 e-4	u
electron molar mass	5.485 799 090 70 e-7	0.000 000 000 16 e-7	kg mol^-1
electron-muon mag. mom. ratio	206.766 9880	0.000 0046	
electron-muon mass ratio	4.836 331 70 e-3	0.000 000 11 e-3	
electron-neutron mag. mom. ratio	960.920 50	0.000 23	
electron-neutron mass ratio	5.438 673 4428 e-4	0.000 000 0027 e-4	
electron-proton mag. mom. ratio	-658.210 6866	0.000 0020	
electron-proton mass ratio	5.446 170 213 52 e-4	0.000 000 000 52 e-4	
electron-tau mass ratio	2.875 92 e-4	0.000 26 e-4	
electron to alpha particle mass ratio	1.370 933 554 798 e-4	0.000 000 000 045 e-4	
electron to shielded helion mag. mom. ratio	864.058 257	0.000 010	
electron to shielded proton mag. mom. ratio	-658.227 5971	0.000 0072	
electron-triton mass ratio	1.819 200 062 203 e-4	0.000 000 000 084 e-4	
electron volt	1.602 176 6208 e-19	0.000 000 0098 e-19	J
electron volt-atomic mass unit relationship	1.073 544 1105 e-9	0.000 000 0066 e-9	u
electron volt-hartree relationship	3.674 932 248 e-2	0.000 000 023 e-2	E_h
electron volt-hertz relationship	2.417 989 262 e14	0.000 000 015 e14	Hz
electron volt-inverse meter relationship	8.065 544 005 e5	0.000 000 050 e5	m^-1
electron volt-joule relationship	1.602 176 6208 e-19	0.000 000 0098 e-19	J
electron volt-kelvin relationship	1.160 452 21 e4	0.000 000 67 e4	K
electron volt-kilogram relationship	1.782 661 907 e-36	0.000 000 011 e-36	kg
elementary charge	1.602 176 6208 e-19	0.000 000 0098 e-19	C
elementary charge over h	2.417 989 262 e14	0.000 000 015 e14	A J^-1
Faraday constant	96 485.332 89	0.000 59	C mol^-1
Faraday constant for conventional electric current	96 485.3251	0.0012	C_90 mol^-1
Fermi coupling constant	1.166 3787 e-5	0.000 0006 e-5	GeV^-2
fine-structure constant	7.297 352 5664 e-3	0.000 000 0017 e-3	
first radiation constant	3.741 771 790 e-16	0.000 000 046 e-16	W m^2
first radiation constant for spectral radiance	1.191 042 953 e-16	0.000 000 015 e-16	W m^2 sr^-1
hartree-atomic mass unit relationship	2.921 262 3197 e-8	0.000 000 0013 e-8	u
hartree-electron volt relationship	27.211 386 02	0.000 000 17	eV
Hartree energy	4.359 744 650 e-18	0.000 000 054 e-18	J
Hartree energy in eV	27.211 386 02	0.000 000 17	eV
hartree-hertz relationship	6.579 683 920 711 e15	0.000 000 000 039 e15	Hz
hartree-inverse meter relationship	2.194 746 313 702 e7	0.000 000 000 013 e7	m^-1
hartree-joule relationship	4.359 744 650 e-18	0.000 000 054 e-18	J
hartree-kelvin relationship	3.157 7513 e5	0.000 0018 e5	K
hartree-kilogram relationship	4.850 870 129 e-35	0.000 000 060 e-35	kg
helion-electron mass ratio	5495.885 279 22	0.000 000 27	
helion g factor	-4.255 250 616	0.000 000 050	
helion mag. mom.	-1.074 617 522 e-26	0.000 000 014 e-26	J T^-1
helion mag. mom. to Bohr magneton ratio	-1.158 740 958 e-3	0.000 000 014 e-3	
helion mag. mom. to nuclear magneton ratio	-2.127 625 308	0.000 000 025	
helion mass	5.006 412 700 e-27	0.000 000 062 e-27	kg
helion mass energy equivalent	4.499 539 341 e-10	0.000 000 055 e-10	J
helion mass energy equivalent in MeV	2808.391 586	0.000 017	MeV
helion mass in u	3.014 932 246 73	0.000 000 000 12	u
helion molar mass	3.014 932 246 73 e-3	0.000 000 000 12 e-3	kg mol^-1
helion-proton mass ratio	2.993 152 670 46	0.000 000 000 29	
hertz-atomic mass unit relationship	4.439 821 6616 e-24	0.000 000 0020 e-24	u
hertz-electron volt relationship	4.135 667 662 e-15	0.000 000 025 e-15	eV
hertz-hartree relationship	1.519 829 846 0088 e-16	0.000 000 000 0090 e-16	E_h
hertz-inverse meter relationship	3.335 640 951... e-9	(exact)	m^-1
hertz-joule relationship	6.626 070 040 e-34	0.000 000 081 e-34	J
hertz-kelvin relationship	4.799 2447 e-11	0.000 0028 e-11	K
hertz-kilogram relationship	7.372 497 201 e-51	0.000 000 091 e-51	kg
inverse fine-structure constant	137.035 999 139	0.000 000 031	
inverse meter-atomic mass unit relationship	1.331 025 049 00 e-15	0.000 000 000 61 e-15	u
inverse meter-electron volt relationship	1.239 841 9739 e-6	0.000 000 0076 e-6	eV
inverse meter-hartree relationship	4.556 335 252 767 e-8	0.000 000 000 027 e-8	E_h
inverse meter-hertz relationship	299 792 458	(exact)	Hz
inverse meter-joule relationship	1.986 445 824 e-25	0.000 000 024 e-25	J
inverse meter-kelvin relationship	1.438 777 36 e-2	0.000 000 83 e-2	K
inverse meter-kilogram relationship	2.210 219 057 e-42	0.000 000 027 e-42	kg
inverse of conductance quantum	12 906.403 7278	0.000 0029	ohm
Josephson constant	483 597.8525 e9	0.0030 e9	Hz V^-1
joule-atomic mass unit relationship	6.700 535 363 e9	0.000 000 082 e9	u
joule-electron volt relationship	6.241 509 126 e18	0.000 000 038 e18	eV
joule-hartree relationship	2.293 712 317 e17	0.000 000 028 e17	E_h
joule-hertz relationship	1.509 190 205 e33	0.000 000 019 e33	Hz
joule-inverse meter relationship	5.034 116 651 e24	0.000 000 062 e24	m^-1
joule-kelvin relationship	7.242 9731 e22	0.000 0042 e22	K
joule-kilogram relationship	1.112 650 056... e-17	(exact)	kg
kelvin-atomic mass unit relationship	9.251 0842 e-14	0.000 0053 e-14	u
kelvin-electron volt relationship	8.617 3303 e-5	0.000 0050 e-5	eV
kelvin-hartree relationship	3.166 8105 e-6	0.000 0018 e-6	E_h
kelvin-hertz relationship	2.083 6612 e10	0.000 0012 e10	Hz
kelvin-inverse meter relationship	69.503 457	0.000 040	m^-1
kelvin-joule relationship	1.380 648 52 e-23	0.000 000 79 e-23	J
kelvin-kilogram relationship	1.536 178 65 e-40	0.000 000 88 e-40	kg
kilogram-atomic mass unit relationship	6.022 140 857 e26	0.000 000 074 e26	u
kilogram-electron volt relationship	5.609 588 650 e35	0.000 000 034 e35	eV
kilogram-hartree relationship	2.061 485 823 e34	0.000 000 025 e34	E_h
kilogram-hertz relationship	1.356 392 512 e50	0.000 000 017 e50	Hz
kilogram-inverse meter relationship	4.524 438 411 e41	0.000 000 056 e41	m^-1
kilogram-joule relationship	8.987 551 787... e16	(exact)	J
kilogram-kelvin relationship	6.509 6595 e39	0.000 0037 e39	K
lattice parameter of silicon	543.102 0504 e-12	0.000 0089 e-12	m
Loschmidt constant (273.15 K, 100 kPa)	2.651 6467 e25	0.000 0015 e25	m^-3
Loschmidt constant (273.15 K, 101.325 kPa)	2.686 7811 e25	0.000 0015 e25	m^-3
mag. constant	12.566 370 614... e-7	(exact)	N A^-2
mag. flux quantum	2.067 833 831 e-15	0.000 000 013 e-15	Wb
molar gas constant	8.314 4598	0.000 0048	J mol^-1 K^-1
molar mass constant	1 e-3	(exact)	kg mol^-1
molar mass of carbon-12	12 e-3	(exact)	kg mol^-1
molar Planck constant	3.990 312 7110 e-10	0.000 000 0018 e-10	J s mol^-1
molar Planck constant times c	0.119 626 565 582	0.000 000 000 054	J m mol^-1
molar volume of ideal gas (273.15 K, 100 kPa)	22.710 947 e-3	0.000 013 e-3	m^3 mol^-1
molar volume of ideal gas (273.15 K, 101.325 kPa)	22.413 962 e-3	0.000 013 e-3	m^3 mol^-1
molar volume of silicon	12.058 832 14 e-6	0.000 000 61 e-6	m^3 mol^-1
Mo x unit	1.002 099 52 e-13	0.000 000 53 e-13	m
muon Compton wavelength	11.734 441 11 e-15	0.000 000 26 e-15	m

muon Compton wavelength over 2 pi	1.867 594 308 e-15	0.000 000 042 e-15	m
muon-electron mass ratio	206.768 2826	0.000 0046	
muon g factor	-2.002 331 8418	0.000 000 0013	
muon mag. mom.	-4.490 448 26 e-26	0.000 000 10 e-26	J T^-1
muon mag. mom. anomaly	1.165 920 89 e-3	0.000 000 63 e-3	
muon mag. mom. to Bohr magneton ratio	-4.841 970 48 e-3	0.000 000 11 e-3	
muon mag. mom. to nuclear magneton ratio	-8.890 597 05	0.000 000 20	
muon mass	1.883 531 594 e-28	0.000 000 048 e-28	kg
muon mass energy equivalent	1.692 833 774 e-11	0.000 000 043 e-11	J
muon mass energy equivalent in MeV	105.658 3745	0.000 0024	MeV
muon mass in u	0.113 428 9257	0.000 000 0025	u
muon molar mass	0.113 428 9257 e-3	0.000 000 0025 e-3	kg mol^-1
muon-neutron mass ratio	0.112 454 5167	0.000 000 0025	
muon-proton mag. mom. ratio	-3.183 345 142	0.000 000 071	
muon-proton mass ratio	0.112 609 5262	0.000 000 0025	
muon-tau mass ratio	5.946 49 e-2	0.000 54 e-2	
natural unit of action	1.054 571 800 e-34	0.000 000 013 e-34	J s
natural unit of action in eV s	6.582 119 514 e-16	0.000 000 040 e-16	eV s
natural unit of energy	8.187 105 65 e-14	0.000 000 10 e-14	J
natural unit of energy in MeV	0.510 998 9461	0.000 000 0031	MeV
natural unit of length	386.159 267 64 e-15	0.000 000 18 e-15	m
natural unit of mass	9.109 383 56 e-31	0.000 000 11 e-31	kg
natural unit of mom.um	2.730 924 488 e-22	0.000 000 034 e-22	kg m s^-1
natural unit of mom.um in MeV/c	0.510 998 9461	0.000 000 0031	MeV/c
natural unit of time	1.288 088 667 12 e-21	0.000 000 000 58 e-21	s
natural unit of velocity	299 792 458	(exact)	m s^-1
neutron Compton wavelength	1.319 590 904 81 e-15	0.000 000 000 88 e-15	m
neutron Compton wavelength over 2 pi	0.210 019 415 36 e-15	0.000 000 000 14 e-15	m
neutron-electron mag. mom. ratio	1.040 668 82 e-3	0.000 000 25 e-3	
neutron-electron mass ratio	1838.683 661 58	0.000 000 90	
neutron g factor	-3.826 085 45	0.000 000 90	
neutron gyromag. ratio	1.832 471 72 e8	0.000 000 43 e8	s^-1 T^-1
neutron gyromag. ratio over 2 pi	29.164 6933	0.000 0069	MHz T^-1
neutron mag. mom.	-0.966 236 50 e-26	0.000 000 23 e-26	J T^-1
neutron mag. mom. to Bohr magneton ratio	-1.041 875 63 e-3	0.000 000 25 e-3	
neutron mag. mom. to nuclear magneton ratio	-1.913 042 73	0.000 000 45	
neutron mass	1.674 927 471 e-27	0.000 000 021 e-27	kg
neutron mass energy equivalent	1.505 349 739 e-10	0.000 000 019 e-10	J
neutron mass energy equivalent in MeV	939.565 4133	0.000 0058	MeV
neutron mass in u	1.008 664 915 88	0.000 000 000 49	u
neutron molar mass	1.008 664 915 88 e-3	0.000 000 000 49 e-3	kg mol^-1
neutron-muon mass ratio	8.892 484 08	0.000 000 20	
neutron-proton mag. mom. ratio	-0.684 979 34	0.000 000 16	
neutron-proton mass difference	2.305 573 77 e-30	0.000 000 85 e-30	
neutron-proton mass difference energy equivalent	2.072 146 37 e-13	0.000 000 76 e-13	
neutron-proton mass difference energy equivalent in MeV	1.293 332 05	0.000 000 48	
neutron-proton mass difference in u	0.001 388 449 00	0.000 000 000 51	
neutron-proton mass ratio	1.001 378 418 98	0.000 000 000 51	
neutron-tau mass ratio	0.528 790	0.000 048	
neutron to shielded proton mag. mom. ratio	-0.684 996 94	0.000 000 16	
Newtonian constant of gravitation	6.674 08 e-11	0.000 31 e-11	m^3 kg^-1 s^-2
Newtonian constant of gravitation over h-bar c	6.708 61 e-39	0.000 31 e-39	(GeV/c^2)^-2
nuclear magneton	5.050 783 699 e-27	0.000 000 031 e-27	J T^-1
nuclear magneton in eV/T	3.152 451 2550 e-8	0.000 000 0015 e-8	eV T^-1
nuclear magneton in inverse meters per tesla	2.542 623 432 e-2	0.000 000 016 e-2	m^-1 T^-1
nuclear magneton in K/T	3.658 2690 e-4	0.000 0021 e-4	K T^-1
nuclear magneton in MHz/T	7.622 593 285	0.000 000 047	MHz T^-1
Planck constant	6.626 070 040 e-34	0.000 000 081 e-34	J s
Planck constant in eV s	4.135 667 662 e-15	0.000 000 025 e-15	eV s
Planck constant over 2 pi	1.054 571 800 e-34	0.000 000 013 e-34	J s
Planck constant over 2 pi in eV s	6.582 119 514 e-16	0.000 000 040 e-16	eV s
Planck constant over 2 pi times c in MeV fm	197.326 9788	0.000 0012	MeV fm
Planck length	1.616 229 e-35	0.000 038 e-35	m
Planck mass	2.176 470 e-8	0.000 051 e-8	kg
Planck mass energy equivalent in GeV	1.220 910 e19	0.000 029 e19	GeV
Planck temperature	1.416 808 e32	0.000 033 e32	K
Planck time	5.391 16 e-44	0.000 13 e-44	s
proton charge to mass quotient	9.578 833 226 e7	0.000 000 059 e7	C kg^-1
proton Compton wavelength	1.321 409 853 96 e-15	0.000 000 000 61 e-15	m
proton Compton wavelength over 2 pi	0.210 308 910 109 e-15	0.000 000 000 097 e-15	m
proton-electron mass ratio	1836.152 673 89	0.000 000 17	
proton g factor	5.585 694 702	0.000 000 017	
proton gyromag. ratio	2.675 221 900 e8	0.000 000 018 e8	s^-1 T^-1
proton gyromag. ratio over 2 pi	42.577 478 92	0.000 000 29	MHz T^-1
proton mag. mom.	1.410 606 7873 e-26	0.000 000 0097 e-26	J T^-1
proton mag. mom. to Bohr magneton ratio	1.521 032 2053 e-3	0.000 000 0046 e-3	
proton mag. mom. to nuclear magneton ratio	2.792 847 3508	0.000 000 0085	
proton mag. shielding correction	25.691 e-6	0.011 e-6	
proton mass	1.672 621 898 e-27	0.000 000 021 e-27	kg
proton mass energy equivalent	1.503 277 593 e-10	0.000 000 018 e-10	J
proton mass energy equivalent in MeV	938.272 0813	0.000 0058	MeV
proton mass in u	1.007 276 466 879	0.000 000 000 091	u
proton molar mass	1.007 276 466 879 e-3	0.000 000 000 091 e-3	kg mol^-1
proton-muon mass ratio	8.880 243 38	0.000 000 20	
proton-neutron mag. mom. ratio	-1.459 898 05	0.000 000 34	
proton-neutron mass ratio	0.998 623 478 44	0.000 000 000 51	
proton rms charge radius	0.8751 e-15	0.0061 e-15	m
proton-tau mass ratio	0.528 063	0.000 048	
quantum of circulation	3.636 947 5486 e-4	0.000 000 0017 e-4	m^2 s^-1
quantum of circulation times 2	7.273 895 0972 e-4	0.000 000 0033 e-4	m^2 s^-1
Rydberg constant	10 973 731.568 508	0.000 065	m^-1
Rydberg constant times c in Hz	3.289 841 960 355 e15	0.000 000 000 019 e15	Hz
Rydberg constant times hc in eV	13.605 693 009	0.000 000 084	eV
Rydberg constant times hc in J	2.179 872 325 e-18	0.000 000 027 e-18	J
Sackur-Tetrode constant (1 K, 100 kPa)	-1.151 7084	0.000 0014	
Sackur-Tetrode constant (1 K, 101.325 kPa)	-1.164 8714	0.000 0014	
second radiation constant	1.438 777 36 e-2	0.000 000 83 e-2	m K
shielded helion gyromag. ratio	2.037 894 585 e8	0.000 000 027 e8	s^-1 T^-1
shielded helion gyromag. ratio over 2 pi	32.434 099 66	0.000 000 43	MHz T^-1
shielded helion mag. mom.	-1.074 553 080 e-26	0.000 000 014 e-26	J T^-1
shielded helion mag. mom. to Bohr magneton ratio	-1.158 671 471 e-3	0.000 000 014 e-3	
shielded helion mag. mom. to nuclear magneton ratio	-2.127 497 720	0.000 000 025	
shielded helion to proton mag. mom. ratio	-0.761 766 5603	0.000 000 0092	

shielded helion to shielded proton mag. mom. ratio	-0.761 786 1313	0.000 000 0033	
shielded proton gyromag. ratio	2.675 153 171 e8	0.000 000 033 e8	s^-1 T^-1
shielded proton gyromag. ratio over 2 pi	42.576 385 07	0.000 000 53	MHz T^-1
shielded proton mag. mom.	1.410 570 547 e-26	0.000 000 018 e-26	J T^-1
shielded proton mag. mom. to Bohr magneton ratio	1.520 993 128 e-3	0.000 000 017 e-3	
shielded proton mag. mom. to nuclear magneton ratio	2.792 775 600	0.000 000 030	
speed of light in vacuum	299 792 458	(exact)	m s^-1
standard acceleration of gravity	9.806 65	(exact)	m s^-2
standard atmosphere	101 325	(exact)	Pa
standard-state pressure	100 000	(exact)	Pa
Stefan-Boltzmann constant	5.670 367 e-8	0.000 013 e-8	W m^-2 K^-4
tau Compton wavelength	0.697 787 e-15	0.000 063 e-15	m
tau Compton wavelength over 2 pi	0.111 056 e-15	0.000 010 e-15	m
tau-electron mass ratio	3477.15	0.31	
tau mass	3.167 47 e-27	0.000 29 e-27	kg
tau mass energy equivalent	2.846 78 e-10	0.000 26 e-10	J
tau mass energy equivalent in MeV	1776.82	0.16	MeV
tau mass in u	1.907 49	0.000 17	u
tau molar mass	1.907 49 e-3	0.000 17 e-3	kg mol^-1
tau-muon mass ratio	16.8167	0.0015	
tau-neutron mass ratio	1.891 11	0.000 17	
tau-proton mass ratio	1.893 72	0.000 17	
Thomson cross section	0.665 245 871 58 e-28	0.000 000 000 91 e-28	m^2
triton-electron mass ratio	5496.921 535 88	0.000 000 26	
triton g factor	5.957 924 920	0.000 000 028	
triton mag. mom.	1.504 609 503 e-26	0.000 000 012 e-26	J T^-1
triton mag. mom. to Bohr magneton ratio	1.622 393 6616 e-3	0.000 000 0076 e-3	
triton mag. mom. to nuclear magneton ratio	2.978 962 460	0.000 000 014	
triton mass	5.007 356 665 e-27	0.000 000 062 e-27	kg
triton mass energy equivalent	4.500 387 735 e-10	0.000 000 055 e-10	J
triton mass energy equivalent in MeV	2808.921 112	0.000 017	MeV
triton mass in u	3.015 500 716 32	0.000 000 000 11	u
triton molar mass	3.015 500 716 32 e-3	0.000 000 000 11 e-3	kg mol^-1
triton-proton mass ratio	2.993 717 033 48	0.000 000 000 22	
unified atomic mass unit	1.660 539 040 e-27	0.000 000 020 e-27	kg
von Klitzing constant	25 812.807 4555	0.000 0059	ohm
weak mixing angle	0.2223	0.0021	
Wien frequency displacement law constant	5.878 9238 e10	0.000 0034 e10	Hz K^-1
Wien wavelength displacement law constant	2.897 7729 e-3	0.000 0017 e-3	m K

8.3 2010 Fundamental Physical Constants — Complete Listing

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Fundamental Physical Constants --- Complete Listing
2018 CODATA adjustment
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From:  http://physics.nist.gov/constants
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Quantity	Value	Uncertainty	Unit
alpha particle-electron mass ratio	7294.299 541 42	0.000 000 24	
alpha particle mass	6.644 657 3357 e-27	0.000 000 0020 e-27	kg
alpha particle mass energy equivalent	5.971 920 1914 e-10	0.000 000 0018 e-10	J
alpha particle mass energy equivalent in MeV	3727.379 4066	0.000 0011	MeV
alpha particle mass in u	4.001 506 179 127	0.000 000 000 063	u
alpha particle molar mass	4.001 506 1777 e-3	0.000 000 0012 e-3	kg mol^-1
alpha particle-proton mass ratio	3.972 599 690 09	0.000 000 000 22	
alpha particle relative atomic mass	4.001 506 179 127	0.000 000 000 063	
Angstrom star	1.000 014 95 e-10	0.000 000 90 e-10	m
atomic mass constant	1.660 539 066 60 e-27	0.000 000 000 50 e-27	kg
atomic mass constant energy equivalent	1.492 418 085 60 e-10	0.000 000 000 45 e-10	J
atomic mass constant energy equivalent in MeV	931.494 102 42	0.000 000 28	MeV
atomic mass unit-electron volt relationship	9.314 941 0242 e8	0.000 000 0028 e8	eV
atomic mass unit-hartree relationship	3.423 177 6874 e7	0.000 000 0010 e7	E_h
atomic mass unit-hertz relationship	2.252 342 718 71 e23	0.000 000 000 68 e23	Hz
atomic mass unit-inverse meter relationship	7.513 006 6104 e14	0.000 000 0023 e14	m^-1
atomic mass unit-joule relationship	1.492 418 085 60 e-10	0.000 000 000 45 e-10	J
atomic mass unit-kelvin relationship	1.080 954 019 16 e13	0.000 000 000 33 e13	K
atomic mass unit-kilogram relationship	1.660 539 066 60 e-27	0.000 000 000 50 e-27	kg
atomic unit of 1st hyperpolarizability	3.206 361 3061 e-53	0.000 000 0015 e-53	C^3 m^3 J^-2
atomic unit of 2nd hyperpolarizability	6.235 379 9905 e-65	0.000 000 0038 e-65	C^4 m^4 J^-3
atomic unit of action	1.054 571 817... e-34	(exact)	J s
atomic unit of charge	1.602 176 634 e-19	(exact)	C
atomic unit of charge density	1.081 202 384 57 e12	0.000 000 000 49 e12	C m^-3
atomic unit of current	6.623 618 237 510 e-3	0.000 000 000 013 e-3	A
atomic unit of electric dipole mom.	8.478 353 6255 e-30	0.000 000 0013 e-30	C m
atomic unit of electric field	5.142 206 747 63 e11	0.000 000 000 78 e11	V m^-1
atomic unit of electric field gradient	9.717 362 4292 e21	0.000 000 0029 e21	V m^-2
atomic unit of electric polarizability	1.648 777 274 36 e-41	0.000 000 000 50 e-41	C^2 m^2 J^-1
atomic unit of electric potential	27.211 386 245 988	0.000 000 000 053	V
atomic unit of electric quadrupole mom.	4.486 551 5246 e-40	0.000 000 0014 e-40	C m^2
atomic unit of energy	4.359 744 722 2071 e-18	0.000 000 000 0085 e-18	J
atomic unit of force	8.238 723 4983 e-8	0.000 000 0012 e-8	N
atomic unit of length	5.291 772 109 03 e-11	0.000 000 000 80 e-11	m
atomic unit of mag. dipole mom.	1.854 802 015 66 e-23	0.000 000 000 56 e-23	J T^-1
atomic unit of mag. flux density	2.350 517 567 58 e5	0.000 000 000 71 e5	T
atomic unit of magnetizability	7.891 036 6008 e-29	0.000 000 0048 e-29	J T^-2
atomic unit of mass	9.109 383 7015 e-31	0.000 000 0028 e-31	kg
atomic unit of momentum	1.992 851 914 10 e-24	0.000 000 000 30 e-24	kg m s^-1
atomic unit of permittivity	1.112 650 055 45 e-10	0.000 000 000 17 e-10	F m^-1
atomic unit of time	2.418 884 326 5857 e-17	0.000 000 000 0047 e-17	s
atomic unit of velocity	2.187 691 263 64 e6	0.000 000 000 33 e6	m s^-1
Avogadro constant	6.022 140 76 e23	(exact)	mol^-1
Bohr magneton	9.274 010 0783 e-24	0.000 000 0028 e-24	J T^-1
Bohr magneton in eV/T	5.788 381 8060 e-5	0.000 000 0017 e-5	eV T^-1
Bohr magneton in Hz/T	1.399 624 493 61 e10	0.000 000 000 42 e10	Hz T^-1
Bohr magneton in inverse meter per tesla	46.686 447 783	0.000 000 014	m^-1 T^-1
Bohr magneton in K/T	0.671 713 815 63	0.000 000 000 20	K T^-1
Bohr radius	5.291 772 109 03 e-11	0.000 000 000 80 e-11	m
Boltzmann constant	1.380 649 e-23	(exact)	J K^-1
Boltzmann constant in eV/K	8.617 333 262... e-5	(exact)	eV K^-1
Boltzmann constant in Hz/K	2.083 661 912... e10	(exact)	Hz K^-1
Boltzmann constant in inverse meter per kelvin	69.503 480 04...	(exact)	m^-1 K^-1
characteristic impedance of vacuum	376.730 313 668	0.000 000 057	ohm
classical electron radius	2.817 940 3262 e-15	0.000 000 0013 e-15	m
Compton wavelength	2.426 310 238 67 e-12	0.000 000 000 73 e-12	m
conductance quantum	7.748 091 729... e-5	(exact)	S
conventional value of ampere-90	1.000 000 088 87...	(exact)	A
conventional value of coulomb-90	1.000 000 088 87...	(exact)	C
conventional value of farad-90	0.999 999 982 20...	(exact)	F
conventional value of henry-90	1.000 000 017 79...	(exact)	H
conventional value of Josephson constant	483 597.9 e9	(exact)	Hz V^-1
conventional value of ohm-90	1.000 000 017 79...	(exact)	ohm
conventional value of volt-90	1.000 000 106 66...	(exact)	V
conventional value of von Klitzing constant	25 812.807	(exact)	ohm
conventional value of watt-90	1.000 000 195 53...	(exact)	W
Copper x unit	1.002 076 97 e-13	0.000 000 28 e-13	m
deuteron-electron mag. mom. ratio	-4.664 345 551 e-4	0.000 000 012 e-4	
deuteron-electron mass ratio	3670.482 967 88	0.000 000 13	
deuteron g factor	0.857 438 2338	0.000 000 0022	
deuteron mag. mom.	4.330 735 094 e-27	0.000 000 011 e-27	J T^-1
deuteron mag. mom. to Bohr magneton ratio	4.669 754 570 e-4	0.000 000 012 e-4	
deuteron mag. mom. to nuclear magneton ratio	0.857 438 2338	0.000 000 0022	
deuteron mass	3.343 583 7724 e-27	0.000 000 0010 e-27	kg
deuteron mass energy equivalent	3.005 063 231 02 e-10	0.000 000 000 91 e-10	J
deuteron mass energy equivalent in MeV	1875.612 942 57	0.000 000 57	MeV
deuteron mass in u	2.013 553 212 745	0.000 000 000 040	u
deuteron molar mass	2.013 553 212 05 e-3	0.000 000 000 61 e-3	kg mol^-1
deuteron-neutron mag. mom. ratio	-0.448 206 53	0.000 000 11	
deuteron-proton mag. mom. ratio	0.307 012 209 39	0.000 000 000 79	
deuteron-proton mass ratio	1.999 007 501 39	0.000 000 000 11	
deuteron relative atomic mass	2.013 553 212 745	0.000 000 000 040	
deuteron rms charge radius	2.127 99 e-15	0.000 74 e-15	m
electron charge to mass quotient	-1.758 820 010 76 e11	0.000 000 000 53 e11	C kg^-1
electron-deuteron mag. mom. ratio	-2143.923 4915	0.000 0056	
electron-deuteron mass ratio	2.724 437 107 462 e-4	0.000 000 000 096 e-4	
electron g factor	-2.002 319 304 362 56	0.000 000 000 000 35	
electron gyromag. ratio	1.760 859 630 23 e11	0.000 000 000 53 e11	s^-1 T^-1
electron gyromag. ratio in MHz/T	28 024.951 4242	0.000 0085	MHz T^-1
electron-helion mass ratio	1.819 543 074 573 e-4	0.000 000 000 079 e-4	

electron mag. mom.	-9.284 764 7043 e-24	0.000 000 0028 e-24	J T^-1
electron mag. mom. anomaly	1.159 652 181 28 e-3	0.000 000 000 18 e-3	
electron mag. mom. to Bohr magneton ratio	-1.001 159 652 181 28	0.000 000 000 000 18	
electron mag. mom. to nuclear magneton ratio	-1838.281 971 88	0.000 000 11	
electron mass	9.109 383 7015 e-31	0.000 000 0028 e-31	kg
electron mass energy equivalent	8.187 105 7769 e-14	0.000 000 0025 e-14	J
electron mass energy equivalent in MeV	0.510 998 950 00	0.000 000 000 15	MeV
electron mass in u	5.485 799 090 65 e-4	0.000 000 000 16 e-4	u
electron molar mass	5.485 799 0888 e-7	0.000 000 0017 e-7	kg mol^-1
electron-muon mag. mom. ratio	206.766 9883	0.000 0046	
electron-muon mass ratio	4.836 331 69 e-3	0.000 000 11 e-3	
electron-neutron mag. mom. ratio	960.920 50	0.000 23	
electron-neutron mass ratio	5.438 673 4424 e-4	0.000 000 0026 e-4	
electron-proton mag. mom. ratio	-658.210 687 89	0.000 000 20	
electron-proton mass ratio	5.446 170 214 87 e-4	0.000 000 000 33 e-4	
electron relative atomic mass	5.485 799 090 65 e-4	0.000 000 000 16 e-4	
electron-tau mass ratio	2.875 85 e-4	0.000 19 e-4	
electron to alpha particle mass ratio	1.370 933 554 787 e-4	0.000 000 000 045 e-4	
electron to shielded helion mag. mom. ratio	864.058 257	0.000 010	
electron to shielded proton mag. mom. ratio	-658.227 5971	0.000 0072	
electron-triton mass ratio	1.819 200 062 251 e-4	0.000 000 000 090 e-4	
electron volt	1.602 176 634 e-19	(exact)	J
electron volt-atomic mass unit relationship	1.073 544 102 33 e-9	0.000 000 000 32 e-9	u
electron volt-hartree relationship	3.674 932 217 5655 e-2	0.000 000 000 0071 e-2	E_h
electron volt-hertz relationship	2.417 989 242... e14	(exact)	Hz
electron volt-inverse meter relationship	8.065 543 937... e5	(exact)	m^-1
electron volt-joule relationship	1.602 176 634 e-19	(exact)	J
electron volt-kelvin relationship	1.160 451 812... e4	(exact)	K
electron volt-kilogram relationship	1.782 661 921... e-36	(exact)	kg
elementary charge	1.602 176 634 e-19	(exact)	C
elementary charge over h-bar	1.519 267 447... e15	(exact)	A J^-1
Faraday constant	96 485.332 12...	(exact)	C mol^-1
Fermi coupling constant	1.166 3787 e-5	0.000 0006 e-5	GeV^-2
fine-structure constant	7.297 352 5693 e-3	0.000 000 0011 e-3	
first radiation constant	3.741 771 852... e-16	(exact)	W m^2
first radiation constant for spectral radiance	1.191 042 972... e-16	(exact)	W m^2 sr^-1
hartree-atomic mass unit relationship	2.921 262 322 05 e-8	0.000 000 000 88 e-8	u
hartree-electron volt relationship	27.211 386 245 988	0.000 000 000 053	eV
Hartree energy	4.359 744 722 2071 e-18	0.000 000 000 0085 e-18	J
Hartree energy in eV	27.211 386 245 988	0.000 000 000 053	eV
hartree-hertz relationship	6.579 683 920 502 e15	0.000 000 000 013 e15	Hz
hartree-inverse meter relationship	2.194 746 313 6320 e7	0.000 000 000 0043 e7	m^-1
hartree-joule relationship	4.359 744 722 2071 e-18	0.000 000 000 0085 e-18	J
hartree-kelvin relationship	3.157 750 248 0407 e5	0.000 000 000 0061 e5	K
hartree-kilogram relationship	4.850 870 209 5432 e-35	0.000 000 000 0094 e-35	kg
helion-electron mass ratio	5495.885 280 07	0.000 000 24	
helion g factor	-4.255 250 615	0.000 000 050	
helion mag. mom.	-1.074 617 532 e-26	0.000 000 013 e-26	J T^-1
helion mag. mom. to Bohr magneton ratio	-1.158 740 958 e-3	0.000 000 014 e-3	
helion mag. mom. to nuclear magneton ratio	-2.127 625 307	0.000 000 025	
helion mass	5.006 412 7796 e-27	0.000 000 0015 e-27	kg
helion mass energy equivalent	4.499 539 4125 e-10	0.000 000 0014 e-10	J
helion mass energy equivalent in MeV	2808.391 607 43	0.000 000 85	MeV
helion mass in u	3.014 932 247 175	0.000 000 000 097	u
helion molar mass	3.014 932 246 13 e-3	0.000 000 000 91 e-3	kg mol^-1
helion-proton mass ratio	2.993 152 671 67	0.000 000 000 13	
helion relative atomic mass	3.014 932 247 175	0.000 000 000 097	
helion shielding shift	5.996 743 e-5	0.000 010 e-5	
hertz-atomic mass unit relationship	4.439 821 6652 e-24	0.000 000 0013 e-24	u
hertz-electron volt relationship	4.135 667 696... e-15	(exact)	eV
hertz-hartree relationship	1.519 829 846 0570 e-16	0.000 000 000 0029 e-16	E_h
hertz-inverse meter relationship	3.335 640 951... e-9	(exact)	m^-1
hertz-joule relationship	6.626 070 15 e-34	(exact)	J
hertz-kelvin relationship	4.799 243 073... e-11	(exact)	K
hertz-kilogram relationship	7.372 497 323... e-51	(exact)	kg
hyperfine transition frequency of Cs-133	9 192 631 770	(exact)	Hz
inverse fine-structure constant	137.035 999 084	0.000 000 021	
inverse meter-atomic mass unit relationship	1.331 025 050 10 e-15	0.000 000 000 40 e-15	u
inverse meter-electron volt relationship	1.239 841 984... e-6	(exact)	eV
inverse meter-hartree relationship	4.556 335 252 9120 e-8	0.000 000 000 0088 e-8	E_h
inverse meter-hertz relationship	299 792 458	(exact)	Hz
inverse meter-joule relationship	1.986 445 857... e-25	(exact)	J
inverse meter-kelvin relationship	1.438 776 877... e-2	(exact)	K
inverse meter-kilogram relationship	2.210 219 094... e-42	(exact)	kg
inverse of conductance quantum	12 906.403 72...	(exact)	ohm
Josephson constant	483 597.848 4... e9	(exact)	Hz V^-1
joule-atomic mass unit relationship	6.700 535 2565 e9	0.000 000 0020 e9	u
joule-electron volt relationship	6.241 509 074... e18	(exact)	eV
joule-hartree relationship	2.293 712 278 3963 e17	0.000 000 000 0045 e17	E_h
joule-hertz relationship	1.509 190 179... e33	(exact)	Hz
joule-inverse meter relationship	5.034 116 567... e24	(exact)	m^-1
joule-kelvin relationship	7.242 970 516... e22	(exact)	K
joule-kilogram relationship	1.112 650 056... e-17	(exact)	kg
kelvin-atomic mass unit relationship	9.251 087 3014 e-14	0.000 000 0028 e-14	u
kelvin-electron volt relationship	8.617 333 262... e-5	(exact)	eV
kelvin-hartree relationship	3.166 811 563 4556 e-6	0.000 000 000 0061 e-6	E_h
kelvin-hertz relationship	2.083 661 912... e10	(exact)	Hz
kelvin-inverse meter relationship	69.503 480 04...	(exact)	m^-1
kelvin-joule relationship	1.380 649 e-23	(exact)	J
kelvin-kilogram relationship	1.536 179 187... e-40	(exact)	kg
kilogram-atomic mass unit relationship	6.022 140 7621 e26	0.000 000 0018 e26	u
kilogram-electron volt relationship	5.609 588 603... e35	(exact)	eV
kilogram-hartree relationship	2.061 485 788 7409 e34	0.000 000 000 0040 e34	E_h
kilogram-hertz relationship	1.356 392 489... e50	(exact)	Hz
kilogram-inverse meter relationship	4.524 438 335... e41	(exact)	m^-1
kilogram-joule relationship	8.987 551 787... e16	(exact)	J
kilogram-kelvin relationship	6.509 657 260... e39	(exact)	K
lattice parameter of silicon	5.431 020 511 e-10	0.000 000 089 e-10	m
lattice spacing of ideal Si (220)	1.920 155 716 e-10	0.000 000 032 e-10	m
Loschmidt constant (273.15 K, 100 kPa)	2.651 645 804... e25	(exact)	m^-3
Loschmidt constant (273.15 K, 101.325 kPa)	2.686 780 111... e25	(exact)	m^-3
luminous efficacy	683	(exact)	lm W^-1
mag. flux quantum	2.067 833 848... e-15	(exact)	Wb

molar gas constant	8.314 462 618...	(exact)	J mol^-1 K^-1
molar mass constant	0.999 999 999 65 e-3	0.000 000 000 30 e-3	kg mol^-1
molar mass of carbon-12	11.999 999 9958 e-3	0.000 000 0036 e-3	kg mol^-1
molar Planck constant	3.990 312 712... e-10	(exact)	J Hz^-1 mol^-1
molar volume of ideal gas (273.15 K, 100 kPa)	22.710 954 64... e-3	(exact)	m^3 mol^-1
molar volume of ideal gas (273.15 K, 101.325 kPa)	22.413 969 54... e-3	(exact)	m^3 mol^-1
molar volume of silicon	1.205 883 199 e-5	0.000 000 060 e-5	m^3 mol^-1
Molybdenum x unit	1.002 099 52 e-13	0.000 000 53 e-13	m
muon Compton wavelength	1.173 444 110 e-14	0.000 000 026 e-14	m
muon-electron mass ratio	206.768 2830	0.000 0046	
muon g factor	-2.002 331 8418	0.000 000 0013	
muon mag. mom.	-4.490 448 30 e-26	0.000 000 10 e-26	J T^-1
muon mag. mom. anomaly	1.165 920 89 e-3	0.000 000 63 e-3	
muon mag. mom. to Bohr magneton ratio	-4.841 970 47 e-3	0.000 000 11 e-3	
muon mag. mom. to nuclear magneton ratio	-8.890 597 03	0.000 000 20	
muon mass	1.883 531 627 e-28	0.000 000 042 e-28	kg
muon mass energy equivalent	1.692 833 804 e-11	0.000 000 038 e-11	J
muon mass energy equivalent in MeV	105.658 3755	0.000 0023	MeV
muon mass in u	0.113 428 9259	0.000 000 0025	u
muon molar mass	1.134 289 259 e-4	0.000 000 025 e-4	kg mol^-1
muon-neutron mass ratio	0.112 454 5170	0.000 000 0025	
muon-proton mag. mom. ratio	-3.183 345 142	0.000 000 071	
muon-proton mass ratio	0.112 609 5264	0.000 000 0025	
muon-tau mass ratio	5.946 35 e-2	0.000 40 e-2	
natural unit of action	1.054 571 817... e-34	(exact)	J s
natural unit of action in eV s	6.582 119 569... e-16	(exact)	eV s
natural unit of energy	8.187 105 7769 e-14	0.000 000 0025 e-14	J
natural unit of energy in MeV	0.510 998 950 00	0.000 000 000 15	MeV
natural unit of length	3.861 592 6796 e-13	0.000 000 0012 e-13	m
natural unit of mass	9.109 383 7015 e-31	0.000 000 0028 e-31	kg
natural unit of momentum	2.730 924 530 75 e-22	0.000 000 000 82 e-22	kg m s^-1
natural unit of momentum in MeV/c	0.510 998 950 00	0.000 000 000 15	MeV/c
natural unit of time	1.288 088 668 19 e-21	0.000 000 000 39 e-21	s
natural unit of velocity	299 792 458	(exact)	m s^-1
neutron Compton wavelength	1.319 590 905 81 e-15	0.000 000 000 75 e-15	m
neutron-electron mag. mom. ratio	1.040 668 82 e-3	0.000 000 25 e-3	
neutron-electron mass ratio	1838.683 661 73	0.000 000 89	
neutron g factor	-3.826 085 45	0.000 000 90	
neutron gyromag. ratio	1.832 471 71 e8	0.000 000 43 e8	s^-1 T^-1
neutron gyromag. ratio in MHz/T	29.164 6931	0.000 0069	MHz T^-1
neutron mag. mom.	-9.662 3651 e-27	0.000 0023 e-27	J T^-1
neutron mag. mom. to Bohr magneton ratio	-1.041 875 63 e-3	0.000 000 25 e-3	
neutron mag. mom. to nuclear magneton ratio	-1.913 042 73	0.000 000 45	
neutron mass	1.674 927 498 04 e-27	0.000 000 000 95 e-27	kg
neutron mass energy equivalent	1.505 349 762 87 e-10	0.000 000 000 86 e-10	J
neutron mass energy equivalent in MeV	939.565 420 52	0.000 000 54	MeV
neutron mass in u	1.008 664 915 95	0.000 000 000 49	u
neutron molar mass	1.008 664 915 60 e-3	0.000 000 000 57 e-3	kg mol^-1
neutron-muon mass ratio	8.892 484 06	0.000 000 20	
neutron-proton mag. mom. ratio	-0.684 979 34	0.000 000 16	
neutron-proton mass difference	2.305 574 35 e-30	0.000 000 82 e-30	kg
neutron-proton mass difference energy equivalent	2.072 146 89 e-13	0.000 000 74 e-13	J
neutron-proton mass difference energy equivalent in MeV	1.293 332 36	0.000 000 46	MeV
neutron-proton mass difference in u	1.388 449 33 e-3	0.000 000 49 e-3	u
neutron-proton mass ratio	1.001 378 419 31	0.000 000 000 49	
neutron relative atomic mass	1.008 664 915 95	0.000 000 000 49	
neutron-tau mass ratio	0.528 779	0.000 036	
neutron to shielded proton mag. mom. ratio	-0.684 996 94	0.000 000 16	
Newtonian constant of gravitation	6.674 30 e-11	0.000 15 e-11	m^3 kg^-1 s^-2
Newtonian constant of gravitation over h-bar c	6.708 83 e-39	0.000 15 e-39	(GeV/c^2)^-2
nuclear magneton	5.050 783 7461 e-27	0.000 000 0015 e-27	J T^-1
nuclear magneton in eV/T	3.152 451 258 44 e-8	0.000 000 000 96 e-8	eV T^-1
nuclear magneton in inverse meter per tesla	2.542 623 413 53 e-2	0.000 000 000 78 e-2	m^-1 T^-1
nuclear magneton in K/T	3.658 267 7756 e-4	0.000 000 0011 e-4	K T^-1
nuclear magneton in MHz/T	7.622 593 2291	0.000 000 0023	MHz T^-1
Planck constant	6.626 070 15 e-34	(exact)	J Hz^-1
Planck constant in eV/Hz	4.135 667 696... e-15	(exact)	eV Hz^-1
Planck length	1.616 255 e-35	0.000 018 e-35	m
Planck mass	2.176 434 e-8	0.000 024 e-8	kg
Planck mass energy equivalent in GeV	1.220 890 e19	0.000 014 e19	GeV
Planck temperature	1.416 784 e32	0.000 016 e32	K
Planck time	5.391 247 e-44	0.000 060 e-44	s
proton charge to mass quotient	9.578 833 1560 e7	0.000 000 0029 e7	C kg^-1
proton Compton wavelength	1.321 409 855 39 e-15	0.000 000 000 40 e-15	m
proton-electron mass ratio	1836.152 673 43	0.000 000 11	
proton g factor	5.585 694 6893	0.000 000 0016	
proton gyromag. ratio	2.675 221 8744 e8	0.000 000 0011 e8	s^-1 T^-1
proton gyromag. ratio in MHz/T	42.577 478 518	0.000 000 018	MHz T^-1
proton mag. mom.	1.410 606 797 36 e-26	0.000 000 000 60 e-26	J T^-1
proton mag. mom. to Bohr magneton ratio	1.521 032 202 30 e-3	0.000 000 000 46 e-3	
proton mag. mom. to nuclear magneton ratio	2.792 847 344 63	0.000 000 000 82	
proton mag. shielding correction	2.5689 e-5	0.0011 e-5	
proton mass	1.672 621 923 69 e-27	0.000 000 000 51 e-27	kg
proton mass energy equivalent	1.503 277 615 98 e-10	0.000 000 000 46 e-10	J
proton mass energy equivalent in MeV	938.272 088 16	0.000 000 29	MeV
proton mass in u	1.007 276 466 621	0.000 000 000 053	u
proton molar mass	1.007 276 466 27 e-3	0.000 000 000 31 e-3	kg mol^-1
proton-muon mass ratio	8.880 243 37	0.000 000 20	
proton-neutron mag. mom. ratio	-1.459 898 05	0.000 000 34	
proton-neutron mass ratio	0.998 623 478 12	0.000 000 000 49	
proton relative atomic mass	1.007 276 466 621	0.000 000 000 053	
proton rms charge radius	8.414 e-16	0.019 e-16	m
proton-tau mass ratio	0.528 051	0.000 036	
quantum of circulation	3.636 947 5516 e-4	0.000 000 0011 e-4	m^2 s^-1
quantum of circulation times 2	7.273 895 1032 e-4	0.000 000 0022 e-4	m^2 s^-1
reduced Compton wavelength	3.861 592 6796 e-13	0.000 000 0012 e-13	m
reduced muon Compton wavelength	1.867 594 306 e-15	0.000 000 042 e-15	m
reduced neutron Compton wavelength	2.100 194 1552 e-16	0.000 000 0012 e-16	m
reduced Planck constant	1.054 571 817... e-34	(exact)	J s
reduced Planck constant in eV s	6.582 119 569... e-16	(exact)	eV s
reduced Planck constant times c in MeV fm	197.326 980 4...	(exact)	MeV fm
reduced proton Compton wavelength	2.103 089 103 36 e-16	0.000 000 000 64 e-16	m
reduced tau Compton wavelength	1.110 538 e-16	0.000 075 e-16	m

Rydberg constant	10 973 731.568 160	0.000 021	m^-1
Rydberg constant times c in Hz	3.289 841 960 2508 e15	0.000 000 000 0064 e15	Hz
Rydberg constant times hc in eV	13.605 693 122 994	0.000 000 000 026	eV
Rydberg constant times hc in J	2.179 872 361 1035 e-18	0.000 000 000 0042 e-18	J
Sackur-Tetrode constant (1 K, 100 kPa)	-1.151 707 537 06	0.000 000 000 45	
Sackur-Tetrode constant (1 K, 101.325 kPa)	-1.164 870 523 58	0.000 000 000 45	
second radiation constant	1.438 776 877... e-2	(exact)	m K
shielded helion gyromag. ratio	2.037 894 569 e8	0.000 000 024 e8	s^-1 T^-1
shielded helion gyromag. ratio in MHz/T	32.434 099 42	0.000 000 38	MHz T^-1
shielded helion mag. mom.	-1.074 553 090 e-26	0.000 000 013 e-26	J T^-1
shielded helion mag. mom. to Bohr magneton ratio	-1.158 671 471 e-3	0.000 000 014 e-3	
shielded helion mag. mom. to nuclear magneton ratio	-2.127 497 719	0.000 000 025	
shielded helion to proton mag. mom. ratio	-0.761 766 5618	0.000 000 0089	
shielded helion to shielded proton mag. mom. ratio	-0.761 786 1313	0.000 000 0033	
shielded proton gyromag. ratio	2.675 153 151 e8	0.000 000 029 e8	s^-1 T^-1
shielded proton gyromag. ratio in MHz/T	42.576 384 74	0.000 000 46	MHz T^-1
shielded proton mag. mom.	1.410 570 560 e-26	0.000 000 015 e-26	J T^-1
shielded proton mag. mom. to Bohr magneton ratio	1.520 993 128 e-3	0.000 000 017 e-3	
shielded proton mag. mom. to nuclear magneton ratio	2.792 775 599	0.000 000 030	
shielding difference of d and p in HD	2.0200 e-8	0.0020 e-8	
shielding difference of t and p in HT	2.4140 e-8	0.0020 e-8	
speed of light in vacuum	299 792 458	(exact)	m s^-1
standard acceleration of gravity	9.806 65	(exact)	m s^-2
standard atmosphere	101 325	(exact)	Pa
standard-state pressure	100 000	(exact)	Pa
Stefan-Boltzmann constant	5.670 374 419... e-8	(exact)	W m^-2 K^-4
tau Compton wavelength	6.977 71 e-16	0.000 47 e-16	m
tau-electron mass ratio	3477.23	0.23	
tau energy equivalent	1776.86	0.12	MeV
tau mass	3.167 54 e-27	0.000 21 e-27	kg
tau mass energy equivalent	2.846 84 e-10	0.000 19 e-10	J
tau mass in u	1.907 54	0.000 13	u
tau molar mass	1.907 54 e-3	0.000 13 e-3	kg mol^-1
tau-muon mass ratio	16.8170	0.0011	
tau-neutron mass ratio	1.891 15	0.000 13	
tau-proton mass ratio	1.893 76	0.000 13	
Thomson cross section	6.652 458 7321 e-29	0.000 000 0060 e-29	m^2
triton-electron mass ratio	5496.921 535 73	0.000 000 27	
triton g factor	5.957 924 931	0.000 000 012	
triton mag. mom.	1.504 609 5202 e-26	0.000 000 0030 e-26	J T^-1
triton mag. mom. to Bohr magneton ratio	1.622 393 6651 e-3	0.000 000 0032 e-3	
triton mag. mom. to nuclear magneton ratio	2.978 962 4656	0.000 000 0059	
triton mass	5.007 356 7446 e-27	0.000 000 0015 e-27	kg
triton mass energy equivalent	4.500 387 8060 e-10	0.000 000 0014 e-10	J
triton mass energy equivalent in MeV	2808.921 132 98	0.000 000 85	MeV
triton mass in u	3.015 500 716 21	0.000 000 000 12	u
triton molar mass	3.015 500 715 17 e-3	0.000 000 000 92 e-3	kg mol^-1
triton-proton mass ratio	2.993 717 034 14	0.000 000 000 15	
triton relative atomic mass	3.015 500 716 21	0.000 000 000 12	
triton to proton mag. mom. ratio	1.066 639 9191	0.000 000 0021	
unified atomic mass unit	1.660 539 066 60 e-27	0.000 000 000 50 e-27	kg
vacuum electric permittivity	8.854 187 8128 e-12	0.000 000 0013 e-12	F m^-1
vacuum mag. permeability	1.256 637 062 12 e-6	0.000 000 000 19 e-6	N A^-2
von Klitzing constant	25 812.807 45...	(exact)	ohm
weak mixing angle	0.222 90	0.000 30	
Wien frequency displacement law constant	5.878 925 757... e10	(exact)	Hz K^-1
Wien wavelength displacement law constant	2.897 771 955... e-3	(exact)	m K
W to Z mass ratio	0.881 53	0.000 17	

8.4 2010 Fundamental Physical Constants — Complete Listing

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             Fundamental Physical Constants --- Complete Listing
             2022 CODATA adjustment


  From:  http://physics.nist.gov/constants



  Quantity                                                       Value                 Uncertainty           Unit
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alpha particle-electron mass ratio                          7294.299 541 71          0.000 000 17
alpha particle mass                                         6.644 657 3450 e-27      0.000 000 0021 e-27      kg
alpha particle mass energy equivalent                       5.971 920 1997 e-10      0.000 000 0019 e-10      J
alpha particle mass energy equivalent in MeV                3727.379 4118            0.000 0012               MeV
alpha particle mass in u                                    4.001 506 179 129        0.000 000 000 062        u
alpha particle molar mass                                   4.001 506 1833 e-3       0.000 000 0012 e-3       kg mol^-1
alpha particle-proton mass ratio                            3.972 599 690 252        0.000 000 000 070
alpha particle relative atomic mass                         4.001 506 179 129        0.000 000 000 062
alpha particle rms charge radius                            1.6785 e-15              0.0021 e-15              m
Angstrom star                                               1.000 014 95 e-10        0.000 000 90 e-10        m
atomic mass constant                                        1.660 539 068 92 e-27    0.000 000 000 52 e-27    kg
atomic mass constant energy equivalent                      1.492 418 087 68 e-10    0.000 000 000 46 e-10    J
atomic mass constant energy equivalent in MeV               931.494 103 72           0.000 000 29             MeV
atomic mass unit-electron volt relationship                 9.314 941 0372 e8        0.000 000 0029 e8        eV
atomic mass unit-hartree relationship                       3.423 177 6922 e7        0.000 000 0011 e7        E_h
atomic mass unit-hertz relationship                         2.252 342 721 85 e23     0.000 000 000 70 e23     Hz
atomic mass unit-inverse meter relationship                 7.513 006 6209 e14       0.000 000 0023 e14       m^-1
atomic mass unit-joule relationship                         1.492 418 087 68 e-10    0.000 000 000 46 e-10    J
atomic mass unit-kelvin relationship                        1.080 954 020 67 e13     0.000 000 000 34 e13     K
atomic mass unit-kilogram relationship                      1.660 539 068 92 e-27    0.000 000 000 52 e-27    kg
atomic unit of 1st hyperpolarizability                      3.206 361 2996 e-53      0.000 000 0015 e-53      C^3 m^3 J^-2
atomic unit of 2nd hyperpolarizability                      6.235 379 9735 e-65      0.000 000 0039 e-65      C^4 m^4 J^-3
atomic unit of action                                       1.054 571 817... e-34    (exact)                  J s
atomic unit of charge                                       1.602 176 634 e-19       (exact)                  C
atomic unit of charge density                               1.081 202 386 77 e12     0.000 000 000 51 e12     C m^-3
atomic unit of current                                      6.623 618 237 5082 e-3   0.000 000 000 0072 e-3   A
atomic unit of electric dipole mom.                         8.478 353 6198 e-30      0.000 000 0013 e-30      C m
atomic unit of electric field                               5.142 206 751 12 e11     0.000 000 000 80 e11     V m^-1
atomic unit of electric field gradient                      9.717 362 4424 e21       0.000 000 0030 e21       V m^-2
atomic unit of electric polarizability                      1.648 777 272 12 e-41    0.000 000 000 51 e-41    C^2 m^2 J^-1
atomic unit of electric potential                           27.211 386 245 981       0.000 000 000 030        V
atomic unit of electric quadrupole mom.                     4.486 551 5185 e-40      0.000 000 0014 e-40      C m^2
atomic unit of energy                                       4.359 744 722 2060 e-18  0.000 000 000 0048 e-18  J
atomic unit of force                                        8.238 723 5038 e-8       0.000 000 0013 e-8       N
atomic unit of length                                       5.291 772 105 44 e-11    0.000 000 000 82 e-11    m
atomic unit of mag. dipole mom.                             1.854 802 013 15 e-23    0.000 000 000 58 e-23    J T^-1
atomic unit of mag. flux density                            2.350 517 570 77 e5      0.000 000 000 73 e5      T
atomic unit of magnetizability                              7.891 036 5794 e-29      0.000 000 0049 e-29      J T^-2
atomic unit of mass                                         9.109 383 7139 e-31      0.000 000 0028 e-31      kg
atomic unit of momentum                                     1.992 851 915 45 e-24    0.000 000 000 31 e-24    kg m s^-1
atomic unit of permittivity                                 1.112 650 056 20 e-10    0.000 000 000 17 e-10    F m^-1
atomic unit of time                                         2.418 884 326 5864 e-17  0.000 000 000 0026 e-17  s
atomic unit of velocity                                     2.187 691 262 16 e6      0.000 000 000 34 e6      m s^-1
Avogadro constant                                           6.022 140 76 e23         (exact)                  mol^-1
Bohr magneton                                               9.274 010 0657 e-24      0.000 000 0029 e-24      J T^-1
Bohr magneton in eV/T                                       5.788 381 7982 e-5       0.000 000 0018 e-5       eV T^-1
Bohr magneton in Hz/T                                       1.399 624 491 71 e10     0.000 000 000 44 e10     Hz T^-1
Bohr magneton in inverse meter per tesla                    46.686 447 719           0.000 000 015            m^-1 T^-1
Bohr magneton in K/T                                        0.671 713 814 72         0.000 000 000 21         K T^-1
Bohr radius                                                 5.291 772 105 44 e-11    0.000 000 000 82 e-11    m
Boltzmann constant                                          1.380 649 e-23           (exact)                  J K^-1
Boltzmann constant in eV/K                                  8.617 333 262... e-5     (exact)                  eV K^-1
Boltzmann constant in Hz/K                                  2.083 661 912... e10     (exact)                  Hz K^-1
Boltzmann constant in inverse meter per kelvin              69.503 480 04...         (exact)                  m^-1 K^-1
characteristic impedance of vacuum                          376.730 313 412          0.000 000 059            ohm
classical electron radius                                   2.817 940 3205 e-15      0.000 000 0013 e-15      m
Compton wavelength                                          2.426 310 235 38 e-12    0.000 000 000 76 e-12    m
conductance quantum                                         7.748 091 729... e-5     (exact)                  S
conventional value of ampere-90                             1.000 000 088 87...      (exact)                  A
conventional value of coulomb-90                            1.000 000 088 87...      (exact)                  C
conventional value of farad-90                              0.999 999 982 20...      (exact)                  F
conventional value of henry-90                              1.000 000 017 79...      (exact)                  H
conventional value of Josephson constant                    483 597.9 e9             (exact)                  Hz V^-1
conventional value of ohm-90                                1.000 000 017 79...      (exact)                  ohm
conventional value of volt-90                               1.000 000 106 66...      (exact)                  V
conventional value of von Klitzing constant                 25 812.807               (exact)                  ohm
conventional value of watt-90                               1.000 000 195 53...      (exact)                  W
Copper x unit                                               1.002 076 97 e-13        0.000 000 28 e-13        m
deuteron-electron mag. mom. ratio                           -4.664 345 550 e-4       0.000 000 012 e-4
deuteron-electron mass ratio                                3670.482 967 655         0.000 000 063
deuteron g factor                                           0.857 438 2335           0.000 000 0022
deuteron mag. mom.                                          4.330 735 087 e-27       0.000 000 011 e-27       J T^-1
deuteron mag. mom. to Bohr magneton ratio                   4.669 754 568 e-4        0.000 000 012 e-4
deuteron mag. mom. to nuclear magneton ratio                0.857 438 2335           0.000 000 0022
deuteron mass                                               3.343 583 7768 e-27      0.000 000 0010 e-27      kg
deuteron mass energy equivalent                             3.005 063 234 91 e-10    0.000 000 000 94 e-10    J
deuteron mass energy equivalent in MeV                      1875.612 945 00          0.000 000 58             MeV
deuteron mass in u                                          2.013 553 212 544        0.000 000 000 015        u
deuteron molar mass                                         2.013 553 214 66 e-3     0.000 000 000 63 e-3     kg mol^-1
deuteron-neutron mag. mom. ratio                            -0.448 206 52            0.000 000 11
deuteron-proton mag. mom. ratio                             0.307 012 209 30         0.000 000 000 79
deuteron-proton mass ratio                                  1.999 007 501 2699       0.000 000 000 0084
deuteron relative atomic mass                               2.013 553 212 544        0.000 000 000 015
deuteron rms charge radius                                  2.127 78 e-15            0.000 27 e-15            m
electron charge to mass quotient                            -1.758 820 008 38 e11    0.000 000 000 55 e11     C kg^-1
electron-deuteron mag. mom. ratio                           -2143.923 4921           0.000 0056
electron-deuteron mass ratio                                2.724 437 107 629 e-4    0.000 000 000 047 e-4
electron g factor                                           -2.002 319 304 360 92    0.000 000 000 000 36
electron gyromag. ratio                                     1.760 859 627 84 e11     0.000 000 000 55 e11     s^-1 T^-1
electron gyromag. ratio in MHz/T                            28 024.951 3861          0.000 0087               MHz T^-1
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electron-helion mass ratio	1.819 543 074 649 e-4	0.000 000 000 053 e-4	
electron mag. mom.	-9.284 764 6917 e-24	0.000 000 0029 e-24	J T^-1
electron mag. mom. anomaly	1.159 652 180 46 e-3	0.000 000 000 18 e-3	
electron mag. mom. to Bohr magneton ratio	-1.001 159 652 180 46	0.000 000 000 000 18	
electron mag. mom. to nuclear magneton ratio	-1838.281 971 877	0.000 000 032	
electron mass	9.109 383 7139 e-31	0.000 000 0028 e-31	kg
electron mass energy equivalent	8.187 105 7880 e-14	0.000 000 0026 e-14	J
electron mass energy equivalent in MeV	0.510 998 950 69	0.000 000 000 16	MeV
electron mass in u	5.485 799 090 441 e-4	0.000 000 000 097 e-4	u
electron molar mass	5.485 799 0962 e-7	0.000 000 0017 e-7	kg mol^-1
electron-muon mag. mom. ratio	206.766 9881	0.000 0046	
electron-muon mass ratio	4.836 331 70 e-3	0.000 000 11 e-3	
electron-neutron mag. mom. ratio	960.920 48	0.000 23	
electron-neutron mass ratio	5.438 673 4416 e-4	0.000 000 0022 e-4	
electron-proton mag. mom. ratio	-658.210 687 89	0.000 000 19	
electron-proton mass ratio	5.446 170 214 889 e-4	0.000 000 000 094 e-4	
electron relative atomic mass	5.485 799 090 441 e-4	0.000 000 000 097 e-4	
electron-tau mass ratio	2.875 85 e-4	0.000 19 e-4	
electron to alpha particle mass ratio	1.370 933 554 733 e-4	0.000 000 000 032 e-4	
electron to shielded helion mag. mom. ratio	864.058 239 86	0.000 000 70	
electron to shielded proton mag. mom. ratio	-658.227 5856	0.000 0027	
electron-triton mass ratio	1.819 200 062 327 e-4	0.000 000 000 068 e-4	
electron volt	1.602 176 634 e-19	(exact)	J
electron volt-atomic mass unit relationship	1.073 544 100 83 e-9	0.000 000 000 33 e-9	u
electron volt-hartree relationship	3.674 932 217 5665 e-2	0.000 000 000 0040 e-2	E_h
electron volt-hertz relationship	2.417 989 242... e14	(exact)	Hz
electron volt-inverse meter relationship	8.065 543 937... e5	(exact)	m^-1
electron volt-joule relationship	1.602 176 634 e-19	(exact)	J
electron volt-kelvin relationship	1.160 451 812... e4	(exact)	K
electron volt-kilogram relationship	1.782 661 921... e-36	(exact)	kg
elementary charge	1.602 176 634 e-19	(exact)	C
elementary charge over h-bar	1.519 267 447... e15	(exact)	A J^-1
Faraday constant	96 485.332 12...	(exact)	C mol^-1
Fermi coupling constant	1.166 3787 e-5	0.000 0006 e-5	GeV^-2
fine-structure constant	7.297 352 5643 e-3	0.000 000 0011 e-3	
first radiation constant	3.741 771 852... e-16	(exact)	W m^2
first radiation constant for spectral radiance	1.191 042 972... e-16	(exact)	W m^2 sr^-1
hartree-atomic mass unit relationship	2.921 262 317 97 e-8	0.000 000 000 91 e-8	u
hartree-electron volt relationship	27.211 386 245 981	0.000 000 000 030	eV
Hartree energy	4.359 744 722 2060 e-18	0.000 000 000 0048 e-18	J
Hartree energy in eV	27.211 386 245 981	0.000 000 000 030	eV
hartree-hertz relationship	6.579 683 920 4999 e15	0.000 000 000 0072 e15	Hz
hartree-inverse meter relationship	2.194 746 313 6314 e7	0.000 000 000 0024 e7	m^-1
hartree-joule relationship	4.359 744 722 2060 e-18	0.000 000 000 0048 e-18	J
hartree-kelvin relationship	3.157 750 248 0398 e5	0.000 000 000 0034 e5	K
hartree-kilogram relationship	4.850 870 209 5419 e-35	0.000 000 000 0053 e-35	kg
helion-electron mass ratio	5495.885 279 84	0.000 000 16	
helion g factor	-4.255 250 6995	0.000 000 0034	
helion mag. mom.	-1.074 617 551 98 e-26	0.000 000 000 93 e-26	J T^-1
helion mag. mom. to Bohr magneton ratio	-1.158 740 980 83 e-3	0.000 000 000 94 e-3	
helion mag. mom. to nuclear magneton ratio	-2.127 625 3498	0.000 000 0017	
helion mass	5.006 412 7862 e-27	0.000 000 0016 e-27	kg
helion mass energy equivalent	4.499 539 4185 e-10	0.000 000 0014 e-10	J
helion mass energy equivalent in MeV	2808.391 611 12	0.000 000 88	MeV
helion mass in u	3.014 932 246 932	0.000 000 000 074	u
helion molar mass	3.014 932 250 10 e-3	0.000 000 000 94 e-3	kg mol^-1
helion-proton mass ratio	2.993 152 671 552	0.000 000 000 070	
helion relative atomic mass	3.014 932 246 932	0.000 000 000 074	
helion shielding shift	5.996 7029 e-5	0.000 0023 e-5	
hertz-atomic mass unit relationship	4.439 821 6590 e-24	0.000 000 0014 e-24	u
hertz-electron volt relationship	4.135 667 696... e-15	(exact)	eV
hertz-hartree relationship	1.519 829 846 0574 e-16	0.000 000 000 0017 e-16	E_h
hertz-inverse meter relationship	3.335 640 951... e-9	(exact)	m^-1
hertz-joule relationship	6.626 070 15 e-34	(exact)	J
hertz-kelvin relationship	4.799 243 073... e-11	(exact)	K
hertz-kilogram relationship	7.372 497 323... e-51	(exact)	kg
hyperfine transition frequency of Cs-133	9 192 631 770	(exact)	Hz
inverse fine-structure constant	137.035 999 177	0.000 000 021	
inverse meter-atomic mass unit relationship	1.331 025 048 24 e-15	0.000 000 000 41 e-15	u
inverse meter-electron volt relationship	1.239 841 984... e-6	(exact)	eV
inverse meter-hartree relationship	4.556 335 252 9132 e-8	0.000 000 000 0050 e-8	E_h
inverse meter-hertz relationship	299 792 458	(exact)	Hz
inverse meter-joule relationship	1.986 445 857... e-25	(exact)	J
inverse meter-kelvin relationship	1.438 776 877... e-2	(exact)	K
inverse meter-kilogram relationship	2.210 219 094... e-42	(exact)	kg
inverse of conductance quantum	12 906.403 72...	(exact)	ohm
Josephson constant	483 597.848 4... e9	(exact)	Hz V^-1
joule-atomic mass unit relationship	6.700 535 2471 e9	0.000 000 0021 e9	u
joule-electron volt relationship	6.241 509 074... e18	(exact)	eV
joule-hartree relationship	2.293 712 278 3969 e17	0.000 000 000 0025 e17	E_h
joule-hertz relationship	1.509 190 179... e33	(exact)	Hz
joule-inverse meter relationship	5.034 116 567... e24	(exact)	m^-1
joule-kelvin relationship	7.242 970 516... e22	(exact)	K
joule-kilogram relationship	1.112 650 056... e-17	(exact)	kg
kelvin-atomic mass unit relationship	9.251 087 2884 e-14	0.000 000 0029 e-14	u
kelvin-electron volt relationship	8.617 333 262... e-5	(exact)	eV
kelvin-hartree relationship	3.166 811 563 4564 e-6	0.000 000 000 0035 e-6	E_h
kelvin-hertz relationship	2.083 661 912... e10	(exact)	Hz
kelvin-inverse meter relationship	69.503 480 04...	(exact)	m^-1
kelvin-joule relationship	1.380 649 e-23	(exact)	J
kelvin-kilogram relationship	1.536 179 187... e-40	(exact)	kg
kilogram-atomic mass unit relationship	6.022 140 7537 e26	0.000 000 0019 e26	u
kilogram-electron volt relationship	5.609 588 603... e35	(exact)	eV
kilogram-hartree relationship	2.061 485 788 7415 e34	0.000 000 000 0022 e34	E_h
kilogram-hertz relationship	1.356 392 489... e50	(exact)	Hz
kilogram-inverse meter relationship	4.524 438 335... e41	(exact)	m^-1
kilogram-joule relationship	8.987 551 787... e16	(exact)	J
kilogram-kelvin relationship	6.509 657 260... e39	(exact)	K
lattice parameter of silicon	5.431 020 511 e-10	0.000 000 089 e-10	m
lattice spacing of ideal Si (220)	1.920 155 716 e-10	0.000 000 032 e-10	m
Loschmidt constant (273.15 K, 100 kPa)	2.651 645 804... e25	(exact)	m^-3
Loschmidt constant (273.15 K, 101.325 kPa)	2.686 780 111... e25	(exact)	m^-3
luminous efficacy	683	(exact)	lm W^-1

mag. flux quantum	2.067 833 848... e-15	(exact)	Wb
molar gas constant	8.314 462 618...	(exact)	J mol^-1 K^-1
molar mass constant	1.000 000 001 05 e-3	0.000 000 000 31 e-3	kg mol^-1
molar mass of carbon-12	12.000 000 0126 e-3	0.000 000 0037 e-3	kg mol^-1
molar Planck constant	3.990 312 712... e-10	(exact)	J Hz^-1 mol^-1
molar volume of ideal gas (273.15 K, 100 kPa)	22.710 954 64... e-3	(exact)	m^3 mol^-1
molar volume of ideal gas (273.15 K, 101.325 kPa)	22.413 969 54... e-3	(exact)	m^3 mol^-1
molar volume of silicon	1.205 883 199 e-5	0.000 000 060 e-5	m^3 mol^-1
Molybdenum x unit	1.002 099 52 e-13	0.000 000 53 e-13	m
muon Compton wavelength	1.173 444 110 e-14	0.000 000 026 e-14	m
muon-electron mass ratio	206.768 2827	0.000 0046	
muon g factor	-2.002 331 841 23	0.000 000 000 82	
muon mag. mom.	-4.490 448 30 e-26	0.000 000 10 e-26	J T^-1
muon mag. mom. anomaly	1.165 920 62 e-3	0.000 000 41 e-3	
muon mag. mom. to Bohr magneton ratio	-4.841 970 48 e-3	0.000 000 11 e-3	
muon mag. mom. to nuclear magneton ratio	-8.890 597 04	0.000 000 20	
muon mass	1.883 531 627 e-28	0.000 000 042 e-28	kg
muon mass energy equivalent	1.692 833 804 e-11	0.000 000 038 e-11	J
muon mass energy equivalent in MeV	105.658 3755	0.000 0023	MeV
muon mass in u	0.113 428 9257	0.000 000 0025	u
muon molar mass	1.134 289 258 e-4	0.000 000 025 e-4	kg mol^-1
muon-neutron mass ratio	0.112 454 5168	0.000 000 0025	
muon-proton mag. mom. ratio	-3.183 345 146	0.000 000 071	
muon-proton mass ratio	0.112 609 5262	0.000 000 0025	
muon-tau mass ratio	5.946 35 e-2	0.000 40 e-2	
natural unit of action	1.054 571 817... e-34	(exact)	J s
natural unit of action in eV s	6.582 119 569... e-16	(exact)	eV s
natural unit of energy	8.187 105 7880 e-14	0.000 000 0026 e-14	J
natural unit of energy in MeV	0.510 998 950 69	0.000 000 000 16	MeV
natural unit of length	3.861 592 6744 e-13	0.000 000 0012 e-13	m
natural unit of mass	9.109 383 7139 e-31	0.000 000 0028 e-31	kg
natural unit of momentum	2.730 924 534 46 e-22	0.000 000 000 85 e-22	kg m s^-1
natural unit of momentum in MeV/c	0.510 998 950 69	0.000 000 000 16	MeV/c
natural unit of time	1.288 088 666 44 e-21	0.000 000 000 40 e-21	s
natural unit of velocity	299 792 458	(exact)	m s^-1
neutron Compton wavelength	1.319 590 903 82 e-15	0.000 000 000 67 e-15	m
neutron-electron mag. mom. ratio	1.040 668 84 e-3	0.000 000 24 e-3	
neutron-electron mass ratio	1838.683 662 00	0.000 000 74	
neutron g factor	-3.826 085 52	0.000 000 90	
neutron gyromag. ratio	1.832 471 74 e8	0.000 000 43 e8	s^-1 T^-1
neutron gyromag. ratio in MHz/T	29.164 6935	0.000 0069	MHz T^-1
neutron mag. mom.	-9.662 3653 e-27	0.000 0023 e-27	J T^-1
neutron mag. mom. to Bohr magneton ratio	-1.041 875 65 e-3	0.000 000 25 e-3	
neutron mag. mom. to nuclear magneton ratio	-1.913 042 76	0.000 000 45	
neutron mass	1.674 927 500 56 e-27	0.000 000 000 85 e-27	kg
neutron mass energy equivalent	1.505 349 765 14 e-10	0.000 000 000 76 e-10	J
neutron mass energy equivalent in MeV	939.565 421 94	0.000 000 48	MeV
neutron mass in u	1.008 664 916 06	0.000 000 000 40	u
neutron molar mass	1.008 664 917 12 e-3	0.000 000 000 51 e-3	kg mol^-1
neutron-muon mass ratio	8.892 484 08	0.000 000 20	
neutron-proton mag. mom. ratio	-0.684 979 35	0.000 000 16	
neutron-proton mass difference	2.305 574 61 e-30	0.000 000 67 e-30	kg
neutron-proton mass difference energy equivalent	2.072 147 12 e-13	0.000 000 60 e-13	J
neutron-proton mass difference energy equivalent in MeV	1.293 332 51	0.000 000 38	MeV
neutron-proton mass difference in u	1.388 449 48 e-3	0.000 000 40 e-3	u
neutron-proton mass ratio	1.001 378 419 46	0.000 000 000 40	
neutron relative atomic mass	1.008 664 916 06	0.000 000 000 40	
neutron-tau mass ratio	0.528 779	0.000 036	
neutron to shielded proton mag. mom. ratio	-0.684 996 94	0.000 000 16	
Newtonian constant of gravitation	6.674 30 e-11	0.000 15 e-11	m^3 kg^-1 s^-2
Newtonian constant of gravitation over h-bar c	6.708 83 e-39	0.000 15 e-39	(GeV/c^2)^-2
nuclear magneton	5.050 783 7393 e-27	0.000 000 0016 e-27	J T^-1
nuclear magneton in eV/T	3.152 451 254 17 e-8	0.000 000 000 98 e-8	eV T^-1
nuclear magneton in inverse meter per tesla	2.542 623 410 09 e-2	0.000 000 000 79 e-2	m^-1 T^-1
nuclear magneton in K/T	3.658 267 7706 e-4	0.000 000 0011 e-4	K T^-1
nuclear magneton in MHz/T	7.622 593 2188	0.000 000 0024	MHz T^-1
Planck constant	6.626 070 15 e-34	(exact)	J Hz^-1
Planck constant in eV/Hz	4.135 667 696... e-15	(exact)	eV Hz^-1
Planck length	1.616 255 e-35	0.000 018 e-35	m
Planck mass	2.176 434 e-8	0.000 024 e-8	kg
Planck mass energy equivalent in GeV	1.220 890 e19	0.000 014 e19	GeV
Planck temperature	1.416 784 e32	0.000 016 e32	K
Planck time	5.391 247 e-44	0.000 060 e-44	s
proton charge to mass quotient	9.578 833 1430 e7	0.000 000 0030 e7	C kg^-1
proton Compton wavelength	1.321 409 853 60 e-15	0.000 000 000 41 e-15	m
proton-electron mass ratio	1836.152 673 426	0.000 000 032	
proton g factor	5.585 694 6893	0.000 000 0016	
proton gyromag. ratio	2.675 221 8708 e8	0.000 000 0011 e8	s^-1 T^-1
proton gyromag. ratio in MHz/T	42.577 478 461	0.000 000 018	MHz T^-1
proton mag. mom.	1.410 606 795 45 e-26	0.000 000 000 60 e-26	J T^-1
proton mag. mom. to Bohr magneton ratio	1.521 032 202 30 e-3	0.000 000 000 45 e-3	
proton mag. mom. to nuclear magneton ratio	2.792 847 344 63	0.000 000 000 82	
proton mag. shielding correction	2.567 15 e-5	0.000 41 e-5	
proton mass	1.672 621 925 95 e-27	0.000 000 000 52 e-27	kg
proton mass energy equivalent	1.503 277 618 02 e-10	0.000 000 000 47 e-10	J
proton mass energy equivalent in MeV	938.272 089 43	0.000 000 29	MeV
proton mass in u	1.007 276 466 5789	0.000 000 000 0083	u
proton molar mass	1.007 276 467 64 e-3	0.000 000 000 31 e-3	kg mol^-1
proton-muon mass ratio	8.880 243 38	0.000 000 20	
proton-neutron mag. mom. ratio	-1.459 898 02	0.000 000 34	
proton-neutron mass ratio	0.998 623 477 97	0.000 000 000 40	
proton relative atomic mass	1.007 276 466 5789	0.000 000 000 0083	
proton rms charge radius	8.4075 e-16	0.0064 e-16	m
proton-tau mass ratio	0.528 051	0.000 036	
quantum of circulation	3.636 947 5467 e-4	0.000 000 0011 e-4	m^2 s^-1
quantum of circulation times 2	7.273 895 0934 e-4	0.000 000 0023 e-4	m^2 s^-1
reduced Compton wavelength	3.861 592 6744 e-13	0.000 000 0012 e-13	m
reduced muon Compton wavelength	1.867 594 306 e-15	0.000 000 042 e-15	m
reduced neutron Compton wavelength	2.100 194 1520 e-16	0.000 000 0011 e-16	m
reduced Planck constant	1.054 571 817... e-34	(exact)	J s
reduced Planck constant in eV s	6.582 119 569... e-16	(exact)	eV s
reduced Planck constant times c in MeV fm	197.326 980 4...	(exact)	MeV fm
reduced proton Compton wavelength	2.103 089 100 51 e-16	0.000 000 000 66 e-16	m

reduced tau Compton wavelength	1.110 538 e-16	0.000 075 e-16	m
Rydberg constant	10 973 731.568 157	0.000 012	m^-1
Rydberg constant times c in Hz	3.289 841 960 2500 e15	0.000 000 000 0036 e15	Hz
Rydberg constant times hc in eV	13.605 693 122 990	0.000 000 000 015	eV
Rydberg constant times hc in J	2.179 872 361 1030 e-18	0.000 000 000 0024 e-18	J
Sackur-Tetrode constant (1 K, 100 kPa)	-1.151 707 534 96	0.000 000 000 47	
Sackur-Tetrode constant (1 K, 101.325 kPa)	-1.164 870 521 49	0.000 000 000 47	
second radiation constant	1.438 776 877... e-2	(exact)	m K
shielded helion gyromag. ratio	2.037 894 6078 e8	0.000 000 0018 e8	s^-1 T^-1
shielded helion gyromag. ratio in MHz/T	32.434 100 033	0.000 000 028	MHz T^-1
shielded helion mag. mom.	-1.074 553 110 35 e-26	0.000 000 000 93 e-26	J T^-1
shielded helion mag. mom. to Bohr magneton ratio	-1.158 671 494 57 e-3	0.000 000 000 94 e-3	
shielded helion mag. mom. to nuclear magneton ratio	-2.127 497 7624	0.000 000 0017	
shielded helion to proton mag. mom. ratio	-0.761 766 577 21	0.000 000 000 66	
shielded helion to shielded proton mag. mom. ratio	-0.761 786 1334	0.000 000 0031	
shielded proton gyromag. ratio	2.675 153 194 e8	0.000 000 011 e8	s^-1 T^-1
shielded proton gyromag. ratio in MHz/T	42.576 385 43	0.000 000 17	MHz T^-1
shielded proton mag. mom.	1.410 570 5830 e-26	0.000 000 0058 e-26	J T^-1
shielded proton mag. mom. to Bohr magneton ratio	1.520 993 1551 e-3	0.000 000 0062 e-3	
shielded proton mag. mom. to nuclear magneton ratio	2.792 775 648	0.000 000 011	
shielding difference of d and p in HD	1.987 70 e-8	0.000 10 e-8	
shielding difference of t and p in HT	2.394 50 e-8	0.000 20 e-8	
speed of light in vacuum	299 792 458	(exact)	m s^-1
standard acceleration of gravity	9.806 65	(exact)	m s^-2
standard atmosphere	101 325	(exact)	Pa
standard-state pressure	100 000	(exact)	Pa
Stefan-Boltzmann constant	5.670 374 419... e-8	(exact)	W m^-2 K^-4
tau Compton wavelength	6.977 71 e-16	0.000 47 e-16	m
tau-electron mass ratio	3477.23	0.23	
tau energy equivalent	1776.86	0.12	MeV
tau mass	3.167 54 e-27	0.000 21 e-27	kg
tau mass energy equivalent	2.846 84 e-10	0.000 19 e-10	J
tau mass in u	1.907 54	0.000 13	u
tau molar mass	1.907 54 e-3	0.000 13 e-3	kg mol^-1
tau-muon mass ratio	16.8170	0.0011	
tau-neutron mass ratio	1.891 15	0.000 13	
tau-proton mass ratio	1.893 76	0.000 13	
Thomson cross section	6.652 458 7051 e-29	0.000 000 0062 e-29	m^2
triton-electron mass ratio	5496.921 535 51	0.000 000 21	
triton g factor	5.957 924 930	0.000 000 012	
triton mag. mom.	1.504 609 5178 e-26	0.000 000 0030 e-26	J T^-1
triton mag. mom. to Bohr magneton ratio	1.622 393 6648 e-3	0.000 000 0032 e-3	
triton mag. mom. to nuclear magneton ratio	2.978 962 4650	0.000 000 0059	
triton mass	5.007 356 7512 e-27	0.000 000 0016 e-27	kg
triton mass energy equivalent	4.500 387 8119 e-10	0.000 000 0014 e-10	J
triton mass energy equivalent in MeV	2808.921 136 68	0.000 000 88	MeV
triton mass in u	3.015 500 715 97	0.000 000 000 10	u
triton molar mass	3.015 500 719 13 e-3	0.000 000 000 94 e-3	kg mol^-1
triton-proton mass ratio	2.993 717 034 03	0.000 000 000 10	
triton relative atomic mass	3.015 500 715 97	0.000 000 000 10	
triton to proton mag. mom. ratio	1.066 639 9189	0.000 000 0021	
unified atomic mass unit	1.660 539 068 92 e-27	0.000 000 000 52 e-27	kg
vacuum electric permittivity	8.854 187 8188 e-12	0.000 000 0014 e-12	F m^-1
vacuum mag. permeability	1.256 637 061 27 e-6	0.000 000 000 20 e-6	N A^-2
von Klitzing constant	25 812.807 45...	(exact)	ohm
weak mixing angle	0.223 05	0.000 23	
Wien frequency displacement law constant	5.878 925 757... e10	(exact)	Hz K^-1
Wien wavelength displacement law constant	2.897 771 955... e-3	(exact)	m K
W to Z mass ratio	0.881 45	0.000 13	

Chapter 9

codata.h

The C API is defined in the header file *codata.h*.

```
/* SPDX-License-Identifier: MIT */

#ifndef CODATA_H
#define CODATA_H
#if _MSC_VER
#define ADD_IMPORT __declspec(dllimport)
#else
#define ADD_IMPORT
#endif

extern char* codata_get_version(void);
extern char* codata_version(void);

typedef struct codata_constant_type{
    char name[65];
    double value;
    double uncertainty;
    char unit[33];
}cct;

//----------------------------------------------------------------------
// CODATA_CONSTANTS_2010
//----------------------------------------------------------------------
//{{{
ADD_IMPORT extern const int YEAR_2010;
ADD_IMPORT extern const cct LATTICE_SPACING_OF_SILICON_2010;
ADD_IMPORT extern const cct ALPHA_PARTICLE_ELECTRON_MASS_RATIO_2010;
ADD_IMPORT extern const cct ALPHA_PARTICLE_MASS_2010;
ADD_IMPORT extern const cct ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_2010;
ADD_IMPORT extern const cct ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_IN_MEV_2010;
ADD_IMPORT extern const cct ALPHA_PARTICLE_MASS_IN_U_2010;
ADD_IMPORT extern const cct ALPHA_PARTICLE_MOLAR_MASS_2010;
ADD_IMPORT extern const cct ALPHA_PARTICLE_PROTON_MASS_RATIO_2010;
ADD_IMPORT extern const cct ANGSTROM_STAR_2010;
ADD_IMPORT extern const cct ATOMIC_MASS_CONSTANT_2010;
ADD_IMPORT extern const cct ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_2010;
ADD_IMPORT extern const cct ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_IN_MEV_2010;
ADD_IMPORT extern const cct ATOMIC_MASS_UNIT_ELECTRON_VOLT_RELATIONSHIP_2010;
ADD_IMPORT extern const cct ATOMIC_MASS_UNIT_HARTREE_RELATIONSHIP_2010;
ADD_IMPORT extern const cct ATOMIC_MASS_UNIT_HERTZ_RELATIONSHIP_2010;
ADD_IMPORT extern const cct ATOMIC_MASS_UNIT_INVERSE_METER_RELATIONSHIP_2010;
ADD_IMPORT extern const cct ATOMIC_MASS_UNIT_JOULE_RELATIONSHIP_2010;
ADD_IMPORT extern const cct ATOMIC_MASS_UNIT_KELVIN_RELATIONSHIP_2010;
ADD_IMPORT extern const cct ATOMIC_MASS_UNIT_KILOGRAM_RELATIONSHIP_2010;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_1ST_HYPERPOLARIZABILITY_2010;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_2ND_HYPERPOLARIZABILITY_2010;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_ACTION_2010;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_CHARGE_2010;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_CHARGE_DENSITY_2010;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_CURRENT_2010;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_ELECTRIC_DIPOLE_MOM_2010;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_ELECTRIC_FIELD_2010;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_ELECTRIC_FIELD_GRADIENT_2010;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_ELECTRIC_POLARIZABILITY_2010;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_ELECTRIC_POTENTIAL_2010;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_ELECTRIC_QUADRUPOLE_MOM_2010;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_ENERGY_2010;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_FORCE_2010;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_LENGTH_2010;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_MAG_DIPOLE_MOM_2010;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_MAG_FLUX_DENSITY_2010;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_MAGNETIZABILITY_2010;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_MASS_2010;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_MOMUM_2010;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_PERMITTIVITY_2010;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_TIME_2010;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_VELOCITY_2010;
ADD_IMPORT extern const cct AVOGADRO_CONSTANT_2010;
ADD_IMPORT extern const cct BOHR_MAGNETON_2010;
ADD_IMPORT extern const cct BOHR_MAGNETON_IN_EV_T_2010;
ADD_IMPORT extern const cct BOHR_MAGNETON_IN_HZ_T_2010;
ADD_IMPORT extern const cct BOHR_MAGNETON_IN_INVERSE_METERS_PER_TESLA_2010;
ADD_IMPORT extern const cct BOHR_MAGNETON_IN_K_T_2010;
ADD_IMPORT extern const cct BOHR_RADIUS_2010;
ADD_IMPORT extern const cct BOLTZMANN_CONSTANT_2010;
```

```
ADD_IMPORT extern const cct BOLTZMANN_CONSTANT_IN_EV_K_2010;
ADD_IMPORT extern const cct BOLTZMANN_CONSTANT_IN_HZ_K_2010;
ADD_IMPORT extern const cct BOLTZMANN_CONSTANT_IN_INVERSE_METERS_PER_KELVIN_2010;
ADD_IMPORT extern const cct CHARACTERISTIC_IMPEDANCE_OF_VACUUM_2010;
ADD_IMPORT extern const cct CLASSICAL_ELECTRON_RADIUS_2010;
ADD_IMPORT extern const cct COMPTON_WAVELENGTH_2010;
ADD_IMPORT extern const cct COMPTON_WAVELENGTH_OVER_2_PI_2010;
ADD_IMPORT extern const cct CONDUCTANCE_QUANTUM_2010;
ADD_IMPORT extern const cct CONVENTIONAL_VALUE_OF_JOSEPHSON_CONSTANT_2010;
ADD_IMPORT extern const cct CONVENTIONAL_VALUE_OF_VON_KLITZING_CONSTANT_2010;
ADD_IMPORT extern const cct CU_X_UNIT_2010;
ADD_IMPORT extern const cct DEUTERON_ELECTRON_MAG_MOM_RATIO_2010;
ADD_IMPORT extern const cct DEUTERON_ELECTRON_MASS_RATIO_2010;
ADD_IMPORT extern const cct DEUTERON_G_FACTOR_2010;
ADD_IMPORT extern const cct DEUTERON_MAG_MOM_2010;
ADD_IMPORT extern const cct DEUTERON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010;
ADD_IMPORT extern const cct DEUTERON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010;
ADD_IMPORT extern const cct DEUTERON_MASS_2010;
ADD_IMPORT extern const cct DEUTERON_MASS_ENERGY_EQUIVALENT_2010;
ADD_IMPORT extern const cct DEUTERON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010;
ADD_IMPORT extern const cct DEUTERON_MASS_IN_U_2010;
ADD_IMPORT extern const cct DEUTERON_MOLAR_MASS_2010;
ADD_IMPORT extern const cct DEUTERON_NEUTRON_MAG_MOM_RATIO_2010;
ADD_IMPORT extern const cct DEUTERON_PROTON_MAG_MOM_RATIO_2010;
ADD_IMPORT extern const cct DEUTERON_PROTON_MASS_RATIO_2010;
ADD_IMPORT extern const cct DEUTERON_RMS_CHARGE_RADIUS_2010;
ADD_IMPORT extern const cct ELECTRIC_CONSTANT_2010;
ADD_IMPORT extern const cct ELECTRON_CHARGE_TO_MASS_QUOTIENT_2010;
ADD_IMPORT extern const cct ELECTRON_DEUTERON_MAG_MOM_RATIO_2010;
ADD_IMPORT extern const cct ELECTRON_DEUTERON_MASS_RATIO_2010;
ADD_IMPORT extern const cct ELECTRON_G_FACTOR_2010;
ADD_IMPORT extern const cct ELECTRON_GYROMAG_RATIO_2010;
ADD_IMPORT extern const cct ELECTRON_GYROMAG_RATIO_OVER_2_PI_2010;
ADD_IMPORT extern const cct ELECTRON_HELION_MASS_RATIO_2010;
ADD_IMPORT extern const cct ELECTRON_MAG_MOM_2010;
ADD_IMPORT extern const cct ELECTRON_MAG_MOM_ANOMALY_2010;
ADD_IMPORT extern const cct ELECTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010;
ADD_IMPORT extern const cct ELECTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010;
ADD_IMPORT extern const cct ELECTRON_MASS_2010;
ADD_IMPORT extern const cct ELECTRON_MASS_ENERGY_EQUIVALENT_2010;
ADD_IMPORT extern const cct ELECTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010;
ADD_IMPORT extern const cct ELECTRON_MASS_IN_U_2010;
ADD_IMPORT extern const cct ELECTRON_MOLAR_MASS_2010;
ADD_IMPORT extern const cct ELECTRON_MUON_MAG_MOM_RATIO_2010;
ADD_IMPORT extern const cct ELECTRON_MUON_MASS_RATIO_2010;
ADD_IMPORT extern const cct ELECTRON_NEUTRON_MAG_MOM_RATIO_2010;
ADD_IMPORT extern const cct ELECTRON_NEUTRON_MASS_RATIO_2010;
ADD_IMPORT extern const cct ELECTRON_PROTON_MAG_MOM_RATIO_2010;
ADD_IMPORT extern const cct ELECTRON_PROTON_MASS_RATIO_2010;
ADD_IMPORT extern const cct ELECTRON_TAU_MASS_RATIO_2010;
ADD_IMPORT extern const cct ELECTRON_TO_ALPHA_PARTICLE_MASS_RATIO_2010;
ADD_IMPORT extern const cct ELECTRON_TO_SHIELDED_HELION_MAG_MOM_RATIO_2010;
ADD_IMPORT extern const cct ELECTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2010;
ADD_IMPORT extern const cct ELECTRON_TRITON_MASS_RATIO_2010;
ADD_IMPORT extern const cct ELECTRON_VOLT_2010;
ADD_IMPORT extern const cct ELECTRON_VOLT_ATOMIC_MASS_UNIT_RELATIONSHIP_2010;
ADD_IMPORT extern const cct ELECTRON_VOLT_HARTREE_RELATIONSHIP_2010;
ADD_IMPORT extern const cct ELECTRON_VOLT_HERTZ_RELATIONSHIP_2010;
ADD_IMPORT extern const cct ELECTRON_VOLT_INVERSE_METER_RELATIONSHIP_2010;
ADD_IMPORT extern const cct ELECTRON_VOLT_JOULE_RELATIONSHIP_2010;
ADD_IMPORT extern const cct ELECTRON_VOLT_KELVIN_RELATIONSHIP_2010;
ADD_IMPORT extern const cct ELECTRON_VOLT_KILOGRAM_RELATIONSHIP_2010;
ADD_IMPORT extern const cct ELEMENTARY_CHARGE_2010;
ADD_IMPORT extern const cct ELEMENTARY_CHARGE_OVER_H_2010;
ADD_IMPORT extern const cct FARADAY_CONSTANT_2010;
ADD_IMPORT extern const cct FARADAY_CONSTANT_FOR_CONVENTIONAL_ELECTRIC_CURRENT_2010;
ADD_IMPORT extern const cct FERMI_COUPLING_CONSTANT_2010;
ADD_IMPORT extern const cct FINE_STRUCTURE_CONSTANT_2010;
ADD_IMPORT extern const cct FIRST_RADIATION_CONSTANT_2010;
ADD_IMPORT extern const cct FIRST_RADIATION_CONSTANT_FOR_SPECTRAL_RADIANCE_2010;
ADD_IMPORT extern const cct HARTREE_ATOMIC_MASS_UNIT_RELATIONSHIP_2010;
ADD_IMPORT extern const cct HARTREE_ELECTRON_VOLT_RELATIONSHIP_2010;
ADD_IMPORT extern const cct HARTREE_ENERGY_2010;
ADD_IMPORT extern const cct HARTREE_ENERGY_IN_EV_2010;
ADD_IMPORT extern const cct HARTREE_HERTZ_RELATIONSHIP_2010;
ADD_IMPORT extern const cct HARTREE_INVERSE_METER_RELATIONSHIP_2010;
ADD_IMPORT extern const cct HARTREE_JOULE_RELATIONSHIP_2010;
ADD_IMPORT extern const cct HARTREE_KELVIN_RELATIONSHIP_2010;
ADD_IMPORT extern const cct HARTREE_KILOGRAM_RELATIONSHIP_2010;
ADD_IMPORT extern const cct HELION_ELECTRON_MASS_RATIO_2010;
ADD_IMPORT extern const cct HELION_G_FACTOR_2010;
ADD_IMPORT extern const cct HELION_MAG_MOM_2010;
ADD_IMPORT extern const cct HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010;
ADD_IMPORT extern const cct HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010;
ADD_IMPORT extern const cct HELION_MASS_2010;
ADD_IMPORT extern const cct HELION_MASS_ENERGY_EQUIVALENT_2010;
ADD_IMPORT extern const cct HELION_MASS_ENERGY_EQUIVALENT_IN_MEV_2010;
ADD_IMPORT extern const cct HELION_MASS_IN_U_2010;
ADD_IMPORT extern const cct HELION_MOLAR_MASS_2010;
ADD_IMPORT extern const cct HELION_PROTON_MASS_RATIO_2010;
ADD_IMPORT extern const cct HERTZ_ATOMIC_MASS_UNIT_RELATIONSHIP_2010;
ADD_IMPORT extern const cct HERTZ_ELECTRON_VOLT_RELATIONSHIP_2010;
ADD_IMPORT extern const cct HERTZ_HARTREE_RELATIONSHIP_2010;
ADD_IMPORT extern const cct HERTZ_INVERSE_METER_RELATIONSHIP_2010;
ADD_IMPORT extern const cct HERTZ_JOULE_RELATIONSHIP_2010;
ADD_IMPORT extern const cct HERTZ_KELVIN_RELATIONSHIP_2010;
ADD_IMPORT extern const cct HERTZ_KILOGRAM_RELATIONSHIP_2010;
ADD_IMPORT extern const cct INVERSE_FINE_STRUCTURE_CONSTANT_2010;
ADD_IMPORT extern const cct INVERSE_METER_ATOMIC_MASS_UNIT_RELATIONSHIP_2010;
ADD_IMPORT extern const cct INVERSE_METER_ELECTRON_VOLT_RELATIONSHIP_2010;
ADD_IMPORT extern const cct INVERSE_METER_HARTREE_RELATIONSHIP_2010;
ADD_IMPORT extern const cct INVERSE_METER_HERTZ_RELATIONSHIP_2010;
ADD_IMPORT extern const cct INVERSE_METER_JOULE_RELATIONSHIP_2010;
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ADD_IMPORT extern const cct INVERSE_METER_KELVIN_RELATIONSHIP_2010;
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ADD_IMPORT extern const cct INVERSE_OF_CONDUCTANCE_QUANTUM_2010;
ADD_IMPORT extern const cct JOSEPHSON_CONSTANT_2010;
ADD_IMPORT extern const cct JOULE_ATOMIC_MASS_UNIT_RELATIONSHIP_2010;
ADD_IMPORT extern const cct JOULE_ELECTRON_VOLT_RELATIONSHIP_2010;
ADD_IMPORT extern const cct JOULE_HARTREE_RELATIONSHIP_2010;
ADD_IMPORT extern const cct JOULE_HERTZ_RELATIONSHIP_2010;
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ADD_IMPORT extern const cct JOULE_KILOGRAM_RELATIONSHIP_2010;
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ADD_IMPORT extern const cct KILOGRAM_HARTREE_RELATIONSHIP_2010;
ADD_IMPORT extern const cct KILOGRAM_HERTZ_RELATIONSHIP_2010;
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ADD_IMPORT extern const cct KILOGRAM_JOULE_RELATIONSHIP_2010;
ADD_IMPORT extern const cct KILOGRAM_KELVIN_RELATIONSHIP_2010;
ADD_IMPORT extern const cct LATTICE_PARAMETER_OF_SILICON_2010;
ADD_IMPORT extern const cct LOSCHMIDT_CONSTANT_273_15_K_100_KPA_2010;
ADD_IMPORT extern const cct LOSCHMIDT_CONSTANT_273_15_K_101_325_KPA_2010;
ADD_IMPORT extern const cct MAG_CONSTANT_2010;
ADD_IMPORT extern const cct MAG_FLUX_QUANTUM_2010;
ADD_IMPORT extern const cct MOLAR_GAS_CONSTANT_2010;
ADD_IMPORT extern const cct MOLAR_MASS_CONSTANT_2010;
ADD_IMPORT extern const cct MOLAR_MASS_OF_CARBON_12_2010;
ADD_IMPORT extern const cct MOLAR_PLANCK_CONSTANT_2010;
ADD_IMPORT extern const cct MOLAR_PLANCK_CONSTANT_TIMES_C_2010;
ADD_IMPORT extern const cct MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_100_KPA_2010;
ADD_IMPORT extern const cct MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_101_325_KPA_2010;
ADD_IMPORT extern const cct MOLAR_VOLUME_OF_SILICON_2010;
ADD_IMPORT extern const cct MO_X_UNIT_2010;
ADD_IMPORT extern const cct MUON_COMPTON_WAVELENGTH_2010;
ADD_IMPORT extern const cct MUON_COMPTON_WAVELENGTH_OVER_2_PI_2010;
ADD_IMPORT extern const cct MUON_ELECTRON_MASS_RATIO_2010;
ADD_IMPORT extern const cct MUON_G_FACTOR_2010;
ADD_IMPORT extern const cct MUON_MAG_MOM_2010;
ADD_IMPORT extern const cct MUON_MAG_MOM_ANOMALY_2010;
ADD_IMPORT extern const cct MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010;
ADD_IMPORT extern const cct MUON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010;
ADD_IMPORT extern const cct MUON_MASS_2010;
ADD_IMPORT extern const cct MUON_MASS_ENERGY_EQUIVALENT_2010;
ADD_IMPORT extern const cct MUON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010;
ADD_IMPORT extern const cct MUON_MASS_IN_U_2010;
ADD_IMPORT extern const cct MUON_MOLAR_MASS_2010;
ADD_IMPORT extern const cct MUON_NEUTRON_MASS_RATIO_2010;
ADD_IMPORT extern const cct MUON_PROTON_MAG_MOM_RATIO_2010;
ADD_IMPORT extern const cct MUON_PROTON_MASS_RATIO_2010;
ADD_IMPORT extern const cct MUON_TAU_MASS_RATIO_2010;
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ADD_IMPORT extern const cct NATURAL_UNIT_OF_ACTION_IN_EV_S_2010;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_ENERGY_2010;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_ENERGY_IN_MEV_2010;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_LENGTH_2010;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_MASS_2010;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_MOMUM_2010;
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ADD_IMPORT extern const cct NEUTRON_ELECTRON_MAG_MOM_RATIO_2010;
ADD_IMPORT extern const cct NEUTRON_ELECTRON_MASS_RATIO_2010;
ADD_IMPORT extern const cct NEUTRON_G_FACTOR_2010;
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ADD_IMPORT extern const cct NEUTRON_MAG_MOM_2010;
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ADD_IMPORT extern const cct NEUTRON_MASS_IN_U_2010;
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ADD_IMPORT extern const cct NEUTRON_PROTON_MASS_DIFFERENCE_IN_U_2010;
ADD_IMPORT extern const cct NEUTRON_PROTON_MASS_RATIO_2010;
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ADD_IMPORT extern const cct NEUTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2010;
ADD_IMPORT extern const cct NEWTONIAN_CONSTANT_OF_GRAVITATION_2010;
ADD_IMPORT extern const cct NEWTONIAN_CONSTANT_OF_GRAVITATION_OVER_H_BAR_C_2010;
ADD_IMPORT extern const cct NUCLEAR_MAGNETON_2010;
ADD_IMPORT extern const cct NUCLEAR_MAGNETON_IN_EV_T_2010;
ADD_IMPORT extern const cct NUCLEAR_MAGNETON_IN_INVERSE_METERS_PER_TESLA_2010;
ADD_IMPORT extern const cct NUCLEAR_MAGNETON_IN_K_T_2010;
ADD_IMPORT extern const cct NUCLEAR_MAGNETON_IN_MHZ_T_2010;
ADD_IMPORT extern const cct PLANCK_CONSTANT_2010;
ADD_IMPORT extern const cct PLANCK_CONSTANT_IN_EV_S_2010;
ADD_IMPORT extern const cct PLANCK_CONSTANT_OVER_2_PI_2010;
ADD_IMPORT extern const cct PLANCK_CONSTANT_OVER_2_PI_IN_EV_S_2010;
ADD_IMPORT extern const cct PLANCK_CONSTANT_OVER_2_PI_TIMES_C_IN_MEV_FM_2010;
ADD_IMPORT extern const cct PLANCK_LENGTH_2010;
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ADD_IMPORT extern const cct PLANCK_MASS_ENERGY_EQUIVALENT_IN_GEV_2010;
ADD_IMPORT extern const cct PLANCK_TEMPERATURE_2010;
ADD_IMPORT extern const cct PLANCK_TIME_2010;
ADD_IMPORT extern const cct PROTON_CHARGE_TO_MASS_QUOTIENT_2010;
ADD_IMPORT extern const cct PROTON_COMPTON_WAVELENGTH_2010;
ADD_IMPORT extern const cct PROTON_COMPTON_WAVELENGTH_OVER_2_PI_2010;
ADD_IMPORT extern const cct PROTON_ELECTRON_MASS_RATIO_2010;
ADD_IMPORT extern const cct PROTON_G_FACTOR_2010;
ADD_IMPORT extern const cct PROTON_GYROMAG_RATIO_2010;
ADD_IMPORT extern const cct PROTON_GYROMAG_RATIO_OVER_2_PI_2010;
ADD_IMPORT extern const cct PROTON_MAG_MOM_2010;
ADD_IMPORT extern const cct PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010;
ADD_IMPORT extern const cct PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010;
ADD_IMPORT extern const cct PROTON_MAG_SHIELDING_CORRECTION_2010;
ADD_IMPORT extern const cct PROTON_MASS_2010;
ADD_IMPORT extern const cct PROTON_MASS_ENERGY_EQUIVALENT_2010;
ADD_IMPORT extern const cct PROTON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010;
ADD_IMPORT extern const cct PROTON_MASS_IN_U_2010;
ADD_IMPORT extern const cct PROTON_MOLAR_MASS_2010;
ADD_IMPORT extern const cct PROTON_MUON_MASS_RATIO_2010;
ADD_IMPORT extern const cct PROTON_NEUTRON_MAG_MOM_RATIO_2010;
ADD_IMPORT extern const cct PROTON_NEUTRON_MASS_RATIO_2010;
ADD_IMPORT extern const cct PROTON_RMS_CHARGE_RADIUS_2010;
ADD_IMPORT extern const cct PROTON_TAU_MASS_RATIO_2010;
ADD_IMPORT extern const cct QUANTUM_OF_CIRCULATION_2010;
ADD_IMPORT extern const cct QUANTUM_OF_CIRCULATION_TIMES_2_2010;
ADD_IMPORT extern const cct RYDBERG_CONSTANT_2010;
ADD_IMPORT extern const cct RYDBERG_CONSTANT_TIMES_C_IN_HZ_2010;
ADD_IMPORT extern const cct RYDBERG_CONSTANT_TIMES_HC_IN_EV_2010;
ADD_IMPORT extern const cct RYDBERG_CONSTANT_TIMES_HC_IN_J_2010;
ADD_IMPORT extern const cct SACKUR_TETRODE_CONSTANT_1_K_100_KPA_2010;
ADD_IMPORT extern const cct SACKUR_TETRODE_CONSTANT_1_K_101_325_KPA_2010;
ADD_IMPORT extern const cct SECOND_RADIATION_CONSTANT_2010;
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ADD_IMPORT extern const cct SHIELDED_HELION_GYROMAG_RATIO_OVER_2_PI_2010;
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ADD_IMPORT extern const cct SHIELDED_HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010;
ADD_IMPORT extern const cct SHIELDED_HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010;
ADD_IMPORT extern const cct SHIELDED_HELION_TO_PROTON_MAG_MOM_RATIO_2010;
ADD_IMPORT extern const cct SHIELDED_HELION_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2010;
ADD_IMPORT extern const cct SHIELDED_PROTON_GYROMAG_RATIO_2010;
ADD_IMPORT extern const cct SHIELDED_PROTON_GYROMAG_RATIO_OVER_2_PI_2010;
ADD_IMPORT extern const cct SHIELDED_PROTON_MAG_MOM_2010;
ADD_IMPORT extern const cct SHIELDED_PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010;
ADD_IMPORT extern const cct SHIELDED_PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010;
ADD_IMPORT extern const cct SPEED_OF_LIGHT_IN_VACUUM_2010;
ADD_IMPORT extern const cct STANDARD_ACCELERATION_OF_GRAVITY_2010;
ADD_IMPORT extern const cct STANDARD_ATMOSPHERE_2010;
ADD_IMPORT extern const cct STANDARD_STATE_PRESSURE_2010;
ADD_IMPORT extern const cct STEFAN_BOLTZMANN_CONSTANT_2010;
ADD_IMPORT extern const cct TAU_COMPTON_WAVELENGTH_2010;
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ADD_IMPORT extern const cct TAU_MASS_ENERGY_EQUIVALENT_2010;
ADD_IMPORT extern const cct TAU_MASS_ENERGY_EQUIVALENT_IN_MEV_2010;
ADD_IMPORT extern const cct TAU_MASS_IN_U_2010;
ADD_IMPORT extern const cct TAU_MOLAR_MASS_2010;
ADD_IMPORT extern const cct TAU_MUON_MASS_RATIO_2010;
ADD_IMPORT extern const cct TAU_NEUTRON_MASS_RATIO_2010;
ADD_IMPORT extern const cct TAU_PROTON_MASS_RATIO_2010;
ADD_IMPORT extern const cct THOMSON_CROSS_SECTION_2010;
ADD_IMPORT extern const cct TRITON_ELECTRON_MASS_RATIO_2010;
ADD_IMPORT extern const cct TRITON_G_FACTOR_2010;
ADD_IMPORT extern const cct TRITON_MAG_MOM_2010;
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ADD_IMPORT extern const cct TRITON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010;
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ADD_IMPORT extern const cct TRITON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010;
ADD_IMPORT extern const cct TRITON_MASS_IN_U_2010;
ADD_IMPORT extern const cct TRITON_MOLAR_MASS_2010;
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ADD_IMPORT extern const cct UNIFIED_ATOMIC_MASS_UNIT_2010;
ADD_IMPORT extern const cct VON_KLITZING_CONSTANT_2010;
ADD_IMPORT extern const cct WEAK_MIXING_ANGLE_2010;
ADD_IMPORT extern const cct WIEN_FREQUENCY_DISPLACEMENT_LAW_CONSTANT_2010;
ADD_IMPORT extern const cct WIEN_WAVELENGTH_DISPLACEMENT_LAW_CONSTANT_2010;
//}}}

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// CODATA_CONSTANTS_2014
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ADD_IMPORT extern const cct LATTICE_SPACING_OF_SILICON_2014;
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ADD_IMPORT extern const cct ALPHA_PARTICLE_MASS_2014;
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ADD_IMPORT extern const cct ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_IN_MEV_2014;
ADD_IMPORT extern const cct ALPHA_PARTICLE_MASS_IN_U_2014;
ADD_IMPORT extern const cct ALPHA_PARTICLE_MOLAR_MASS_2014;
ADD_IMPORT extern const cct ALPHA_PARTICLE_PROTON_MASS_RATIO_2014;
ADD_IMPORT extern const cct ANGSTROM_STAR_2014;
ADD_IMPORT extern const cct ATOMIC_MASS_CONSTANT_2014;
ADD_IMPORT extern const cct ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_2014;
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ADD_IMPORT extern const cct ATOMIC_MASS_UNIT_ELECTRON_VOLT_RELATIONSHIP_2014;
ADD_IMPORT extern const cct ATOMIC_MASS_UNIT_HARTREE_RELATIONSHIP_2014;
ADD_IMPORT extern const cct ATOMIC_MASS_UNIT_HERTZ_RELATIONSHIP_2014;
ADD_IMPORT extern const cct ATOMIC_MASS_UNIT_INVERSE_METER_RELATIONSHIP_2014;
ADD_IMPORT extern const cct ATOMIC_MASS_UNIT_JOULE_RELATIONSHIP_2014;
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ADD_IMPORT extern const cct ATOMIC_MASS_UNIT_KELVIN_RELATIONSHIP_2014;
ADD_IMPORT extern const cct ATOMIC_MASS_UNIT_KILOGRAM_RELATIONSHIP_2014;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_1ST_HYPERPOLARIZABILITY_2014;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_2ND_HYPERPOLARIZABILITY_2014;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_ACTION_2014;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_CHARGE_2014;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_CHARGE_DENSITY_2014;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_CURRENT_2014;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_ELECTRIC_DIPOLE_MOM_2014;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_ELECTRIC_FIELD_2014;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_ELECTRIC_FIELD_GRADIENT_2014;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_ELECTRIC_POLARIZABILITY_2014;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_ELECTRIC_POTENTIAL_2014;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_ELECTRIC_QUADRUPOLE_MOM_2014;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_ENERGY_2014;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_FORCE_2014;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_LENGTH_2014;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_MAG_DIPOLE_MOM_2014;
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ADD_IMPORT extern const cct BOLTZMANN_CONSTANT_2014;
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ADD_IMPORT extern const cct DEUTERON_G_FACTOR_2014;
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ADD_IMPORT extern const cct DEUTERON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014;
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ADD_IMPORT extern const cct DEUTERON_MASS_IN_U_2014;
ADD_IMPORT extern const cct DEUTERON_MOLAR_MASS_2014;
ADD_IMPORT extern const cct DEUTERON_NEUTRON_MAG_MOM_RATIO_2014;
ADD_IMPORT extern const cct DEUTERON_PROTON_MAG_MOM_RATIO_2014;
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ADD_IMPORT extern const cct ELECTRON_HELION_MASS_RATIO_2014;
ADD_IMPORT extern const cct ELECTRON_MAG_MOM_2014;
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ADD_IMPORT extern const cct ELECTRON_MASS_IN_U_2014;
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ADD_IMPORT extern const cct ELECTRON_NEUTRON_MAG_MOM_RATIO_2014;
ADD_IMPORT extern const cct ELECTRON_NEUTRON_MASS_RATIO_2014;
ADD_IMPORT extern const cct ELECTRON_PROTON_MAG_MOM_RATIO_2014;
ADD_IMPORT extern const cct ELECTRON_PROTON_MASS_RATIO_2014;
ADD_IMPORT extern const cct ELECTRON_TAU_MASS_RATIO_2014;
ADD_IMPORT extern const cct ELECTRON_TO_ALPHA_PARTICLE_MASS_RATIO_2014;
ADD_IMPORT extern const cct ELECTRON_TO_SHIELDED_HELION_MAG_MOM_RATIO_2014;
ADD_IMPORT extern const cct ELECTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2014;
ADD_IMPORT extern const cct ELECTRON_TRITON_MASS_RATIO_2014;
ADD_IMPORT extern const cct ELECTRON_VOLT_2014;
ADD_IMPORT extern const cct ELECTRON_VOLT_ATOMIC_MASS_UNIT_RELATIONSHIP_2014;
ADD_IMPORT extern const cct ELECTRON_VOLT_HARTREE_RELATIONSHIP_2014;
ADD_IMPORT extern const cct ELECTRON_VOLT_HERTZ_RELATIONSHIP_2014;
ADD_IMPORT extern const cct ELECTRON_VOLT_INVERSE_METER_RELATIONSHIP_2014;
ADD_IMPORT extern const cct ELECTRON_VOLT_JOULE_RELATIONSHIP_2014;
ADD_IMPORT extern const cct ELECTRON_VOLT_KELVIN_RELATIONSHIP_2014;
ADD_IMPORT extern const cct ELECTRON_VOLT_KILOGRAM_RELATIONSHIP_2014;
ADD_IMPORT extern const cct ELEMENTARY_CHARGE_2014;
ADD_IMPORT extern const cct ELEMENTARY_CHARGE_OVER_H_2014;
ADD_IMPORT extern const cct FARADAY_CONSTANT_2014;
ADD_IMPORT extern const cct FARADAY_CONSTANT_FOR_CONVENTIONAL_ELECTRIC_CURRENT_2014;
ADD_IMPORT extern const cct FERMI_COUPLING_CONSTANT_2014;
ADD_IMPORT extern const cct FINE_STRUCTURE_CONSTANT_2014;
ADD_IMPORT extern const cct FIRST_RADIATION_CONSTANT_2014;
ADD_IMPORT extern const cct FIRST_RADIATION_CONSTANT_FOR_SPECTRAL_RADIANCE_2014;
```

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ADD_IMPORT extern const cct HARTREE_ATOMIC_MASS_UNIT_RELATIONSHIP_2014;
ADD_IMPORT extern const cct HARTREE_ELECTRON_VOLT_RELATIONSHIP_2014;
ADD_IMPORT extern const cct HARTREE_ENERGY_2014;
ADD_IMPORT extern const cct HARTREE_ENERGY_IN_EV_2014;
ADD_IMPORT extern const cct HARTREE_HERTZ_RELATIONSHIP_2014;
ADD_IMPORT extern const cct HARTREE_INVERSE_METER_RELATIONSHIP_2014;
ADD_IMPORT extern const cct HARTREE_JOULE_RELATIONSHIP_2014;
ADD_IMPORT extern const cct HARTREE_KELVIN_RELATIONSHIP_2014;
ADD_IMPORT extern const cct HARTREE_KILOGRAM_RELATIONSHIP_2014;
ADD_IMPORT extern const cct HELION_ELECTRON_MASS_RATIO_2014;
ADD_IMPORT extern const cct HELION_G_FACTOR_2014;
ADD_IMPORT extern const cct HELION_MAG_MOM_2014;
ADD_IMPORT extern const cct HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014;
ADD_IMPORT extern const cct HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014;
ADD_IMPORT extern const cct HELION_MASS_2014;
ADD_IMPORT extern const cct HELION_MASS_ENERGY_EQUIVALENT_2014;
ADD_IMPORT extern const cct HELION_MASS_ENERGY_EQUIVALENT_IN_MEV_2014;
ADD_IMPORT extern const cct HELION_MASS_IN_U_2014;
ADD_IMPORT extern const cct HELION_MOLAR_MASS_2014;
ADD_IMPORT extern const cct HELION_PROTON_MASS_RATIO_2014;
ADD_IMPORT extern const cct HERTZ_ATOMIC_MASS_UNIT_RELATIONSHIP_2014;
ADD_IMPORT extern const cct HERTZ_ELECTRON_VOLT_RELATIONSHIP_2014;
ADD_IMPORT extern const cct HERTZ_HARTREE_RELATIONSHIP_2014;
ADD_IMPORT extern const cct HERTZ_INVERSE_METER_RELATIONSHIP_2014;
ADD_IMPORT extern const cct HERTZ_JOULE_RELATIONSHIP_2014;
ADD_IMPORT extern const cct HERTZ_KELVIN_RELATIONSHIP_2014;
ADD_IMPORT extern const cct HERTZ_KILOGRAM_RELATIONSHIP_2014;
ADD_IMPORT extern const cct INVERSE_FINE_STRUCTURE_CONSTANT_2014;
ADD_IMPORT extern const cct INVERSE_METER_ATOMIC_MASS_UNIT_RELATIONSHIP_2014;
ADD_IMPORT extern const cct INVERSE_METER_ELECTRON_VOLT_RELATIONSHIP_2014;
ADD_IMPORT extern const cct INVERSE_METER_HARTREE_RELATIONSHIP_2014;
ADD_IMPORT extern const cct INVERSE_METER_HERTZ_RELATIONSHIP_2014;
ADD_IMPORT extern const cct INVERSE_METER_JOULE_RELATIONSHIP_2014;
ADD_IMPORT extern const cct INVERSE_METER_KELVIN_RELATIONSHIP_2014;
ADD_IMPORT extern const cct INVERSE_METER_KILOGRAM_RELATIONSHIP_2014;
ADD_IMPORT extern const cct INVERSE_OF_CONDUCTANCE_QUANTUM_2014;
ADD_IMPORT extern const cct JOSEPHSON_CONSTANT_2014;
ADD_IMPORT extern const cct JOULE_ATOMIC_MASS_UNIT_RELATIONSHIP_2014;
ADD_IMPORT extern const cct JOULE_ELECTRON_VOLT_RELATIONSHIP_2014;
ADD_IMPORT extern const cct JOULE_HARTREE_RELATIONSHIP_2014;
ADD_IMPORT extern const cct JOULE_HERTZ_RELATIONSHIP_2014;
ADD_IMPORT extern const cct JOULE_INVERSE_METER_RELATIONSHIP_2014;
ADD_IMPORT extern const cct JOULE_KELVIN_RELATIONSHIP_2014;
ADD_IMPORT extern const cct JOULE_KILOGRAM_RELATIONSHIP_2014;
ADD_IMPORT extern const cct KELVIN_ATOMIC_MASS_UNIT_RELATIONSHIP_2014;
ADD_IMPORT extern const cct KELVIN_ELECTRON_VOLT_RELATIONSHIP_2014;
ADD_IMPORT extern const cct KELVIN_HARTREE_RELATIONSHIP_2014;
ADD_IMPORT extern const cct KELVIN_HERTZ_RELATIONSHIP_2014;
ADD_IMPORT extern const cct KELVIN_INVERSE_METER_RELATIONSHIP_2014;
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ADD_IMPORT extern const cct KELVIN_KILOGRAM_RELATIONSHIP_2014;
ADD_IMPORT extern const cct KILOGRAM_ATOMIC_MASS_UNIT_RELATIONSHIP_2014;
ADD_IMPORT extern const cct KILOGRAM_ELECTRON_VOLT_RELATIONSHIP_2014;
ADD_IMPORT extern const cct KILOGRAM_HARTREE_RELATIONSHIP_2014;
ADD_IMPORT extern const cct KILOGRAM_HERTZ_RELATIONSHIP_2014;
ADD_IMPORT extern const cct KILOGRAM_INVERSE_METER_RELATIONSHIP_2014;
ADD_IMPORT extern const cct KILOGRAM_JOULE_RELATIONSHIP_2014;
ADD_IMPORT extern const cct KILOGRAM_KELVIN_RELATIONSHIP_2014;
ADD_IMPORT extern const cct LATTICE_PARAMETER_OF_SILICON_2014;
ADD_IMPORT extern const cct LOSCHMIDT_CONSTANT_273_15_K_100_KPA_2014;
ADD_IMPORT extern const cct LOSCHMIDT_CONSTANT_273_15_K_101_325_KPA_2014;
ADD_IMPORT extern const cct MAG_CONSTANT_2014;
ADD_IMPORT extern const cct MAG_FLUX_QUANTUM_2014;
ADD_IMPORT extern const cct MOLAR_GAS_CONSTANT_2014;
ADD_IMPORT extern const cct MOLAR_MASS_CONSTANT_2014;
ADD_IMPORT extern const cct MOLAR_MASS_OF_CARBON_12_2014;
ADD_IMPORT extern const cct MOLAR_PLANCK_CONSTANT_2014;
ADD_IMPORT extern const cct MOLAR_PLANCK_CONSTANT_TIMES_C_2014;
ADD_IMPORT extern const cct MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_100_KPA_2014;
ADD_IMPORT extern const cct MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_101_325_KPA_2014;
ADD_IMPORT extern const cct MOLAR_VOLUME_OF_SILICON_2014;
ADD_IMPORT extern const cct MO_X_UNIT_2014;
ADD_IMPORT extern const cct MUON_COMPTON_WAVELENGTH_2014;
ADD_IMPORT extern const cct MUON_COMPTON_WAVELENGTH_OVER_2_PI_2014;
ADD_IMPORT extern const cct MUON_ELECTRON_MASS_RATIO_2014;
ADD_IMPORT extern const cct MUON_G_FACTOR_2014;
ADD_IMPORT extern const cct MUON_MAG_MOM_2014;
ADD_IMPORT extern const cct MUON_MAG_MOM_ANOMALY_2014;
ADD_IMPORT extern const cct MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014;
ADD_IMPORT extern const cct MUON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014;
ADD_IMPORT extern const cct MUON_MASS_2014;
ADD_IMPORT extern const cct MUON_MASS_ENERGY_EQUIVALENT_2014;
ADD_IMPORT extern const cct MUON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014;
ADD_IMPORT extern const cct MUON_MASS_IN_U_2014;
ADD_IMPORT extern const cct MUON_MOLAR_MASS_2014;
ADD_IMPORT extern const cct MUON_NEUTRON_MASS_RATIO_2014;
ADD_IMPORT extern const cct MUON_PROTON_MAG_MOM_RATIO_2014;
ADD_IMPORT extern const cct MUON_PROTON_MASS_RATIO_2014;
ADD_IMPORT extern const cct MUON_TAU_MASS_RATIO_2014;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_ACTION_2014;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_ACTION_IN_EV_S_2014;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_ENERGY_2014;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_ENERGY_IN_MEV_2014;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_LENGTH_2014;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_MASS_2014;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_MOMUM_2014;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_MOMUM_IN_MEV_C_2014;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_TIME_2014;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_VELOCITY_2014;
ADD_IMPORT extern const cct NEUTRON_COMPTON_WAVELENGTH_2014;
ADD_IMPORT extern const cct NEUTRON_COMPTON_WAVELENGTH_OVER_2_PI_2014;
ADD_IMPORT extern const cct NEUTRON_ELECTRON_MAG_MOM_RATIO_2014;
ADD_IMPORT extern const cct NEUTRON_ELECTRON_MASS_RATIO_2014;
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ADD_IMPORT extern const cct NEUTRON_G_FACTOR_2014;
ADD_IMPORT extern const cct NEUTRON_GYROMAG_RATIO_2014;
ADD_IMPORT extern const cct NEUTRON_GYROMAG_RATIO_OVER_2_PI_2014;
ADD_IMPORT extern const cct NEUTRON_MAG_MOM_2014;
ADD_IMPORT extern const cct NEUTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014;
ADD_IMPORT extern const cct NEUTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014;
ADD_IMPORT extern const cct NEUTRON_MASS_2014;
ADD_IMPORT extern const cct NEUTRON_MASS_ENERGY_EQUIVALENT_2014;
ADD_IMPORT extern const cct NEUTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014;
ADD_IMPORT extern const cct NEUTRON_MASS_IN_U_2014;
ADD_IMPORT extern const cct NEUTRON_MOLAR_MASS_2014;
ADD_IMPORT extern const cct NEUTRON_MUON_MASS_RATIO_2014;
ADD_IMPORT extern const cct NEUTRON_PROTON_MAG_MOM_RATIO_2014;
ADD_IMPORT extern const cct NEUTRON_PROTON_MASS_DIFFERENCE_2014;
ADD_IMPORT extern const cct NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_2014;
ADD_IMPORT extern const cct NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_IN_MEV_2014;
ADD_IMPORT extern const cct NEUTRON_PROTON_MASS_DIFFERENCE_IN_U_2014;
ADD_IMPORT extern const cct NEUTRON_PROTON_MASS_RATIO_2014;
ADD_IMPORT extern const cct NEUTRON_TAU_MASS_RATIO_2014;
ADD_IMPORT extern const cct NEUTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2014;
ADD_IMPORT extern const cct NEWTONIAN_CONSTANT_OF_GRAVITATION_2014;
ADD_IMPORT extern const cct NEWTONIAN_CONSTANT_OF_GRAVITATION_OVER_H_BAR_C_2014;
ADD_IMPORT extern const cct NUCLEAR_MAGNETON_2014;
ADD_IMPORT extern const cct NUCLEAR_MAGNETON_IN_EV_T_2014;
ADD_IMPORT extern const cct NUCLEAR_MAGNETON_IN_INVERSE_METERS_PER_TESLA_2014;
ADD_IMPORT extern const cct NUCLEAR_MAGNETON_IN_K_T_2014;
ADD_IMPORT extern const cct NUCLEAR_MAGNETON_IN_MHZ_T_2014;
ADD_IMPORT extern const cct PLANCK_CONSTANT_2014;
ADD_IMPORT extern const cct PLANCK_CONSTANT_IN_EV_S_2014;
ADD_IMPORT extern const cct PLANCK_CONSTANT_OVER_2_PI_2014;
ADD_IMPORT extern const cct PLANCK_CONSTANT_OVER_2_PI_IN_EV_S_2014;
ADD_IMPORT extern const cct PLANCK_CONSTANT_OVER_2_PI_TIMES_C_IN_MEV_FM_2014;
ADD_IMPORT extern const cct PLANCK_LENGTH_2014;
ADD_IMPORT extern const cct PLANCK_MASS_2014;
ADD_IMPORT extern const cct PLANCK_MASS_ENERGY_EQUIVALENT_IN_GEV_2014;
ADD_IMPORT extern const cct PLANCK_TEMPERATURE_2014;
ADD_IMPORT extern const cct PLANCK_TIME_2014;
ADD_IMPORT extern const cct PROTON_CHARGE_TO_MASS_QUOTIENT_2014;
ADD_IMPORT extern const cct PROTON_COMPTON_WAVELENGTH_2014;
ADD_IMPORT extern const cct PROTON_COMPTON_WAVELENGTH_OVER_2_PI_2014;
ADD_IMPORT extern const cct PROTON_ELECTRON_MASS_RATIO_2014;
ADD_IMPORT extern const cct PROTON_G_FACTOR_2014;
ADD_IMPORT extern const cct PROTON_GYROMAG_RATIO_2014;
ADD_IMPORT extern const cct PROTON_GYROMAG_RATIO_OVER_2_PI_2014;
ADD_IMPORT extern const cct PROTON_MAG_MOM_2014;
ADD_IMPORT extern const cct PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014;
ADD_IMPORT extern const cct PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014;
ADD_IMPORT extern const cct PROTON_MAG_SHIELDING_CORRECTION_2014;
ADD_IMPORT extern const cct PROTON_MASS_2014;
ADD_IMPORT extern const cct PROTON_MASS_ENERGY_EQUIVALENT_2014;
ADD_IMPORT extern const cct PROTON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014;
ADD_IMPORT extern const cct PROTON_MASS_IN_U_2014;
ADD_IMPORT extern const cct PROTON_MOLAR_MASS_2014;
ADD_IMPORT extern const cct PROTON_MUON_MASS_RATIO_2014;
ADD_IMPORT extern const cct PROTON_NEUTRON_MAG_MOM_RATIO_2014;
ADD_IMPORT extern const cct PROTON_NEUTRON_MASS_RATIO_2014;
ADD_IMPORT extern const cct PROTON_RMS_CHARGE_RADIUS_2014;
ADD_IMPORT extern const cct PROTON_TAU_MASS_RATIO_2014;
ADD_IMPORT extern const cct QUANTUM_OF_CIRCULATION_2014;
ADD_IMPORT extern const cct QUANTUM_OF_CIRCULATION_TIMES_2_2014;
ADD_IMPORT extern const cct RYDBERG_CONSTANT_2014;
ADD_IMPORT extern const cct RYDBERG_CONSTANT_TIMES_C_IN_HZ_2014;
ADD_IMPORT extern const cct RYDBERG_CONSTANT_TIMES_HC_IN_EV_2014;
ADD_IMPORT extern const cct RYDBERG_CONSTANT_TIMES_HC_IN_J_2014;
ADD_IMPORT extern const cct SACKUR_TETRODE_CONSTANT_1_K_100_KPA_2014;
ADD_IMPORT extern const cct SACKUR_TETRODE_CONSTANT_1_K_101_325_KPA_2014;
ADD_IMPORT extern const cct SECOND_RADIATION_CONSTANT_2014;
ADD_IMPORT extern const cct SHIELDED_HELION_GYROMAG_RATIO_2014;
ADD_IMPORT extern const cct SHIELDED_HELION_GYROMAG_RATIO_OVER_2_PI_2014;
ADD_IMPORT extern const cct SHIELDED_HELION_MAG_MOM_2014;
ADD_IMPORT extern const cct SHIELDED_HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014;
ADD_IMPORT extern const cct SHIELDED_HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014;
ADD_IMPORT extern const cct SHIELDED_HELION_TO_PROTON_MAG_MOM_RATIO_2014;
ADD_IMPORT extern const cct SHIELDED_HELION_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2014;
ADD_IMPORT extern const cct SHIELDED_PROTON_GYROMAG_RATIO_2014;
ADD_IMPORT extern const cct SHIELDED_PROTON_GYROMAG_RATIO_OVER_2_PI_2014;
ADD_IMPORT extern const cct SHIELDED_PROTON_MAG_MOM_2014;
ADD_IMPORT extern const cct SHIELDED_PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014;
ADD_IMPORT extern const cct SHIELDED_PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014;
ADD_IMPORT extern const cct SPEED_OF_LIGHT_IN_VACUUM_2014;
ADD_IMPORT extern const cct STANDARD_ACCELERATION_OF_GRAVITY_2014;
ADD_IMPORT extern const cct STANDARD_ATMOSPHERE_2014;
ADD_IMPORT extern const cct STANDARD_STATE_PRESSURE_2014;
ADD_IMPORT extern const cct STEFAN_BOLTZMANN_CONSTANT_2014;
ADD_IMPORT extern const cct TAU_COMPTON_WAVELENGTH_2014;
ADD_IMPORT extern const cct TAU_COMPTON_WAVELENGTH_OVER_2_PI_2014;
ADD_IMPORT extern const cct TAU_ELECTRON_MASS_RATIO_2014;
ADD_IMPORT extern const cct TAU_MASS_2014;
ADD_IMPORT extern const cct TAU_MASS_ENERGY_EQUIVALENT_2014;
ADD_IMPORT extern const cct TAU_MASS_ENERGY_EQUIVALENT_IN_MEV_2014;
ADD_IMPORT extern const cct TAU_MASS_IN_U_2014;
ADD_IMPORT extern const cct TAU_MOLAR_MASS_2014;
ADD_IMPORT extern const cct TAU_MUON_MASS_RATIO_2014;
ADD_IMPORT extern const cct TAU_NEUTRON_MASS_RATIO_2014;
ADD_IMPORT extern const cct TAU_PROTON_MASS_RATIO_2014;
ADD_IMPORT extern const cct THOMSON_CROSS_SECTION_2014;
ADD_IMPORT extern const cct TRITON_ELECTRON_MASS_RATIO_2014;
ADD_IMPORT extern const cct TRITON_G_FACTOR_2014;
ADD_IMPORT extern const cct TRITON_MAG_MOM_2014;
ADD_IMPORT extern const cct TRITON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014;
ADD_IMPORT extern const cct TRITON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014;
ADD_IMPORT extern const cct TRITON_MASS_2014;
ADD_IMPORT extern const cct TRITON_MASS_ENERGY_EQUIVALENT_2014;
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ADD_IMPORT extern const cct TRITON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014;
ADD_IMPORT extern const cct TRITON_MASS_IN_U_2014;
ADD_IMPORT extern const cct TRITON_MOLAR_MASS_2014;
ADD_IMPORT extern const cct TRITON_PROTON_MASS_RATIO_2014;
ADD_IMPORT extern const cct UNIFIED_ATOMIC_MASS_UNIT_2014;
ADD_IMPORT extern const cct VON_KLITZING_CONSTANT_2014;
ADD_IMPORT extern const cct WEAK_MIXING_ANGLE_2014;
ADD_IMPORT extern const cct WIEN_FREQUENCY_DISPLACEMENT_LAW_CONSTANT_2014;
ADD_IMPORT extern const cct WIEN_WAVELENGTH_DISPLACEMENT_LAW_CONSTANT_2014;
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// CODATA_CONSTANTS_2018
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ADD_IMPORT extern const int YEAR_2018;
ADD_IMPORT extern const cct ALPHA_PARTICLE_ELECTRON_MASS_RATIO_2018;
ADD_IMPORT extern const cct ALPHA_PARTICLE_MASS_2018;
ADD_IMPORT extern const cct ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_2018;
ADD_IMPORT extern const cct ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_IN_MEV_2018;
ADD_IMPORT extern const cct ALPHA_PARTICLE_MASS_IN_U_2018;
ADD_IMPORT extern const cct ALPHA_PARTICLE_MOLAR_MASS_2018;
ADD_IMPORT extern const cct ALPHA_PARTICLE_PROTON_MASS_RATIO_2018;
ADD_IMPORT extern const cct ALPHA_PARTICLE_RELATIVE_ATOMIC_MASS_2018;
ADD_IMPORT extern const cct ANGSTROM_STAR_2018;
ADD_IMPORT extern const cct ATOMIC_MASS_CONSTANT_2018;
ADD_IMPORT extern const cct ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_2018;
ADD_IMPORT extern const cct ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_IN_MEV_2018;
ADD_IMPORT extern const cct ATOMIC_MASS_UNIT_ELECTRON_VOLT_RELATIONSHIP_2018;
ADD_IMPORT extern const cct ATOMIC_MASS_UNIT_HARTREE_RELATIONSHIP_2018;
ADD_IMPORT extern const cct ATOMIC_MASS_UNIT_HERTZ_RELATIONSHIP_2018;
ADD_IMPORT extern const cct ATOMIC_MASS_UNIT_INVERSE_METER_RELATIONSHIP_2018;
ADD_IMPORT extern const cct ATOMIC_MASS_UNIT_JOULE_RELATIONSHIP_2018;
ADD_IMPORT extern const cct ATOMIC_MASS_UNIT_KELVIN_RELATIONSHIP_2018;
ADD_IMPORT extern const cct ATOMIC_MASS_UNIT_KILOGRAM_RELATIONSHIP_2018;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_1ST_HYPERPOLARIZABILITY_2018;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_2ND_HYPERPOLARIZABILITY_2018;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_ACTION_2018;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_CHARGE_2018;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_CHARGE_DENSITY_2018;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_CURRENT_2018;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_ELECTRIC_DIPOLE_MOM_2018;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_ELECTRIC_FIELD_2018;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_ELECTRIC_FIELD_GRADIENT_2018;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_ELECTRIC_POLARIZABILITY_2018;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_ELECTRIC_POTENTIAL_2018;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_ELECTRIC_QUADRUPOLE_MOM_2018;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_ENERGY_2018;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_FORCE_2018;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_LENGTH_2018;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_MAG_DIPOLE_MOM_2018;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_MAG_FLUX_DENSITY_2018;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_MAGNETIZABILITY_2018;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_MASS_2018;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_MOMENTUM_2018;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_PERMITTIVITY_2018;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_TIME_2018;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_VELOCITY_2018;
ADD_IMPORT extern const cct AVOGADRO_CONSTANT_2018;
ADD_IMPORT extern const cct BOHR_MAGNETON_2018;
ADD_IMPORT extern const cct BOHR_MAGNETON_IN_EV_T_2018;
ADD_IMPORT extern const cct BOHR_MAGNETON_IN_HZ_T_2018;
ADD_IMPORT extern const cct BOHR_MAGNETON_IN_INVERSE_METER_PER_TESLA_2018;
ADD_IMPORT extern const cct BOHR_MAGNETON_IN_K_T_2018;
ADD_IMPORT extern const cct BOHR_RADIUS_2018;
ADD_IMPORT extern const cct BOLTZMANN_CONSTANT_2018;
ADD_IMPORT extern const cct BOLTZMANN_CONSTANT_IN_EV_K_2018;
ADD_IMPORT extern const cct BOLTZMANN_CONSTANT_IN_HZ_K_2018;
ADD_IMPORT extern const cct BOLTZMANN_CONSTANT_IN_INVERSE_METER_PER_KELVIN_2018;
ADD_IMPORT extern const cct CHARACTERISTIC_IMPEDANCE_OF_VACUUM_2018;
ADD_IMPORT extern const cct CLASSICAL_ELECTRON_RADIUS_2018;
ADD_IMPORT extern const cct COMPTON_WAVELENGTH_2018;
ADD_IMPORT extern const cct CONDUCTANCE_QUANTUM_2018;
ADD_IMPORT extern const cct CONVENTIONAL_VALUE_OF_AMPERE_90_2018;
ADD_IMPORT extern const cct CONVENTIONAL_VALUE_OF_COULOMB_90_2018;
ADD_IMPORT extern const cct CONVENTIONAL_VALUE_OF_FARAD_90_2018;
ADD_IMPORT extern const cct CONVENTIONAL_VALUE_OF_HENRY_90_2018;
ADD_IMPORT extern const cct CONVENTIONAL_VALUE_OF_JOSEPHSON_CONSTANT_2018;
ADD_IMPORT extern const cct CONVENTIONAL_VALUE_OF_OHM_90_2018;
ADD_IMPORT extern const cct CONVENTIONAL_VALUE_OF_VOLT_90_2018;
ADD_IMPORT extern const cct CONVENTIONAL_VALUE_OF_VON_KLITZING_CONSTANT_2018;
ADD_IMPORT extern const cct CONVENTIONAL_VALUE_OF_WATT_90_2018;
ADD_IMPORT extern const cct COPPER_X_UNIT_2018;
ADD_IMPORT extern const cct DEUTERON_ELECTRON_MAG_MOM_RATIO_2018;
ADD_IMPORT extern const cct DEUTERON_ELECTRON_MASS_RATIO_2018;
ADD_IMPORT extern const cct DEUTERON_G_FACTOR_2018;
ADD_IMPORT extern const cct DEUTERON_MAG_MOM_2018;
ADD_IMPORT extern const cct DEUTERON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018;
ADD_IMPORT extern const cct DEUTERON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018;
ADD_IMPORT extern const cct DEUTERON_MASS_2018;
ADD_IMPORT extern const cct DEUTERON_MASS_ENERGY_EQUIVALENT_2018;
ADD_IMPORT extern const cct DEUTERON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018;
ADD_IMPORT extern const cct DEUTERON_MASS_IN_U_2018;
ADD_IMPORT extern const cct DEUTERON_MOLAR_MASS_2018;
ADD_IMPORT extern const cct DEUTERON_NEUTRON_MAG_MOM_RATIO_2018;
ADD_IMPORT extern const cct DEUTERON_PROTON_MAG_MOM_RATIO_2018;
ADD_IMPORT extern const cct DEUTERON_PROTON_MASS_RATIO_2018;
ADD_IMPORT extern const cct DEUTERON_RELATIVE_ATOMIC_MASS_2018;
ADD_IMPORT extern const cct DEUTERON_RMS_CHARGE_RADIUS_2018;
ADD_IMPORT extern const cct ELECTRON_CHARGE_TO_MASS_QUOTIENT_2018;
ADD_IMPORT extern const cct ELECTRON_DEUTERON_MAG_MOM_RATIO_2018;
ADD_IMPORT extern const cct ELECTRON_DEUTERON_MASS_RATIO_2018;
ADD_IMPORT extern const cct ELECTRON_G_FACTOR_2018;
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ADD_IMPORT extern const cct ELECTRON_GYROMAG_RATIO_2018;
ADD_IMPORT extern const cct ELECTRON_GYROMAG_RATIO_IN_MHZ_T_2018;
ADD_IMPORT extern const cct ELECTRON_HELION_MASS_RATIO_2018;
ADD_IMPORT extern const cct ELECTRON_MAG_MOM_2018;
ADD_IMPORT extern const cct ELECTRON_MAG_MOM_ANOMALY_2018;
ADD_IMPORT extern const cct ELECTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018;
ADD_IMPORT extern const cct ELECTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018;
ADD_IMPORT extern const cct ELECTRON_MASS_2018;
ADD_IMPORT extern const cct ELECTRON_MASS_ENERGY_EQUIVALENT_2018;
ADD_IMPORT extern const cct ELECTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018;
ADD_IMPORT extern const cct ELECTRON_MASS_IN_U_2018;
ADD_IMPORT extern const cct ELECTRON_MOLAR_MASS_2018;
ADD_IMPORT extern const cct ELECTRON_MUON_MAG_MOM_RATIO_2018;
ADD_IMPORT extern const cct ELECTRON_MUON_MASS_RATIO_2018;
ADD_IMPORT extern const cct ELECTRON_NEUTRON_MAG_MOM_RATIO_2018;
ADD_IMPORT extern const cct ELECTRON_NEUTRON_MASS_RATIO_2018;
ADD_IMPORT extern const cct ELECTRON_PROTON_MAG_MOM_RATIO_2018;
ADD_IMPORT extern const cct ELECTRON_PROTON_MASS_RATIO_2018;
ADD_IMPORT extern const cct ELECTRON_RELATIVE_ATOMIC_MASS_2018;
ADD_IMPORT extern const cct ELECTRON_TAU_MASS_RATIO_2018;
ADD_IMPORT extern const cct ELECTRON_TO_ALPHA_PARTICLE_MASS_RATIO_2018;
ADD_IMPORT extern const cct ELECTRON_TO_SHIELDED_HELION_MAG_MOM_RATIO_2018;
ADD_IMPORT extern const cct ELECTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2018;
ADD_IMPORT extern const cct ELECTRON_TRITON_MASS_RATIO_2018;
ADD_IMPORT extern const cct ELECTRON_VOLT_2018;
ADD_IMPORT extern const cct ELECTRON_VOLT_ATOMIC_MASS_UNIT_RELATIONSHIP_2018;
ADD_IMPORT extern const cct ELECTRON_VOLT_HARTREE_RELATIONSHIP_2018;
ADD_IMPORT extern const cct ELECTRON_VOLT_HERTZ_RELATIONSHIP_2018;
ADD_IMPORT extern const cct ELECTRON_VOLT_INVERSE_METER_RELATIONSHIP_2018;
ADD_IMPORT extern const cct ELECTRON_VOLT_JOULE_RELATIONSHIP_2018;
ADD_IMPORT extern const cct ELECTRON_VOLT_KELVIN_RELATIONSHIP_2018;
ADD_IMPORT extern const cct ELECTRON_VOLT_KILOGRAM_RELATIONSHIP_2018;
ADD_IMPORT extern const cct ELEMENTARY_CHARGE_2018;
ADD_IMPORT extern const cct ELEMENTARY_CHARGE_OVER_H_BAR_2018;
ADD_IMPORT extern const cct FARADAY_CONSTANT_2018;
ADD_IMPORT extern const cct FERMI_COUPLING_CONSTANT_2018;
ADD_IMPORT extern const cct FINE_STRUCTURE_CONSTANT_2018;
ADD_IMPORT extern const cct FIRST_RADIATION_CONSTANT_2018;
ADD_IMPORT extern const cct FIRST_RADIATION_CONSTANT_FOR_SPECTRAL_RADIANCE_2018;
ADD_IMPORT extern const cct HARTREE_ATOMIC_MASS_UNIT_RELATIONSHIP_2018;
ADD_IMPORT extern const cct HARTREE_ELECTRON_VOLT_RELATIONSHIP_2018;
ADD_IMPORT extern const cct HARTREE_ENERGY_2018;
ADD_IMPORT extern const cct HARTREE_ENERGY_IN_EV_2018;
ADD_IMPORT extern const cct HARTREE_HERTZ_RELATIONSHIP_2018;
ADD_IMPORT extern const cct HARTREE_INVERSE_METER_RELATIONSHIP_2018;
ADD_IMPORT extern const cct HARTREE_JOULE_RELATIONSHIP_2018;
ADD_IMPORT extern const cct HARTREE_KELVIN_RELATIONSHIP_2018;
ADD_IMPORT extern const cct HARTREE_KILOGRAM_RELATIONSHIP_2018;
ADD_IMPORT extern const cct HELION_ELECTRON_MASS_RATIO_2018;
ADD_IMPORT extern const cct HELION_G_FACTOR_2018;
ADD_IMPORT extern const cct HELION_MAG_MOM_2018;
ADD_IMPORT extern const cct HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018;
ADD_IMPORT extern const cct HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018;
ADD_IMPORT extern const cct HELION_MASS_2018;
ADD_IMPORT extern const cct HELION_MASS_ENERGY_EQUIVALENT_2018;
ADD_IMPORT extern const cct HELION_MASS_ENERGY_EQUIVALENT_IN_MEV_2018;
ADD_IMPORT extern const cct HELION_MASS_IN_U_2018;
ADD_IMPORT extern const cct HELION_MOLAR_MASS_2018;
ADD_IMPORT extern const cct HELION_PROTON_MASS_RATIO_2018;
ADD_IMPORT extern const cct HELION_RELATIVE_ATOMIC_MASS_2018;
ADD_IMPORT extern const cct HELION_SHIELDING_SHIFT_2018;
ADD_IMPORT extern const cct HERTZ_ATOMIC_MASS_UNIT_RELATIONSHIP_2018;
ADD_IMPORT extern const cct HERTZ_ELECTRON_VOLT_RELATIONSHIP_2018;
ADD_IMPORT extern const cct HERTZ_HARTREE_RELATIONSHIP_2018;
ADD_IMPORT extern const cct HERTZ_INVERSE_METER_RELATIONSHIP_2018;
ADD_IMPORT extern const cct HERTZ_JOULE_RELATIONSHIP_2018;
ADD_IMPORT extern const cct HERTZ_KELVIN_RELATIONSHIP_2018;
ADD_IMPORT extern const cct HERTZ_KILOGRAM_RELATIONSHIP_2018;
ADD_IMPORT extern const cct HYPERFINE_TRANSITION_FREQUENCY_OF_CS_133_2018;
ADD_IMPORT extern const cct INVERSE_FINE_STRUCTURE_CONSTANT_2018;
ADD_IMPORT extern const cct INVERSE_METER_ATOMIC_MASS_UNIT_RELATIONSHIP_2018;
ADD_IMPORT extern const cct INVERSE_METER_ELECTRON_VOLT_RELATIONSHIP_2018;
ADD_IMPORT extern const cct INVERSE_METER_HARTREE_RELATIONSHIP_2018;
ADD_IMPORT extern const cct INVERSE_METER_HERTZ_RELATIONSHIP_2018;
ADD_IMPORT extern const cct INVERSE_METER_JOULE_RELATIONSHIP_2018;
ADD_IMPORT extern const cct INVERSE_METER_KELVIN_RELATIONSHIP_2018;
ADD_IMPORT extern const cct INVERSE_METER_KILOGRAM_RELATIONSHIP_2018;
ADD_IMPORT extern const cct INVERSE_OF_CONDUCTANCE_QUANTUM_2018;
ADD_IMPORT extern const cct JOSEPHSON_CONSTANT_2018;
ADD_IMPORT extern const cct JOULE_ATOMIC_MASS_UNIT_RELATIONSHIP_2018;
ADD_IMPORT extern const cct JOULE_ELECTRON_VOLT_RELATIONSHIP_2018;
ADD_IMPORT extern const cct JOULE_HARTREE_RELATIONSHIP_2018;
ADD_IMPORT extern const cct JOULE_HERTZ_RELATIONSHIP_2018;
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ADD_IMPORT extern const cct JOULE_KILOGRAM_RELATIONSHIP_2018;
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ADD_IMPORT extern const cct KELVIN_ELECTRON_VOLT_RELATIONSHIP_2018;
ADD_IMPORT extern const cct KELVIN_HARTREE_RELATIONSHIP_2018;
ADD_IMPORT extern const cct KELVIN_HERTZ_RELATIONSHIP_2018;
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ADD_IMPORT extern const cct KELVIN_KILOGRAM_RELATIONSHIP_2018;
ADD_IMPORT extern const cct KILOGRAM_ATOMIC_MASS_UNIT_RELATIONSHIP_2018;
ADD_IMPORT extern const cct KILOGRAM_ELECTRON_VOLT_RELATIONSHIP_2018;
ADD_IMPORT extern const cct KILOGRAM_HARTREE_RELATIONSHIP_2018;
ADD_IMPORT extern const cct KILOGRAM_HERTZ_RELATIONSHIP_2018;
ADD_IMPORT extern const cct KILOGRAM_INVERSE_METER_RELATIONSHIP_2018;
ADD_IMPORT extern const cct KILOGRAM_JOULE_RELATIONSHIP_2018;
ADD_IMPORT extern const cct KILOGRAM_KELVIN_RELATIONSHIP_2018;
ADD_IMPORT extern const cct LATTICE_PARAMETER_OF_SILICON_2018;
ADD_IMPORT extern const cct LATTICE_SPACING_OF_IDEAL_SI_220_2018;
ADD_IMPORT extern const cct LOSCHMIDT_CONSTANT_273_15_K_100_KPA_2018;
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ADD_IMPORT extern const cct LOSCHMIDT_CONSTANT_273_15_K_101_325_KPA_2018;
ADD_IMPORT extern const cct LUMINOUS_EFFICACY_2018;
ADD_IMPORT extern const cct MAG_FLUX_QUANTUM_2018;
ADD_IMPORT extern const cct MOLAR_GAS_CONSTANT_2018;
ADD_IMPORT extern const cct MOLAR_MASS_CONSTANT_2018;
ADD_IMPORT extern const cct MOLAR_MASS_OF_CARBON_12_2018;
ADD_IMPORT extern const cct MOLAR_PLANCK_CONSTANT_2018;
ADD_IMPORT extern const cct MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_100_KPA_2018;
ADD_IMPORT extern const cct MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_101_325_KPA_2018;
ADD_IMPORT extern const cct MOLAR_VOLUME_OF_SILICON_2018;
ADD_IMPORT extern const cct MOLYBDENUM_X_UNIT_2018;
ADD_IMPORT extern const cct MUON_COMPTON_WAVELENGTH_2018;
ADD_IMPORT extern const cct MUON_ELECTRON_MASS_RATIO_2018;
ADD_IMPORT extern const cct MUON_G_FACTOR_2018;
ADD_IMPORT extern const cct MUON_MAG_MOM_2018;
ADD_IMPORT extern const cct MUON_MAG_MOM_ANOMALY_2018;
ADD_IMPORT extern const cct MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018;
ADD_IMPORT extern const cct MUON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018;
ADD_IMPORT extern const cct MUON_MASS_2018;
ADD_IMPORT extern const cct MUON_MASS_ENERGY_EQUIVALENT_2018;
ADD_IMPORT extern const cct MUON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018;
ADD_IMPORT extern const cct MUON_MASS_IN_U_2018;
ADD_IMPORT extern const cct MUON_MOLAR_MASS_2018;
ADD_IMPORT extern const cct MUON_NEUTRON_MASS_RATIO_2018;
ADD_IMPORT extern const cct MUON_PROTON_MAG_MOM_RATIO_2018;
ADD_IMPORT extern const cct MUON_PROTON_MASS_RATIO_2018;
ADD_IMPORT extern const cct MUON_TAU_MASS_RATIO_2018;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_ACTION_2018;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_ACTION_IN_EV_S_2018;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_ENERGY_2018;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_ENERGY_IN_MEV_2018;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_LENGTH_2018;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_MASS_2018;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_MOMENTUM_2018;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_MOMENTUM_IN_MEV_C_2018;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_TIME_2018;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_VELOCITY_2018;
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ADD_IMPORT extern const cct NEUTRON_ELECTRON_MAG_MOM_RATIO_2018;
ADD_IMPORT extern const cct NEUTRON_ELECTRON_MASS_RATIO_2018;
ADD_IMPORT extern const cct NEUTRON_G_FACTOR_2018;
ADD_IMPORT extern const cct NEUTRON_GYROMAG_RATIO_2018;
ADD_IMPORT extern const cct NEUTRON_GYROMAG_RATIO_IN_MHZ_T_2018;
ADD_IMPORT extern const cct NEUTRON_MAG_MOM_2018;
ADD_IMPORT extern const cct NEUTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018;
ADD_IMPORT extern const cct NEUTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018;
ADD_IMPORT extern const cct NEUTRON_MASS_2018;
ADD_IMPORT extern const cct NEUTRON_MASS_ENERGY_EQUIVALENT_2018;
ADD_IMPORT extern const cct NEUTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018;
ADD_IMPORT extern const cct NEUTRON_MASS_IN_U_2018;
ADD_IMPORT extern const cct NEUTRON_MOLAR_MASS_2018;
ADD_IMPORT extern const cct NEUTRON_MUON_MASS_RATIO_2018;
ADD_IMPORT extern const cct NEUTRON_PROTON_MAG_MOM_RATIO_2018;
ADD_IMPORT extern const cct NEUTRON_PROTON_MASS_DIFFERENCE_2018;
ADD_IMPORT extern const cct NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_2018;
ADD_IMPORT extern const cct NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_IN_MEV_2018;
ADD_IMPORT extern const cct NEUTRON_PROTON_MASS_DIFFERENCE_IN_U_2018;
ADD_IMPORT extern const cct NEUTRON_PROTON_MASS_RATIO_2018;
ADD_IMPORT extern const cct NEUTRON_RELATIVE_ATOMIC_MASS_2018;
ADD_IMPORT extern const cct NEUTRON_TAU_MASS_RATIO_2018;
ADD_IMPORT extern const cct NEUTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2018;
ADD_IMPORT extern const cct NEWTONIAN_CONSTANT_OF_GRAVITATION_2018;
ADD_IMPORT extern const cct NEWTONIAN_CONSTANT_OF_GRAVITATION_OVER_H_BAR_C_2018;
ADD_IMPORT extern const cct NUCLEAR_MAGNETON_2018;
ADD_IMPORT extern const cct NUCLEAR_MAGNETON_IN_EV_T_2018;
ADD_IMPORT extern const cct NUCLEAR_MAGNETON_IN_INVERSE_METER_PER_TESLA_2018;
ADD_IMPORT extern const cct NUCLEAR_MAGNETON_IN_K_T_2018;
ADD_IMPORT extern const cct NUCLEAR_MAGNETON_IN_MHZ_T_2018;
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ADD_IMPORT extern const cct PLANCK_CONSTANT_IN_EV_HZ_2018;
ADD_IMPORT extern const cct PLANCK_LENGTH_2018;
ADD_IMPORT extern const cct PLANCK_MASS_2018;
ADD_IMPORT extern const cct PLANCK_MASS_ENERGY_EQUIVALENT_IN_GEV_2018;
ADD_IMPORT extern const cct PLANCK_TEMPERATURE_2018;
ADD_IMPORT extern const cct PLANCK_TIME_2018;
ADD_IMPORT extern const cct PROTON_CHARGE_TO_MASS_QUOTIENT_2018;
ADD_IMPORT extern const cct PROTON_COMPTON_WAVELENGTH_2018;
ADD_IMPORT extern const cct PROTON_ELECTRON_MASS_RATIO_2018;
ADD_IMPORT extern const cct PROTON_G_FACTOR_2018;
ADD_IMPORT extern const cct PROTON_GYROMAG_RATIO_2018;
ADD_IMPORT extern const cct PROTON_GYROMAG_RATIO_IN_MHZ_T_2018;
ADD_IMPORT extern const cct PROTON_MAG_MOM_2018;
ADD_IMPORT extern const cct PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018;
ADD_IMPORT extern const cct PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018;
ADD_IMPORT extern const cct PROTON_MAG_SHIELDING_CORRECTION_2018;
ADD_IMPORT extern const cct PROTON_MASS_2018;
ADD_IMPORT extern const cct PROTON_MASS_ENERGY_EQUIVALENT_2018;
ADD_IMPORT extern const cct PROTON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018;
ADD_IMPORT extern const cct PROTON_MASS_IN_U_2018;
ADD_IMPORT extern const cct PROTON_MOLAR_MASS_2018;
ADD_IMPORT extern const cct PROTON_MUON_MASS_RATIO_2018;
ADD_IMPORT extern const cct PROTON_NEUTRON_MAG_MOM_RATIO_2018;
ADD_IMPORT extern const cct PROTON_NEUTRON_MASS_RATIO_2018;
ADD_IMPORT extern const cct PROTON_RELATIVE_ATOMIC_MASS_2018;
ADD_IMPORT extern const cct PROTON_RMS_CHARGE_RADIUS_2018;
ADD_IMPORT extern const cct PROTON_TAU_MASS_RATIO_2018;
ADD_IMPORT extern const cct QUANTUM_OF_CIRCULATION_2018;
ADD_IMPORT extern const cct QUANTUM_OF_CIRCULATION_TIMES_2_2018;
ADD_IMPORT extern const cct REDUCED_COMPTON_WAVELENGTH_2018;
ADD_IMPORT extern const cct REDUCED_MUON_COMPTON_WAVELENGTH_2018;
ADD_IMPORT extern const cct REDUCED_NEUTRON_COMPTON_WAVELENGTH_2018;
ADD_IMPORT extern const cct REDUCED_PLANCK_CONSTANT_2018;
ADD_IMPORT extern const cct REDUCED_PLANCK_CONSTANT_IN_EV_S_2018;
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ADD_IMPORT extern const cct REDUCED_PLANCK_CONSTANT_TIMES_C_IN_MEV_FM_2018;
ADD_IMPORT extern const cct REDUCED_PROTON_COMPTON_WAVELENGTH_2018;
ADD_IMPORT extern const cct REDUCED_TAU_COMPTON_WAVELENGTH_2018;
ADD_IMPORT extern const cct RYDBERG_CONSTANT_2018;
ADD_IMPORT extern const cct RYDBERG_CONSTANT_TIMES_C_IN_HZ_2018;
ADD_IMPORT extern const cct RYDBERG_CONSTANT_TIMES_HC_IN_EV_2018;
ADD_IMPORT extern const cct RYDBERG_CONSTANT_TIMES_HC_IN_J_2018;
ADD_IMPORT extern const cct SACKUR_TETRODE_CONSTANT_1_K_100_KPA_2018;
ADD_IMPORT extern const cct SACKUR_TETRODE_CONSTANT_1_K_101_325_KPA_2018;
ADD_IMPORT extern const cct SECOND_RADIATION_CONSTANT_2018;
ADD_IMPORT extern const cct SHIELDED_HELION_GYROMAG_RATIO_2018;
ADD_IMPORT extern const cct SHIELDED_HELION_GYROMAG_RATIO_IN_MHZ_T_2018;
ADD_IMPORT extern const cct SHIELDED_HELION_MAG_MOM_2018;
ADD_IMPORT extern const cct SHIELDED_HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018;
ADD_IMPORT extern const cct SHIELDED_HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018;
ADD_IMPORT extern const cct SHIELDED_HELION_TO_PROTON_MAG_MOM_RATIO_2018;
ADD_IMPORT extern const cct SHIELDED_HELION_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2018;
ADD_IMPORT extern const cct SHIELDED_PROTON_GYROMAG_RATIO_2018;
ADD_IMPORT extern const cct SHIELDED_PROTON_GYROMAG_RATIO_IN_MHZ_T_2018;
ADD_IMPORT extern const cct SHIELDED_PROTON_MAG_MOM_2018;
ADD_IMPORT extern const cct SHIELDED_PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018;
ADD_IMPORT extern const cct SHIELDED_PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018;
ADD_IMPORT extern const cct SHIELDING_DIFFERENCE_OF_D_AND_P_IN_HD_2018;
ADD_IMPORT extern const cct SHIELDING_DIFFERENCE_OF_T_AND_P_IN_HT_2018;
ADD_IMPORT extern const cct SPEED_OF_LIGHT_IN_VACUUM_2018;
ADD_IMPORT extern const cct STANDARD_ACCELERATION_OF_GRAVITY_2018;
ADD_IMPORT extern const cct STANDARD_ATMOSPHERE_2018;
ADD_IMPORT extern const cct STANDARD_STATE_PRESSURE_2018;
ADD_IMPORT extern const cct STEFAN_BOLTZMANN_CONSTANT_2018;
ADD_IMPORT extern const cct TAU_COMPTON_WAVELENGTH_2018;
ADD_IMPORT extern const cct TAU_ELECTRON_MASS_RATIO_2018;
ADD_IMPORT extern const cct TAU_ENERGY_EQUIVALENT_2018;
ADD_IMPORT extern const cct TAU_MASS_2018;
ADD_IMPORT extern const cct TAU_MASS_ENERGY_EQUIVALENT_2018;
ADD_IMPORT extern const cct TAU_MASS_IN_U_2018;
ADD_IMPORT extern const cct TAU_MOLAR_MASS_2018;
ADD_IMPORT extern const cct TAU_MUON_MASS_RATIO_2018;
ADD_IMPORT extern const cct TAU_NEUTRON_MASS_RATIO_2018;
ADD_IMPORT extern const cct TAU_PROTON_MASS_RATIO_2018;
ADD_IMPORT extern const cct THOMSON_CROSS_SECTION_2018;
ADD_IMPORT extern const cct TRITON_ELECTRON_MASS_RATIO_2018;
ADD_IMPORT extern const cct TRITON_G_FACTOR_2018;
ADD_IMPORT extern const cct TRITON_MAG_MOM_2018;
ADD_IMPORT extern const cct TRITON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018;
ADD_IMPORT extern const cct TRITON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018;
ADD_IMPORT extern const cct TRITON_MASS_2018;
ADD_IMPORT extern const cct TRITON_MASS_ENERGY_EQUIVALENT_2018;
ADD_IMPORT extern const cct TRITON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018;
ADD_IMPORT extern const cct TRITON_MASS_IN_U_2018;
ADD_IMPORT extern const cct TRITON_MOLAR_MASS_2018;
ADD_IMPORT extern const cct TRITON_PROTON_MASS_RATIO_2018;
ADD_IMPORT extern const cct TRITON_RELATIVE_ATOMIC_MASS_2018;
ADD_IMPORT extern const cct TRITON_TO_PROTON_MAG_MOM_RATIO_2018;
ADD_IMPORT extern const cct UNIFIED_ATOMIC_MASS_UNIT_2018;
ADD_IMPORT extern const cct VACUUM_ELECTRIC_PERMITTIVITY_2018;
ADD_IMPORT extern const cct VACUUM_MAG_PERMEABILITY_2018;
ADD_IMPORT extern const cct VON_KLITZING_CONSTANT_2018;
ADD_IMPORT extern const cct WEAK_MIXING_ANGLE_2018;
ADD_IMPORT extern const cct WIEN_FREQUENCY_DISPLACEMENT_LAW_CONSTANT_2018;
ADD_IMPORT extern const cct WIEN_WAVELENGTH_DISPLACEMENT_LAW_CONSTANT_2018;
ADD_IMPORT extern const cct W_TO_Z_MASS_RATIO_2018;
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// CODATA_CONSTANTS_2022
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ADD_IMPORT extern const int YEAR;
ADD_IMPORT extern const cct ALPHA_PARTICLE_ELECTRON_MASS_RATIO;
ADD_IMPORT extern const cct ALPHA_PARTICLE_MASS;
ADD_IMPORT extern const cct ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT;
ADD_IMPORT extern const cct ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_IN_MEV;
ADD_IMPORT extern const cct ALPHA_PARTICLE_MASS_IN_U;
ADD_IMPORT extern const cct ALPHA_PARTICLE_MOLAR_MASS;
ADD_IMPORT extern const cct ALPHA_PARTICLE_PROTON_MASS_RATIO;
ADD_IMPORT extern const cct ALPHA_PARTICLE_RELATIVE_ATOMIC_MASS;
ADD_IMPORT extern const cct ALPHA_PARTICLE_RMS_CHARGE_RADIUS;
ADD_IMPORT extern const cct ANGSTROM_STAR;
ADD_IMPORT extern const cct ATOMIC_MASS_CONSTANT;
ADD_IMPORT extern const cct ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT;
ADD_IMPORT extern const cct ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_IN_MEV;
ADD_IMPORT extern const cct ATOMIC_MASS_UNIT_ELECTRON_VOLT_RELATIONSHIP;
ADD_IMPORT extern const cct ATOMIC_MASS_UNIT_HARTREE_RELATIONSHIP;
ADD_IMPORT extern const cct ATOMIC_MASS_UNIT_HERTZ_RELATIONSHIP;
ADD_IMPORT extern const cct ATOMIC_MASS_UNIT_INVERSE_METER_RELATIONSHIP;
ADD_IMPORT extern const cct ATOMIC_MASS_UNIT_JOULE_RELATIONSHIP;
ADD_IMPORT extern const cct ATOMIC_MASS_UNIT_KELVIN_RELATIONSHIP;
ADD_IMPORT extern const cct ATOMIC_MASS_UNIT_KILOGRAM_RELATIONSHIP;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_1ST_HYPERPOLARIZABILITY;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_2ND_HYPERPOLARIZABILITY;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_ACTION;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_CHARGE;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_CHARGE_DENSITY;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_CURRENT;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_ELECTRIC_DIPOLE_MOM;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_ELECTRIC_FIELD;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_ELECTRIC_FIELD_GRADIENT;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_ELECTRIC_POLARIZABILITY;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_ELECTRIC_POTENTIAL;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_ELECTRIC_QUADRUPOLE_MOM;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_ENERGY;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_FORCE;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_LENGTH;
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ADD_IMPORT extern const cct ATOMIC_UNIT_OF_MAG_DIPOLE_MOM;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_MAG_FLUX_DENSITY;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_MAGNETIZABILITY;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_MASS;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_MOMENTUM;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_PERMITTIVITY;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_TIME;
ADD_IMPORT extern const cct ATOMIC_UNIT_OF_VELOCITY;
ADD_IMPORT extern const cct AVOGADRO_CONSTANT;
ADD_IMPORT extern const cct BOHR_MAGNETON;
ADD_IMPORT extern const cct BOHR_MAGNETON_IN_EV_T;
ADD_IMPORT extern const cct BOHR_MAGNETON_IN_HZ_T;
ADD_IMPORT extern const cct BOHR_MAGNETON_IN_INVERSE_METER_PER_TESLA;
ADD_IMPORT extern const cct BOHR_MAGNETON_IN_K_T;
ADD_IMPORT extern const cct BOHR_RADIUS;
ADD_IMPORT extern const cct BOLTZMANN_CONSTANT;
ADD_IMPORT extern const cct BOLTZMANN_CONSTANT_IN_EV_K;
ADD_IMPORT extern const cct BOLTZMANN_CONSTANT_IN_HZ_K;
ADD_IMPORT extern const cct BOLTZMANN_CONSTANT_IN_INVERSE_METER_PER_KELVIN;
ADD_IMPORT extern const cct CHARACTERISTIC_IMPEDANCE_OF_VACUUM;
ADD_IMPORT extern const cct CLASSICAL_ELECTRON_RADIUS;
ADD_IMPORT extern const cct COMPTON_WAVELENGTH;
ADD_IMPORT extern const cct CONDUCTANCE_QUANTUM;
ADD_IMPORT extern const cct CONVENTIONAL_VALUE_OF_AMPERE_90;
ADD_IMPORT extern const cct CONVENTIONAL_VALUE_OF_COULOMB_90;
ADD_IMPORT extern const cct CONVENTIONAL_VALUE_OF_FARAD_90;
ADD_IMPORT extern const cct CONVENTIONAL_VALUE_OF_HENRY_90;
ADD_IMPORT extern const cct CONVENTIONAL_VALUE_OF_JOSEPHSON_CONSTANT;
ADD_IMPORT extern const cct CONVENTIONAL_VALUE_OF_OHM_90;
ADD_IMPORT extern const cct CONVENTIONAL_VALUE_OF_VOLT_90;
ADD_IMPORT extern const cct CONVENTIONAL_VALUE_OF_VON_KLITZING_CONSTANT;
ADD_IMPORT extern const cct CONVENTIONAL_VALUE_OF_WATT_90;
ADD_IMPORT extern const cct COPPER_X_UNIT;
ADD_IMPORT extern const cct DEUTERON_ELECTRON_MAG_MOM_RATIO;
ADD_IMPORT extern const cct DEUTERON_ELECTRON_MASS_RATIO;
ADD_IMPORT extern const cct DEUTERON_G_FACTOR;
ADD_IMPORT extern const cct DEUTERON_MAG_MOM;
ADD_IMPORT extern const cct DEUTERON_MAG_MOM_TO_BOHR_MAGNETON_RATIO;
ADD_IMPORT extern const cct DEUTERON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO;
ADD_IMPORT extern const cct DEUTERON_MASS;
ADD_IMPORT extern const cct DEUTERON_MASS_ENERGY_EQUIVALENT;
ADD_IMPORT extern const cct DEUTERON_MASS_ENERGY_EQUIVALENT_IN_MEV;
ADD_IMPORT extern const cct DEUTERON_MASS_IN_U;
ADD_IMPORT extern const cct DEUTERON_MOLAR_MASS;
ADD_IMPORT extern const cct DEUTERON_NEUTRON_MAG_MOM_RATIO;
ADD_IMPORT extern const cct DEUTERON_PROTON_MAG_MOM_RATIO;
ADD_IMPORT extern const cct DEUTERON_PROTON_MASS_RATIO;
ADD_IMPORT extern const cct DEUTERON_RELATIVE_ATOMIC_MASS;
ADD_IMPORT extern const cct DEUTERON_RMS_CHARGE_RADIUS;
ADD_IMPORT extern const cct ELECTRON_CHARGE_TO_MASS_QUOTIENT;
ADD_IMPORT extern const cct ELECTRON_DEUTERON_MAG_MOM_RATIO;
ADD_IMPORT extern const cct ELECTRON_DEUTERON_MASS_RATIO;
ADD_IMPORT extern const cct ELECTRON_G_FACTOR;
ADD_IMPORT extern const cct ELECTRON_GYROMAG_RATIO;
ADD_IMPORT extern const cct ELECTRON_GYROMAG_RATIO_IN_MHZ_T;
ADD_IMPORT extern const cct ELECTRON_HELION_MASS_RATIO;
ADD_IMPORT extern const cct ELECTRON_MAG_MOM;
ADD_IMPORT extern const cct ELECTRON_MAG_MOM_ANOMALY;
ADD_IMPORT extern const cct ELECTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO;
ADD_IMPORT extern const cct ELECTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO;
ADD_IMPORT extern const cct ELECTRON_MASS;
ADD_IMPORT extern const cct ELECTRON_MASS_ENERGY_EQUIVALENT;
ADD_IMPORT extern const cct ELECTRON_MASS_ENERGY_EQUIVALENT_IN_MEV;
ADD_IMPORT extern const cct ELECTRON_MASS_IN_U;
ADD_IMPORT extern const cct ELECTRON_MOLAR_MASS;
ADD_IMPORT extern const cct ELECTRON_MUON_MAG_MOM_RATIO;
ADD_IMPORT extern const cct ELECTRON_MUON_MASS_RATIO;
ADD_IMPORT extern const cct ELECTRON_NEUTRON_MAG_MOM_RATIO;
ADD_IMPORT extern const cct ELECTRON_NEUTRON_MASS_RATIO;
ADD_IMPORT extern const cct ELECTRON_PROTON_MAG_MOM_RATIO;
ADD_IMPORT extern const cct ELECTRON_PROTON_MASS_RATIO;
ADD_IMPORT extern const cct ELECTRON_RELATIVE_ATOMIC_MASS;
ADD_IMPORT extern const cct ELECTRON_TAU_MASS_RATIO;
ADD_IMPORT extern const cct ELECTRON_TO_ALPHA_PARTICLE_MASS_RATIO;
ADD_IMPORT extern const cct ELECTRON_TO_SHIELDED_HELION_MAG_MOM_RATIO;
ADD_IMPORT extern const cct ELECTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO;
ADD_IMPORT extern const cct ELECTRON_TRITON_MASS_RATIO;
ADD_IMPORT extern const cct ELECTRON_VOLT;
ADD_IMPORT extern const cct ELECTRON_VOLT_ATOMIC_MASS_UNIT_RELATIONSHIP;
ADD_IMPORT extern const cct ELECTRON_VOLT_HARTREE_RELATIONSHIP;
ADD_IMPORT extern const cct ELECTRON_VOLT_HERTZ_RELATIONSHIP;
ADD_IMPORT extern const cct ELECTRON_VOLT_INVERSE_METER_RELATIONSHIP;
ADD_IMPORT extern const cct ELECTRON_VOLT_JOULE_RELATIONSHIP;
ADD_IMPORT extern const cct ELECTRON_VOLT_KELVIN_RELATIONSHIP;
ADD_IMPORT extern const cct ELECTRON_VOLT_KILOGRAM_RELATIONSHIP;
ADD_IMPORT extern const cct ELEMENTARY_CHARGE;
ADD_IMPORT extern const cct ELEMENTARY_CHARGE_OVER_H_BAR;
ADD_IMPORT extern const cct FARADAY_CONSTANT;
ADD_IMPORT extern const cct FERMI_COUPLING_CONSTANT;
ADD_IMPORT extern const cct FINE_STRUCTURE_CONSTANT;
ADD_IMPORT extern const cct FIRST_RADIATION_CONSTANT;
ADD_IMPORT extern const cct FIRST_RADIATION_CONSTANT_FOR_SPECTRAL_RADIANCE;
ADD_IMPORT extern const cct HARTREE_ATOMIC_MASS_UNIT_RELATIONSHIP;
ADD_IMPORT extern const cct HARTREE_ELECTRON_VOLT_RELATIONSHIP;
ADD_IMPORT extern const cct HARTREE_ENERGY;
ADD_IMPORT extern const cct HARTREE_ENERGY_IN_EV;
ADD_IMPORT extern const cct HARTREE_HERTZ_RELATIONSHIP;
ADD_IMPORT extern const cct HARTREE_INVERSE_METER_RELATIONSHIP;
ADD_IMPORT extern const cct HARTREE_JOULE_RELATIONSHIP;
ADD_IMPORT extern const cct HARTREE_KELVIN_RELATIONSHIP;
ADD_IMPORT extern const cct HARTREE_KILOGRAM_RELATIONSHIP;
ADD_IMPORT extern const cct HELION_ELECTRON_MASS_RATIO;
ADD_IMPORT extern const cct HELION_G_FACTOR;
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ADD_IMPORT extern const cct HELION_MAG_MOM;
ADD_IMPORT extern const cct HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO;
ADD_IMPORT extern const cct HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO;
ADD_IMPORT extern const cct HELION_MASS;
ADD_IMPORT extern const cct HELION_MASS_ENERGY_EQUIVALENT;
ADD_IMPORT extern const cct HELION_MASS_ENERGY_EQUIVALENT_IN_MEV;
ADD_IMPORT extern const cct HELION_MASS_IN_U;
ADD_IMPORT extern const cct HELION_MOLAR_MASS;
ADD_IMPORT extern const cct HELION_PROTON_MASS_RATIO;
ADD_IMPORT extern const cct HELION_RELATIVE_ATOMIC_MASS;
ADD_IMPORT extern const cct HELION_SHIELDING_SHIFT;
ADD_IMPORT extern const cct HERTZ_ATOMIC_MASS_UNIT_RELATIONSHIP;
ADD_IMPORT extern const cct HERTZ_ELECTRON_VOLT_RELATIONSHIP;
ADD_IMPORT extern const cct HERTZ_HARTREE_RELATIONSHIP;
ADD_IMPORT extern const cct HERTZ_INVERSE_METER_RELATIONSHIP;
ADD_IMPORT extern const cct HERTZ_JOULE_RELATIONSHIP;
ADD_IMPORT extern const cct HERTZ_KELVIN_RELATIONSHIP;
ADD_IMPORT extern const cct HERTZ_KILOGRAM_RELATIONSHIP;
ADD_IMPORT extern const cct HYPERFINE_TRANSITION_FREQUENCY_OF_CS_133;
ADD_IMPORT extern const cct INVERSE_FINE_STRUCTURE_CONSTANT;
ADD_IMPORT extern const cct INVERSE_METER_ATOMIC_MASS_UNIT_RELATIONSHIP;
ADD_IMPORT extern const cct INVERSE_METER_ELECTRON_VOLT_RELATIONSHIP;
ADD_IMPORT extern const cct INVERSE_METER_HARTREE_RELATIONSHIP;
ADD_IMPORT extern const cct INVERSE_METER_HERTZ_RELATIONSHIP;
ADD_IMPORT extern const cct INVERSE_METER_JOULE_RELATIONSHIP;
ADD_IMPORT extern const cct INVERSE_METER_KELVIN_RELATIONSHIP;
ADD_IMPORT extern const cct INVERSE_METER_KILOGRAM_RELATIONSHIP;
ADD_IMPORT extern const cct INVERSE_OF_CONDUCTANCE_QUANTUM;
ADD_IMPORT extern const cct JOSEPHSON_CONSTANT;
ADD_IMPORT extern const cct JOULE_ATOMIC_MASS_UNIT_RELATIONSHIP;
ADD_IMPORT extern const cct JOULE_ELECTRON_VOLT_RELATIONSHIP;
ADD_IMPORT extern const cct JOULE_HARTREE_RELATIONSHIP;
ADD_IMPORT extern const cct JOULE_HERTZ_RELATIONSHIP;
ADD_IMPORT extern const cct JOULE_INVERSE_METER_RELATIONSHIP;
ADD_IMPORT extern const cct JOULE_KELVIN_RELATIONSHIP;
ADD_IMPORT extern const cct JOULE_KILOGRAM_RELATIONSHIP;
ADD_IMPORT extern const cct KELVIN_ATOMIC_MASS_UNIT_RELATIONSHIP;
ADD_IMPORT extern const cct KELVIN_ELECTRON_VOLT_RELATIONSHIP;
ADD_IMPORT extern const cct KELVIN_HARTREE_RELATIONSHIP;
ADD_IMPORT extern const cct KELVIN_HERTZ_RELATIONSHIP;
ADD_IMPORT extern const cct KELVIN_INVERSE_METER_RELATIONSHIP;
ADD_IMPORT extern const cct KELVIN_JOULE_RELATIONSHIP;
ADD_IMPORT extern const cct KELVIN_KILOGRAM_RELATIONSHIP;
ADD_IMPORT extern const cct KILOGRAM_ATOMIC_MASS_UNIT_RELATIONSHIP;
ADD_IMPORT extern const cct KILOGRAM_ELECTRON_VOLT_RELATIONSHIP;
ADD_IMPORT extern const cct KILOGRAM_HARTREE_RELATIONSHIP;
ADD_IMPORT extern const cct KILOGRAM_HERTZ_RELATIONSHIP;
ADD_IMPORT extern const cct KILOGRAM_INVERSE_METER_RELATIONSHIP;
ADD_IMPORT extern const cct KILOGRAM_JOULE_RELATIONSHIP;
ADD_IMPORT extern const cct KILOGRAM_KELVIN_RELATIONSHIP;
ADD_IMPORT extern const cct LATTICE_PARAMETER_OF_SILICON;
ADD_IMPORT extern const cct LATTICE_SPACING_OF_IDEAL_SI_220;
ADD_IMPORT extern const cct LOSCHMIDT_CONSTANT_273_15_K_100_KPA;
ADD_IMPORT extern const cct LOSCHMIDT_CONSTANT_273_15_K_101_325_KPA;
ADD_IMPORT extern const cct LUMINOUS_EFFICACY;
ADD_IMPORT extern const cct MAG_FLUX_QUANTUM;
ADD_IMPORT extern const cct MOLAR_GAS_CONSTANT;
ADD_IMPORT extern const cct MOLAR_MASS_CONSTANT;
ADD_IMPORT extern const cct MOLAR_MASS_OF_CARBON_12;
ADD_IMPORT extern const cct MOLAR_PLANCK_CONSTANT;
ADD_IMPORT extern const cct MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_100_KPA;
ADD_IMPORT extern const cct MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_101_325_KPA;
ADD_IMPORT extern const cct MOLAR_VOLUME_OF_SILICON;
ADD_IMPORT extern const cct MOLYBDENUM_X_UNIT;
ADD_IMPORT extern const cct MUON_COMPTON_WAVELENGTH;
ADD_IMPORT extern const cct MUON_ELECTRON_MASS_RATIO;
ADD_IMPORT extern const cct MUON_G_FACTOR;
ADD_IMPORT extern const cct MUON_MAG_MOM;
ADD_IMPORT extern const cct MUON_MAG_MOM_ANOMALY;
ADD_IMPORT extern const cct MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO;
ADD_IMPORT extern const cct MUON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO;
ADD_IMPORT extern const cct MUON_MASS;
ADD_IMPORT extern const cct MUON_MASS_ENERGY_EQUIVALENT;
ADD_IMPORT extern const cct MUON_MASS_ENERGY_EQUIVALENT_IN_MEV;
ADD_IMPORT extern const cct MUON_MASS_IN_U;
ADD_IMPORT extern const cct MUON_MOLAR_MASS;
ADD_IMPORT extern const cct MUON_NEUTRON_MASS_RATIO;
ADD_IMPORT extern const cct MUON_PROTON_MAG_MOM_RATIO;
ADD_IMPORT extern const cct MUON_PROTON_MASS_RATIO;
ADD_IMPORT extern const cct MUON_TAU_MASS_RATIO;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_ACTION;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_ACTION_IN_EV_S;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_ENERGY;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_ENERGY_IN_MEV;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_LENGTH;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_MASS;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_MOMENTUM;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_MOMENTUM_IN_MEV_C;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_TIME;
ADD_IMPORT extern const cct NATURAL_UNIT_OF_VELOCITY;
ADD_IMPORT extern const cct NEUTRON_COMPTON_WAVELENGTH;
ADD_IMPORT extern const cct NEUTRON_ELECTRON_MAG_MOM_RATIO;
ADD_IMPORT extern const cct NEUTRON_ELECTRON_MASS_RATIO;
ADD_IMPORT extern const cct NEUTRON_G_FACTOR;
ADD_IMPORT extern const cct NEUTRON_GYROMAG_RATIO;
ADD_IMPORT extern const cct NEUTRON_GYROMAG_RATIO_IN_MHZ_T;
ADD_IMPORT extern const cct NEUTRON_MAG_MOM;
ADD_IMPORT extern const cct NEUTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO;
ADD_IMPORT extern const cct NEUTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO;
ADD_IMPORT extern const cct NEUTRON_MASS;
ADD_IMPORT extern const cct NEUTRON_MASS_ENERGY_EQUIVALENT;
ADD_IMPORT extern const cct NEUTRON_MASS_ENERGY_EQUIVALENT_IN_MEV;
ADD_IMPORT extern const cct NEUTRON_MASS_IN_U;
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ADD_IMPORT extern const cct NEUTRON_MOLAR_MASS;
ADD_IMPORT extern const cct NEUTRON_MUON_MASS_RATIO;
ADD_IMPORT extern const cct NEUTRON_PROTON_MAG_MOM_RATIO;
ADD_IMPORT extern const cct NEUTRON_PROTON_MASS_DIFFERENCE;
ADD_IMPORT extern const cct NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT;
ADD_IMPORT extern const cct NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_IN_MEV;
ADD_IMPORT extern const cct NEUTRON_PROTON_MASS_DIFFERENCE_IN_U;
ADD_IMPORT extern const cct NEUTRON_PROTON_MASS_RATIO;
ADD_IMPORT extern const cct NEUTRON_RELATIVE_ATOMIC_MASS;
ADD_IMPORT extern const cct NEUTRON_TAU_MASS_RATIO;
ADD_IMPORT extern const cct NEUTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO;
ADD_IMPORT extern const cct NEWTONIAN_CONSTANT_OF_GRAVITATION;
ADD_IMPORT extern const cct NEWTONIAN_CONSTANT_OF_GRAVITATION_OVER_H_BAR_C;
ADD_IMPORT extern const cct NUCLEAR_MAGNETON;
ADD_IMPORT extern const cct NUCLEAR_MAGNETON_IN_EV_T;
ADD_IMPORT extern const cct NUCLEAR_MAGNETON_IN_INVERSE_METER_PER_TESLA;
ADD_IMPORT extern const cct NUCLEAR_MAGNETON_IN_K_T;
ADD_IMPORT extern const cct NUCLEAR_MAGNETON_IN_MHZ_T;
ADD_IMPORT extern const cct PLANCK_CONSTANT;
ADD_IMPORT extern const cct PLANCK_CONSTANT_IN_EV_HZ;
ADD_IMPORT extern const cct PLANCK_LENGTH;
ADD_IMPORT extern const cct PLANCK_MASS;
ADD_IMPORT extern const cct PLANCK_MASS_ENERGY_EQUIVALENT_IN_GEV;
ADD_IMPORT extern const cct PLANCK_TEMPERATURE;
ADD_IMPORT extern const cct PLANCK_TIME;
ADD_IMPORT extern const cct PROTON_CHARGE_TO_MASS_QUOTIENT;
ADD_IMPORT extern const cct PROTON_COMPTON_WAVELENGTH;
ADD_IMPORT extern const cct PROTON_ELECTRON_MASS_RATIO;
ADD_IMPORT extern const cct PROTON_G_FACTOR;
ADD_IMPORT extern const cct PROTON_GYROMAG_RATIO;
ADD_IMPORT extern const cct PROTON_GYROMAG_RATIO_IN_MHZ_T;
ADD_IMPORT extern const cct PROTON_MAG_MOM;
ADD_IMPORT extern const cct PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO;
ADD_IMPORT extern const cct PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO;
ADD_IMPORT extern const cct PROTON_MAG_SHIELDING_CORRECTION;
ADD_IMPORT extern const cct PROTON_MASS;
ADD_IMPORT extern const cct PROTON_MASS_ENERGY_EQUIVALENT;
ADD_IMPORT extern const cct PROTON_MASS_ENERGY_EQUIVALENT_IN_MEV;
ADD_IMPORT extern const cct PROTON_MASS_IN_U;
ADD_IMPORT extern const cct PROTON_MOLAR_MASS;
ADD_IMPORT extern const cct PROTON_MUON_MASS_RATIO;
ADD_IMPORT extern const cct PROTON_NEUTRON_MAG_MOM_RATIO;
ADD_IMPORT extern const cct PROTON_NEUTRON_MASS_RATIO;
ADD_IMPORT extern const cct PROTON_RELATIVE_ATOMIC_MASS;
ADD_IMPORT extern const cct PROTON_RMS_CHARGE_RADIUS;
ADD_IMPORT extern const cct PROTON_TAU_MASS_RATIO;
ADD_IMPORT extern const cct QUANTUM_OF_CIRCULATION;
ADD_IMPORT extern const cct QUANTUM_OF_CIRCULATION_TIMES_2;
ADD_IMPORT extern const cct REDUCED_COMPTON_WAVELENGTH;
ADD_IMPORT extern const cct REDUCED_MUON_COMPTON_WAVELENGTH;
ADD_IMPORT extern const cct REDUCED_NEUTRON_COMPTON_WAVELENGTH;
ADD_IMPORT extern const cct REDUCED_PLANCK_CONSTANT;
ADD_IMPORT extern const cct REDUCED_PLANCK_CONSTANT_IN_EV_S;
ADD_IMPORT extern const cct REDUCED_PLANCK_CONSTANT_TIMES_C_IN_MEV_FM;
ADD_IMPORT extern const cct REDUCED_PROTON_COMPTON_WAVELENGTH;
ADD_IMPORT extern const cct REDUCED_TAU_COMPTON_WAVELENGTH;
ADD_IMPORT extern const cct RYDBERG_CONSTANT;
ADD_IMPORT extern const cct RYDBERG_CONSTANT_TIMES_C_IN_HZ;
ADD_IMPORT extern const cct RYDBERG_CONSTANT_TIMES_HC_IN_EV;
ADD_IMPORT extern const cct RYDBERG_CONSTANT_TIMES_HC_IN_J;
ADD_IMPORT extern const cct SACKUR_TETRODE_CONSTANT_1_K_100_KPA;
ADD_IMPORT extern const cct SACKUR_TETRODE_CONSTANT_1_K_101_325_KPA;
ADD_IMPORT extern const cct SECOND_RADIATION_CONSTANT;
ADD_IMPORT extern const cct SHIELDED_HELION_GYROMAG_RATIO;
ADD_IMPORT extern const cct SHIELDED_HELION_GYROMAG_RATIO_IN_MHZ_T;
ADD_IMPORT extern const cct SHIELDED_HELION_MAG_MOM;
ADD_IMPORT extern const cct SHIELDED_HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO;
ADD_IMPORT extern const cct SHIELDED_HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO;
ADD_IMPORT extern const cct SHIELDED_HELION_TO_PROTON_MAG_MOM_RATIO;
ADD_IMPORT extern const cct SHIELDED_HELION_TO_SHIELDED_PROTON_MAG_MOM_RATIO;
ADD_IMPORT extern const cct SHIELDED_PROTON_GYROMAG_RATIO;
ADD_IMPORT extern const cct SHIELDED_PROTON_GYROMAG_RATIO_IN_MHZ_T;
ADD_IMPORT extern const cct SHIELDED_PROTON_MAG_MOM;
ADD_IMPORT extern const cct SHIELDED_PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO;
ADD_IMPORT extern const cct SHIELDED_PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO;
ADD_IMPORT extern const cct SHIELDING_DIFFERENCE_OF_D_AND_P_IN_HD;
ADD_IMPORT extern const cct SHIELDING_DIFFERENCE_OF_T_AND_P_IN_HT;
ADD_IMPORT extern const cct SPEED_OF_LIGHT_IN_VACUUM;
ADD_IMPORT extern const cct STANDARD_ACCELERATION_OF_GRAVITY;
ADD_IMPORT extern const cct STANDARD_ATMOSPHERE;
ADD_IMPORT extern const cct STANDARD_STATE_PRESSURE;
ADD_IMPORT extern const cct STEFAN_BOLTZMANN_CONSTANT;
ADD_IMPORT extern const cct TAU_COMPTON_WAVELENGTH;
ADD_IMPORT extern const cct TAU_ELECTRON_MASS_RATIO;
ADD_IMPORT extern const cct TAU_ENERGY_EQUIVALENT;
ADD_IMPORT extern const cct TAU_MASS;
ADD_IMPORT extern const cct TAU_MASS_ENERGY_EQUIVALENT;
ADD_IMPORT extern const cct TAU_MASS_IN_U;
ADD_IMPORT extern const cct TAU_MOLAR_MASS;
ADD_IMPORT extern const cct TAU_MUON_MASS_RATIO;
ADD_IMPORT extern const cct TAU_NEUTRON_MASS_RATIO;
ADD_IMPORT extern const cct TAU_PROTON_MASS_RATIO;
ADD_IMPORT extern const cct THOMSON_CROSS_SECTION;
ADD_IMPORT extern const cct TRITON_ELECTRON_MASS_RATIO;
ADD_IMPORT extern const cct TRITON_G_FACTOR;
ADD_IMPORT extern const cct TRITON_MAG_MOM;
ADD_IMPORT extern const cct TRITON_MAG_MOM_TO_BOHR_MAGNETON_RATIO;
ADD_IMPORT extern const cct TRITON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO;
ADD_IMPORT extern const cct TRITON_MASS;
ADD_IMPORT extern const cct TRITON_MASS_ENERGY_EQUIVALENT;
ADD_IMPORT extern const cct TRITON_MASS_ENERGY_EQUIVALENT_IN_MEV;
ADD_IMPORT extern const cct TRITON_MASS_IN_U;
ADD_IMPORT extern const cct TRITON_MOLAR_MASS;
```

```
ADD_IMPORT extern const cct TRITON_PROTON_MASS_RATIO;
ADD_IMPORT extern const cct TRITON_RELATIVE_ATOMIC_MASS;
ADD_IMPORT extern const cct TRITON_TO_PROTON_MAG_MOM_RATIO;
ADD_IMPORT extern const cct UNIFIED_ATOMIC_MASS_UNIT;
ADD_IMPORT extern const cct VACUUM_ELECTRIC_PERMITTIVITY;
ADD_IMPORT extern const cct VACUUM_MAG_PERMEABILITY;
ADD_IMPORT extern const cct VON_KLITZING_CONSTANT;
ADD_IMPORT extern const cct WEAK_MIXING_ANGLE;
ADD_IMPORT extern const cct WIEN_FREQUENCY_DISPLACEMENT_LAW_CONSTANT;
ADD_IMPORT extern const cct WIEN_WAVELENGTH_DISPLACEMENT_LAW_CONSTANT;
ADD_IMPORT extern const cct W_TO_Z_MASS_RATIO;
//}}}

#endif
```

Bibliography

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